

Supplemental Materials

Times Square sampling: an adaptive algorithm for free energy estimation

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Part I

Theory

1 Stochastic approximation and iterative importance sampling

In this section we describe in some more detail the stochastic approximation procedure, and illustrate its relation to the idea of “iterative” importance sampling, which is how we originally derived the TSS recursions. This alternate perspective may provide some insight into the algorithm and some of its features, such as the origin of visit control, the choice of coordinates, and history forgetting.

1.1 Stochastic approximation

Stochastic approximation [24], and can be roughly summarized as a stochastic root finding procedure. We consider the procedure with Markovian noise and explain its use in on-the-fly estimation of free energy differences. In recent years, stochastic approximation has received significant attention in the fields of MCMC and MD, specifically adaptive algorithms (see [10, 7, 16] and the survey refs. [4, 3, 1]). For more complete references we refer the reader to [6, 15].

Let $\mathbb{Y} \subset \mathbb{R}^{n_y}$ and $\Theta \subset \mathbb{R}^{n_\theta}$ denote subsets of Euclidean space, and for each $\theta \in \Theta$, $y \in \mathbb{Y}$, let $P_\theta(y, dy')$ denote a Markov transition kernel with unique stationary distribution $p_\theta(y)dy$, where dy is taken to be the Lebesgue measure for simplicity. Suppose we wish to find a root θ^* of $g(\theta) = 0$, where $g : \Theta \rightarrow \mathbb{R}^{n_\theta}$ is defined by an expectation with respect to the stationary distribution $p_\theta(y)dy$:

$$g(\theta) = \mathbb{E}_\theta[G(\mathcal{Y}; \theta)] = \int_{\mathbb{Y}} G(y; \theta) p_\theta(y) dy. \quad (1.1)$$

The function $g(\theta)$ is sometimes referred to as the “mean field” of the observable $G(y; \theta)$.

Fix an initial guess $(\mathcal{Y}^0, \theta^0) \in \mathbb{Y} \times \Theta$, and for $t \geq 0$ define the process

$$\mathcal{Y}^{t+1} \sim P_{\theta^t}(\mathcal{Y}^t, \cdot) \quad (1.2)$$

$$\theta^{t+1} = \theta^t + \frac{1}{t+1} \Gamma G(\mathcal{Y}^{t+1}; \theta^t). \quad (1.3)$$

where Γ is an $n_\theta \times n_\theta$ matrix called the *gain matrix* and for each $t \geq 0$ t^{-1} is the *gain factor*. In the limit $t \rightarrow \infty$, the average behavior of the iterates θ^t is described by the ordinary differential equation (ODE) expressed in terms of the mean field g ,

$$\dot{\theta}(t) = \Gamma g(\theta(t)). \quad (1.4)$$

(Here, the overhead dot notation indicates a time derivative.) This “mean field ODE” can be derived heuristically from (1.3) by rearranging the left hand side to form a difference quotient, formally taking limits $t \rightarrow \infty$, and replacing the right hand side by its average $\Gamma g(\theta(t))$. The theory of stochastic approximation makes this heuristic rigorous, by leveraging stability properties of the ODE to establish convergence of the stochastic iterates θ^t . One can choose a more general gain factor of $t^{-\beta}$ for $\beta \in (0.5, 1]$ and preserve convergence, but $\beta = 1$ is needed to obtain the minimal asymptotic variance of the estimate θ^t . With $\beta = 1$, it can further be shown that the asymptotically optimal gain matrix is $\Gamma = -(J_g(\theta^*))^{-1}$, where $J_g(\theta)$ denotes the Jacobian of g evaluated at θ (see, for example, Section 3.2.3, Proposition 4 of [6]). Under certain regularity conditions on the function G , the transition kernels P_θ , and the gain Γ , one can establish the almost sure convergence of θ^t to θ^* (see [6] or [15]).

For some fixed $\gamma \in \mathcal{P}^e(K)$, the stochastic approximation procedure with Markovian noise can be applied to free energy estimation by taking $\mathcal{Y} = (\mathcal{X}^t, \mathcal{K}^t)$, the free energy F as the parameter θ , and the kernel to be the simulated tempering transition kernel $P_{\gamma, F}$, defined in Section 2.1 of the main text as the composition of two samplers $P_{\gamma, F}^x$ and $P_{\gamma, F}^k$.

An interesting observation is that both $\mathcal{X}$ and $\mathcal{K}$ can be used separately in observables $G = (G_1, \dots, G_K)$ which give the same mean field $g = (g_1, \dots, g_K)$. Namely,

$$G_k((x', k'); F) = 1 - \frac{e^{F_k - H_k(x')}}{\sum_{\ell \in [K]} \gamma_\ell e^{F_\ell - H_\ell(x')}} \quad (1.5)$$

and

$$G_k((x', k'); F) = 1 - \frac{1}{\gamma_k} 1_{\{k\}}(k') \quad (1.6)$$

both have as mean field

$$g_K(F) = 1 - \frac{e^{F_k - F_k^*}}{\sum_{\ell \in [K]} \gamma_\ell e^{F_\ell - F_\ell^*}}. \quad (1.7)$$

Tan applied the Robbins-Monro algorithm to these (and other related) observables and made the crucial observation that the optimal gain matrix is trivial, being given by $\Gamma = I_{K \times K}$ (cf. Theorem 1 in [28]). The stochastic approximation recursion obtained from (1.5) is the Rao-Blackwellization of (1.6) (see [28]). In particular, the recursion obtained from (1.5) is

$$F_k^{t+1} = F_k^t + \frac{1}{t+1} \left(1 - \frac{e^{F_k^t - H_k(\mathcal{X}^t)}}{\sum_{\ell \in [K]} \gamma_\ell e^{F_\ell^t - H_\ell(\mathcal{X}^t)}} \right) \quad (1.8)$$

1.2 Iterative importance sampling

For a fixed pair (γ, F) , the empirical distribution of the rung trajectory $\mathcal{K}^t$ using the sampler $P_{\gamma, F}$ would converge to the marginal distribution $p_{\gamma, F}^{[K]}(k) \mathcal{C}(dk)$ with probability mass function

$$p_{\gamma, F}^{[K]}(k) = \frac{\gamma_k e^{F_k - F_k^*}}{\sum_{\ell \in [K]} \gamma_\ell e^{F_\ell - F_\ell^*}}. \quad (1.9)$$

In other words, under certain ergodicity assumptions,

$$\lim_{t \rightarrow \infty} \frac{1}{t} \sum_{s=1}^t 1_{\{k\}}(\mathcal{K}^s) = \frac{\gamma_k e^{F_k - F_k^*}}{\sum_{\ell \in [K]} \gamma_\ell e^{F_\ell - F_\ell^*}}. \quad (1.10)$$

Because the hypothesized free energies F are known, in principle (1.10) allows for the determination of the exact free energy differences. Unless the hypothesized and exact free energies are close to each other, however, the visitation of some rungs is a rare event, which leads to very high error in the free energy difference estimates.

The free energies can also be estimated using the variable $\mathcal{X}$ by means of importance sampling, where $\mathcal{X}^t$ is drawn from the mixture density

$$p_{\gamma, F}^S(x) \propto \sum_{\ell \in [K]} \gamma_\ell e^{F_\ell - H_\ell(x)}. \quad (1.11)$$

An estimator analogous estimator to (1.10) can be obtained from

$$\lim_{t \rightarrow \infty} \frac{1}{t} \sum_{s=1}^t \frac{\gamma_k e^{F_k - H_k(\mathcal{X}^s)}}{\sum_{\ell \in [K]} \gamma_\ell e^{F_\ell - H_\ell(\mathcal{X}^s)}} = \frac{\gamma_k e^{F_k - F_k^*}}{\sum_{\ell \in [K]} \gamma_\ell e^{F_\ell - F_\ell^*}}. \quad (1.12)$$

Note that sampling from the marginal $p_{\gamma, F}^{[K]}(k)$ with a hypothesized parameter F and rung density γ can be thought of as sampling the same marginal with the *exact* (unknown) free energies F_k^* along with a *tilted* rung density $\tilde{\gamma}_k \propto \gamma_k e^{F_k - F_k^*}$. This interplay between the hypothesized free energies F and rung density γ allows us to use estimates based on $\mathcal{X}$ and $\mathcal{K}$ together to mutual advantage, resulting in the mechanism we call visit control.

Now we introduce the idea of iterative importance sampling (IIS), which turns out to be asymptotically equivalent to (1.3). First we note that the importance sampling estimator obtained from (1.12) behaves very differently from (1.3). When dealt samples drawn from a stationary distribution, (1.12) consistently estimates the free energy difference *regardless* of the nature of the mixture distribution from which the samples are drawn. Switching now to partition function coordinates $Z = -\log(F)$, with fixed but arbitrary hypothesized (γ, Z) , (1.12) can be re-arranged as

$$Z_k^t = \frac{1}{t} \sum_{s=1}^t \frac{e^{-H_k(\mathcal{X}^s)}}{\sum_{\ell \in [K]} \gamma_\ell e^{F_\ell - H_\ell(\mathcal{X}^s)}} \rightarrow \frac{Z_k^*}{\sum_{\ell \in [K]} \gamma_\ell Z_\ell^* / Z_\ell} \quad (1.13)$$

as $t \rightarrow \infty$. The efficiency of this method is acceptable only if the hypothesized partition functions are close to the exact ones. The form of (1.13), though, suggests that this sensitivity can be resolved by adjusting the hypothesized Z iteratively by setting it to the previous estimate of Z^* . Explicitly, we define the following iterative importance sampling (IIS) estimator,

$$Z_k^t = \frac{1}{t} \sum_{s=1}^t \frac{e^{-H_k(\mathcal{X}^s)}}{\sum_{\ell \in [K]} \gamma_\ell e^{-H_\ell(\mathcal{X}^s)} / Z_\ell^{s-1}}. \quad (1.14)$$

The IIS estimator converges with optimal asymptotic efficiency, as follows under certain conditions from Tan's work. The estimator takes the recursive form

$$Z_k^{t+1} = Z_k^t + \frac{1}{t+1} \left(\frac{e^{-H_k(\mathcal{X}^{t+1})}}{\sum_{\ell \in [K]} \gamma_\ell e^{-H_\ell(\mathcal{X}^{t+1})} / Z_\ell^t} - Z_k^t \right), \quad (1.15)$$

which in terms of the free energy $F_k = -\log Z_k$ reads

$$F_k^{t+1} = F_k^t - \log \left(1 + \frac{1}{t+1} \left[\frac{e^{F_k^t - H_k(\mathcal{X}^{t+1})}}{\sum_{\ell \in [K]} \gamma_\ell e^{F_\ell^t - H_\ell(\mathcal{X}^{t+1})}} - 1 \right] \right). \quad (1.16)$$

The iteration (1.16) (and hence (1.15)) is asymptotically equivalent to (1.7), as can be inferred by considering the expansion $\log(1+x) = x + O(x^2)$ and neglecting the $O(x^2)$ term, and all three recursions (1.6) (used in the general form (1.3) with $\Gamma = I$ and F instead of Θ), (1.7), and (1.16) feature the self-adjustment property demonstrated in the example above. Despite the asymptotic equivalence between (1.7) and (1.16), we will adopt the IIS formulation (1.15) (and hence the recursion (1.16)) because it makes transparent the interpretation of (1.14) as an importance sampling estimator when the adaptation of Z^t is frozen (i.e., when the expressions defining Z_k^t in (1.12) and (1.15) are the same). With this interpretation, the global convergence of (1.14) when samples are drawn from stationary mixtures follows from the Strong Law of Large

Numbers, whereas the same convergence is generally more difficult to establish for stochastic approximation recursions like (1.7), and does not hold for estimators driven purely by self-adjustment (such as the one defined through (1.6)). The same perspective on importance sampling will be used again in Section 2.3 to derive a history-forgetting mechanism.

In the following section, we will write the IIS estimator in stochastic approximation form by using (1.3) directly in terms of the partition function $Z = e^{-F}$, and we will favor Z as the coordinate to be used for derivations, as they are the natural coordinates for the IIS form (1.15). On a computer, the state of the parameter will be stored as F rather than Z , in order to cope with the limited range for the exponent of floating-point number formats, and so in our implementation of the TSS algorithm we will use (1.16).

2 Optimality for stochastic approximation

2.1 Changing variables from (F, μ) to (Z, ξ)

We recall the notation from Section 2 of the main text: the variables $(\mathcal{X}, \mathcal{K}) \in \mathcal{S} \times [K]$ are the pair of state and rung variables; the parameter $\theta = (F, \mu, o)$ is estimated throughout the simulation and determines the conditional distribution of $\mathcal{K}$ given $\mathcal{X}$. The parameter θ is equivalently used in Section 2.2.4 of the main text as $\theta = (Z, \xi, o)$, and the two are related by the changes of variables $Z = e^{-F}$ and $\xi = e^{-F} \mu$, or more explicitly as

$$Z_k = e^{-F_k}, \quad \xi_{km} = e^{-F_k} \mu_{km}, \quad k \in [K], m \in [M]. \quad (2.1)$$

We have the original recursions

$$Z_k^{t+1} = Z_k^t + \frac{1}{t+1} \left(\frac{e^{-H_k(\mathcal{X}^{t+1})}}{\sum_{\ell \in [K]} \pi_\ell(\theta^t) e^{-H_\ell(\mathcal{X}^{t+1})} / Z_\ell^t} - Z_k^t \right) \quad (2.2)$$

$$\xi_{km}^{t+1} = \xi_{km}^t + \frac{1}{t+1} \left(\frac{e^{-H_k(\mathcal{X}^{t+1})}}{\sum_{\ell \in [K]} \pi_\ell(\theta^t) e^{-H_\ell(\mathcal{X}^{t+1})} / Z_\ell^t} \Psi_m(\mathcal{X}^{t+1}) - \xi_{km}^t \right) \quad (2.3)$$

$$o_k^{t+1} = o_k^t + \frac{1}{t+1} \left(\frac{1}{\gamma_k(\theta^t)} 1_{\{k\}}(\mathcal{K}^{t+1}) - o_k^t \right) \quad (2.4)$$

with mean field

$$\dot{Z}_k = \frac{Z_k^*}{\sum_{\ell \in [K]} \pi_\ell(\theta(t)) Z_\ell^* / Z_\ell(t)} - Z_k(t) \quad (2.5)$$

$$\dot{\xi}_{km} = \frac{Z_k^*}{\sum_{\ell \in [K]} \pi_\ell(\theta(t)) Z_\ell^* / Z_\ell(t)} \mathbb{E}_k[\Psi_m(\mathcal{X})] - \xi_{km}(t) \quad (2.6)$$

$$\dot{o}_k = \frac{\pi_k(\theta(t))}{\gamma_k(\mu(t))} \frac{Z_k^* / Z_k(t)}{\sum_{\ell \in [K]} \pi_\ell(\theta(t)) Z_\ell^* / Z_\ell(t)} - o_k(t) \quad (2.7)$$

where $\theta(t) = (Z(t), \mu(t), o(t))$. In the variables (F, μ) , the first two recursions are

$$F_k^{t+1} = F_k^t - \log \left(1 + \frac{1}{t+1} \left[\frac{e^{F_k^t - H_k(\mathcal{X}^{t+1})}}{\sum_{\ell \in [K]} \pi_\ell e^{F_\ell^t - H_\ell(\mathcal{X}^{t+1})}} - 1 \right] \right) \quad (2.8)$$

$$\mu_{km}^{t+1} = e^{F_k^{t+1} - F_k^t} \left(\mu_{km}^t + \frac{1}{t+1} \left[\frac{e^{F_k^t - H_k(\mathcal{X}^{t+1})}}{\sum_{\ell \in [K]} \pi_\ell e^{F_\ell^t - H_\ell(\mathcal{X}^{t+1})}} \Psi_m(\mathcal{X}^{t+1}) - \mu_{km}^t \right] \right). \quad (2.9)$$

Observe that

$$\mu_{km}^{t+1} = e^{F_k^{t+1} - F_k^t} \left(\mu_{km}^t + \frac{1}{t+1} \left[\frac{e^{F_k^t - H_k(\mathcal{X}^{t+1})}}{\sum_{\ell \in [K]} \pi_\ell e^{F_\ell^t - H_\ell(\mathcal{X}^{t+1})}} \Psi_m(\mathcal{X}^{t+1}) - \mu_{km}^t \right] \right) \quad (2.10)$$

$$= \frac{\left(\mu_{km}^t + \frac{1}{t+1} \left[\frac{e^{F_k^t - H_k(\mathcal{X}^{t+1})}}{\sum_{\ell \in [K]} \pi_\ell e^{F_\ell^t - H_\ell(\mathcal{X}^{t+1})}} \Psi_m(\mathcal{X}^{t+1}) - \mu_{km}^t \right] \right)}{\left(1 + \frac{1}{t+1} \left[\frac{e^{F_k^t - H_k(\mathcal{X}^{t+1})}}{\sum_{\ell \in [K]} \pi_\ell e^{F_\ell^t - H_\ell(\mathcal{X}^{t+1})}} - 1 \right] \right)} \quad (2.11)$$

$$= \mu_{km}^t + \frac{1}{t+1} \frac{e^{F_k^t - H_k(\mathcal{X}^{t+1})}}{\sum_{\ell \in [K]} \pi_\ell e^{F_\ell^t - H_\ell(\mathcal{X}^{t+1})}} [\Psi_m(\mathcal{X}^{t+1}) - \mu_{km}^t] + O(t^{-2}). \quad (2.12)$$

In the variables (F, μ) , we therefore have the mean fields

$$\dot{F}_k = 1 - \frac{e^{F_k - F_k^*}}{\sum_{\ell \in [K]} \pi_\ell(\theta(t)) e^{F_\ell - F_\ell^*}} \quad (2.13)$$

$$\dot{\mu}_{km} = \frac{e^{F_k - F_k^*}}{\sum_{\ell \in [K]} \pi_\ell(\theta(t)) e^{F_\ell - F_\ell^*}} (\mu_{km}^* - \mu_{km}). \quad (2.14)$$

Using the change of variables (2.1) and the chain rule, we find that the two pairs of equations (2.13), (2.14) and (2.5), (2.6) are equivalent (although the corresponding discrete recursions are not equivalent). We will now identify the inverse of the Jacobian in the (F, μ, o) coordinates.

2.2 Identifying the inverse Jacobian

As discussed in Section 2.1 of the main text, in a stochastic approximation procedure of the form

$$\theta^{t+1} = \theta^t + \frac{1}{t+1} \Gamma G(Y^{t+1}; \theta^t), \quad (2.15)$$

the matrix Γ is called the gain matrix, and the noisy observable G has an associated mean field $g(\theta) = \mathbb{E}_\theta[G(Y; \theta)]$. The matrix that minimizes the asymptotic variance of θ^t (when θ^t converges) is $\Gamma = -J_g(\theta^*)^{-1}$, where $J_g(\theta)$ is the Jacobian of g evaluated at θ (see Section 3.2.3, Proposition 4 of [6]).

Consequently, in order to demonstrate the optimality of the stochastic approximation procedure for (F^t, μ^t) , it suffices to identify the inverse of the Jacobian of the expected gain $(g^F(\theta), g^\mu(\theta))$, with g^F and g^μ defined in Sections 2.1 and 2.2.2 of the main text. Tan accomplished this for the free energy estimates in [28] [Theorem 2]. It turns out this property can be extended to the case of estimating the averages

$$\mu_{km}^* = \mathbb{E}_k[\Psi_m(\mathcal{X})] = \int_S \Psi_m(x) \rho_k(x) dx. \quad (2.16)$$

The key to obtaining the optimal gain matrix is to use the assumption that $\sum_{\ell \in [K]} \pi_\ell(F, \mu, o) = 1$ for all $(F, \mu, o) \in \mathbb{R}^K \times \mathbb{R}^{KM} \times \mathcal{P}^e(K)$, which allows us to avoid computing the derivatives of π with respect to F, μ, o at the fixed point F^*, μ^* ; the Jacobian will turn out to be independent of $o \in \mathbb{R}_{\geq 0}^K$.

Proof [Of Proposition 1 in the main text]

The optimal gain matrix for a stochastic approximation procedure is given by the inverse of the Jacobian evaluated at the unique equilibrium point. We will show that the Jacobian for the mean field $(\bar{F}, \bar{\mu}) = (g^F(F, \mu, o), g^\mu(F, \mu, o))$ is equal to the identity at the equilibrium (F^*, μ^*) . We then show the same property extends to $\dot{o} = g^o(F, \mu, o)$ when

$$\pi_k(F, \mu, o) = \frac{\gamma_k(\mu) e^{F_k^* - F_k}}{\sum_{\ell \in [K]} \gamma_\ell(\mu) e^{F_\ell^* - F_\ell}} \quad (\equiv \pi_k^\circ(F, \mu, o)) \quad (2.17)$$

The stochastic approximation procedure for estimating the averages μ_{km}^* is defined by the system of equations

$$F_k^{t+1} = F_k^t + \frac{1}{t+1} (1 - R_k(\mathcal{X}^{t+1}; F^t, \mu^t, o^t)) \quad (2.18)$$

$$\mu_{km}^{t+1} = \mu^t + \frac{1}{t+1} R_k(\mathcal{X}^{t+1}; F^t, \mu^t, o^t) (\Psi_m(\mathcal{X}^{t+1}) - \mu^t), \quad (2.19)$$

where

$$R_k(x; F, \mu, o) = \frac{e^{F_k - H_k(x)}}{\sum_{\ell=1}^K \pi_\ell(F, \mu, o) e^{F_\ell - H_\ell(x)}}. \quad (2.20)$$

The mean fields are

$$g_k^F(F, \mu, o) = 1 - \frac{e^{F_k - F_k^*}}{\sum_{\ell \in [K]} \pi_\ell(F, \mu, o) e^{F_\ell - F_\ell^*}} \quad (2.21)$$

$$g_{km}^\mu(F, \mu, o) = \frac{e^{F_k - F_k^*}}{\sum_{\ell \in [K]} \pi_\ell(F, \mu, o) e^{F_\ell - F_\ell^*}} (\mu_{km}^* - \mu_{km}). \quad (2.22)$$

Let $g = (g_1^F, \dots, g_K^F, g_{11}^\mu, \dots, g_{KM}^\mu) \in \mathbb{R}^{K+KM}$, and let

$$r_k(F, \mu, o) = \frac{e^{F_k - F_k^*}}{\sum_{\ell \in [K]} \pi_\ell(F, \mu, o) e^{F_\ell - F_\ell^*}}. \quad (2.23)$$

To save space, we omit writing the explicit dependence of π_k on $\theta = (F, \mu, o)$. Note that

$$\frac{\partial r_k}{\partial F_j}(F, \mu, o) = \delta_{kj} \frac{e^{F_k - F_k^*}}{\sum_{\ell \in [K]} \pi_\ell e^{F_\ell - F_\ell^*}} \quad (2.24)$$

$$\begin{aligned} & - \frac{e^{F_k - F_k^*}}{\left(\sum_{\ell \in [K]} \pi_\ell e^{F_\ell - F_\ell^*} \right)^2} \left(\pi_j e^{F_j - F_j^*} + \sum_{\ell \in [K]} \frac{\partial \pi_\ell}{\partial F_j} e^{F_\ell - F_\ell^*} \right) \\ \frac{\partial r_k}{\partial \mu_{jn}}(F, \mu, o) &= - \frac{e^{F_k - F_k^*}}{\left(\sum_{\ell \in [K]} \pi_\ell e^{F_\ell - F_\ell^*} \right)^2} \left(\sum_{\ell \in [K]} \frac{\partial \pi_\ell}{\partial \mu_{jn}} e^{F_\ell - F_\ell^*} \right). \end{aligned} \quad (2.25)$$

We can write the derivatives of the mean field in terms of the partial derivatives of r_k , this time omitting

writing explicitly the dependence of r_k and its derivatives on F, μ :

$$\frac{\partial g_k^F}{\partial F_j}(F, \mu, o) = -\frac{\partial r_k}{\partial F_j} \quad (2.26)$$

$$\frac{\partial g_{km}^\mu}{\partial F_j}(F, \mu, o) = \frac{\partial r_k}{\partial F_j}(\mu_{km}^* - \mu_{km}) \quad (2.27)$$

$$\frac{\partial g_k^F}{\partial \mu_{jn}}(F, \mu) = -\frac{\partial r_k}{\partial \mu_{jn}} \quad (2.28)$$

$$\frac{\partial g_{km}^\mu}{\partial \mu_{jn}}(F, \mu, o) = \frac{\partial r_k}{\partial \mu_{jn}}(\mu_{km}^* - \mu_{km}) - \delta_{kj}\delta_{mn}r_k. \quad (2.29)$$

To determine the optimal gain matrix, we evaluate the Jacobian at the point

$$(F^*, \mu^*, o) = (F_1^*, \dots, F_K^*, \mu_{11}^*, \dots, \mu_{KM}^*, o_1, \dots, o_K),$$

where $o \in \mathbb{R}_{\geq 0}^K$ is arbitrary. To this end, we first evaluate the Jacobian at (F^*, μ, o) for some arbitrary $(\mu, o) \in \mathbb{R}^{KM} \times \mathbb{R}_{\geq 0}^K$, and start by noting that

$$\frac{\partial r_k}{\partial \mu_{jn}}(F^*, \mu) = -\sum_{\ell \in [K]} \frac{\partial \pi_\ell}{\partial \mu_{jn}}(F^*, \mu) = -\frac{\partial}{\partial \mu_{jn}} \sum_{\ell \in [K]} \pi_\ell(F^*, \mu) = 0, \quad (2.30)$$

$$\frac{\partial r_k}{\partial F_j}(F^*, \mu) = \delta_{kj} - \left(\pi_j + \sum_{\ell \in [K]} \frac{\partial \pi_\ell}{\partial F_j}(F^*, \mu) \right) = \delta_{kj} - \pi_j, \quad (2.31)$$

since $\sum_{\ell \in [K]} \pi_\ell(F, \mu, o) = 1$ for all $(F, \mu, o) \in \mathbb{R}^K \times \mathbb{R}^{KM} \times \mathbb{Z}_+^K$. It follows that

$$\frac{\partial g_k^F}{\partial F_j}(F^*, \mu^*, o) = -(\delta_{kj} - \pi_j(F^*, \mu^*, o)) \quad (2.32)$$

$$\frac{\partial g_{km}^\mu}{\partial F_j}(F^*, \mu^*, o) = 0 \quad (2.33)$$

$$\frac{\partial g_k^F}{\partial \mu_{jm}}(F^*, \mu^*, o) = 0 \quad (2.34)$$

$$\frac{\partial g_{km}^\mu}{\partial \mu_{jn}}(F^*, \mu^*, o) = -\delta_{kj}\delta_{mn}. \quad (2.35)$$

Using the notation $D_F g$ and $D_\mu g$ to denote the Jacobian of a function g with respect to F and μ respectively,

we can write the Jacobian J in block matrix form as

$$J(F^*, \mu^*, o) = \begin{pmatrix} D_F g^F & D_\mu g^F \\ D_F g^\mu & D_\mu g^\mu \end{pmatrix} \quad (2.36)$$

$$= \begin{pmatrix} -1 + \pi_1 & \pi_2 & \cdots & \pi_K & 0 & \\ \pi_1 & -1 + \pi_2 & \cdots & \pi_K & & \ddots \\ \vdots & \vdots & \ddots & \pi_K & & 0 \\ \pi_1 & \cdots & & -1 + \pi_K & 0 & 0 \\ 0 & & 0 & 0 & -1 & 0 \\ & \ddots & & & & \ddots \\ 0 & & 0 & 0 & 0 & -1 \end{pmatrix} \quad (2.37)$$

$$= \begin{pmatrix} \mathbf{1}\pi^\top - I_{K \times K} & 0 \\ 0 & -I_{KM \times KM} \end{pmatrix}. \quad (2.38)$$

In the above evaluation of the Jacobian, the dependence on o appears only through π in the matrix $\mathbf{1}\pi^\top - I_{K \times K}$. This matrix admits a one-dimensional left null space spanned by the row eigenvector $\pi^\top = (\pi_1, \dots, \pi_K)$ with eigenvalue 0. Thus $(\mathbf{1}\pi^\top - I_{K \times K})\mathbf{x}$ lies in the orthogonal complement of π for all $\mathbf{x} \in \mathbb{R}^K$. The matrix $\mathbf{1}\pi^\top - I_{K \times K}$ is invertible though if its action is restricted to this complement, which is possible because 2.20 implies $\sum_k \pi_k(1 - R_k) = 0$. The inverse is obtained by simply ignoring the term $\mathbf{1}\pi^\top$ in 2.38. We derive

$$\Gamma^* = -Dg(F^*, \mu^*)^{-1} = \begin{pmatrix} I_{K \times K} & 0_{K \times MK} \\ 0_{MK \times K} & I_{MK \times MK} \end{pmatrix}, \quad (2.39)$$

immediately leading to the recursions (2.18) and (2.19).

The second part of Proposition 1 asserts that if $\pi(\theta) = \pi^\circ(\theta)$, the recursion for o^t also has minimal asymptotic variance. With π° defined by (2.17),

$$\pi_k^\circ(F, \mu) = \gamma_k(\mu) \frac{e^{F_k^* - F_k}}{\sum_{\ell \in [K]} \gamma_\ell(\mu) e^{F_\ell^* - F_\ell}}. \quad (2.40)$$

From equation (2.7), the mean field for o is

$$\dot{o}_k(t) = \frac{\pi_k(\theta(t))}{\gamma_k(\mu(t))} \frac{e^{F_k(t) - F_k^*}}{\sum_{\ell \in [K]} \pi_\ell(\theta(t)) e^{F_\ell(t) - F_\ell^*}} - o_k(t), \quad (2.41)$$

in (F, μ) coordinates. Inserting $\pi(\theta(t)) = \pi^\circ(\theta(t))$ into (2.41), we find

$$\dot{o}_k(t) = 1 - o_k(t), \quad (2.42)$$

for which the Jacobian is trivially given by $-I_{K \times K}$. It follows that, when $\pi = \pi^\circ$, the recursion (2.4) is optimal. $\square$

3 Visit control in the steady state approximation

In this section, we demonstrate the effect of visit control on the mean field of the free energy estimates when a steady-state approximation is used. In other words, we will study $g_k^F(F, \mu, o)$ (defined in (2.13)) under

the assumption that $o = o^*(F, \mu)$, which is equivalent to saying that $o = (o_1, \dots, o_K)$ satisfies $g_k^o(F, \mu, o) = 0$ for all $k \in [K]$. In Section 3.1 below we explicitly write out the calculations to identify $g_k^F(F, \mu, o^*(F, \mu))$. Then, in Section 3.2 we study a general mean field and establish that the Lyapunov function for the system decreases faster in the steady state approximation.

3.1 Uniform distributions computations

Here, we explicitly work out the modified mean field equation for the free energy estimates, part of which are presented in Section 2.2 of the main text. The model in question uses two Hamiltonians defined, for some $0 < \delta < 1/2$, as

$$H_1(x) = \infty \cdot 1_{[-1+\delta, \delta]^c}(x) \quad (3.1)$$

$$H_2(x) = \infty \cdot 1_{[-\delta, 1-\delta]^c}(x), \quad (3.2)$$

where A^c denotes the complement of a set $A \subset \mathbb{R}$. For the purpose of illustrating visit control, we let π denote a general function of the tilts o whose target range is contained in $\mathcal{P}^e(K)$. We define $\Delta^t = F_2^t - F_1^t$ where F_k^t is defined using the linearized estimator presented in Section 2.1 of the main text,

$$F_k^{t+1} = F_k^t + \frac{1}{t+1} \left(1 - \frac{e^{F_k^t - H_k(\chi^{t+1})}}{\sum_{\ell \in [K]} \pi_\ell e^{F_\ell^t - H_\ell(\chi^{t+1})}} \right), \quad (3.3)$$

which gives

$$\Delta^{t+1} = \Delta^t + \frac{1}{t+1} \left(\frac{e^{F_1^t - H_1(\chi^{t+1})}}{\sum_{\ell \in [K]} \pi_\ell e^{F_\ell^t - H_\ell(\chi^{t+1})}} - \frac{e^{F_2^t - H_2(\chi^{t+1})}}{\sum_{\ell \in [K]} \pi_\ell e^{F_\ell^t - H_\ell(\chi^{t+1})}} \right) \quad (3.4)$$

$$= \Delta^t + \frac{1}{t+1} \cdot \frac{e^{F_1^t - H_1(\chi^{t+1})} - e^{F_2^t - H_2(\chi^{t+1})}}{\pi_1 e^{F_1^t - H_1(\chi^{t+1})} + \pi_2 e^{F_2^t - H_2(\chi^{t+1})}} \quad (3.5)$$

$$= \Delta^t + \frac{1}{t+1} \cdot \frac{e^{-H_1(\chi^{t+1})} - e^{\Delta^t - H_2(\chi^{t+1})}}{\pi_1 e^{-H_1(\chi^{t+1})} + \pi_2 e^{\Delta^t - H_2(\chi^{t+1})}} \quad (3.6)$$

$$= \Delta^t + \frac{1}{t+1} G(\chi^{t+1}; \Delta) \quad (3.7)$$

The mean field for Δ can be identified either by computing $\mathbb{E}[G(X; \Delta)]$, or simply by using the expression for the mean field $g_k(F)$ (equation 8 in the main text),

$$g_k(F) = 1 - \frac{e^{F_k - F_k^*}}{\sum_{\ell \in [K]} \pi_\ell e^{F_\ell - F_\ell^*}}. \quad (3.8)$$

We then have, with $\Delta(t) = F_2(t) - F_1(t)$ (i.e., $F(t) = (0, \Delta(t))$) up to an additive shift by any multiple of the vector $(1, 1)$ and $F^* = 0$,

$$\dot{\Delta}(t) = g_2(F(t)) - g_1(F(t)) \quad (3.9)$$

$$= \frac{e^{F_1(t)} - e^{F_2(t)}}{\pi_1 e^{F_1(t)} + \pi_2 e^{F_2(t)}} \quad (3.10)$$

$$= \frac{1 - e^{\Delta(t)}}{\pi_1 + \pi_2 e^{\Delta(t)}}. \quad (3.11)$$

Now we use TSS's choice of π for this example, which is defined for any $\eta > 0$ by

$$\pi_k^{\text{TSS}}(o) = \frac{\gamma_k o_k^{-\eta}}{\sum_{\ell \in [K]} \gamma_\ell o_\ell^{-\eta}}. \quad (3.12)$$

The dependence of $\gamma = (\gamma_1, \gamma_2)$ on μ has been ignored in the above expression, since in the example we take $\gamma = (1/2, 1/2)$. The steady state o^* as a function of F is obtained by solving $g_k^o(F, \mu, o) = 0$, where

$$g_k^o(F, \mu, o) = \frac{1}{\gamma_k} \cdot \frac{\pi_k(o) e^{F_k - F_k^*}}{\pi_1(o) e^{F_1 - F_1^*} + \pi_2(o) e^{F_2 - F_2^*}} - o_k. \quad (3.13)$$

(The dependence on μ of g_k^o is present only to mimic the expressions presented in Section 1.2.) Using (3.12) and $F_k^* = 0$ in (3.13), we find that $(o_k^*)^{\eta+1} \propto e^{F_k}$, and so

$$\pi_k = \gamma_k (o_k^*)^{-\eta} \propto e^{-\beta F_k}, \quad (3.14)$$

where $\beta = \eta/(\eta + 1)$. Using $\pi_1 + \pi_2 = 1$, we get

$$\pi(o^*(F)) = \left(\gamma_1 o_1^{-\eta}, \gamma_2 o_2^{-\eta} \right) = \left(\frac{e^{-\beta F_1}}{e^{-\beta F_1} + e^{-\beta F_2}}, \frac{e^{-\beta F_1}}{e^{-\beta F_2} + e^{-\beta F_2}} \right) \quad (3.15)$$

$$= \left(\frac{1}{1 + e^{-\beta \Delta}}, \frac{e^{-\beta \Delta}}{1 + e^{-\beta \Delta}} \right) \quad (3.16)$$

$$= \left(\frac{e^{\beta \Delta/2}}{e^{\beta \Delta/2} + e^{-\beta \Delta/2}}, \frac{e^{-\beta \Delta/2}}{e^{\beta \Delta/2} + e^{-\beta \Delta/2}} \right) \quad (3.17)$$

$$= \frac{1}{2 \cosh(\beta \Delta/2)} \left(e^{\beta \Delta/2}, e^{-\beta \Delta/2} \right) \quad (3.18)$$

Finally, using o^* in (3.9) yields

$$\frac{1 - e^\Delta}{\pi_1 + \pi_2 e^\Delta} = \frac{e^{-\Delta/2} - e^{\Delta/2}}{\pi_1 e^{-\Delta/2} + \pi_2 e^{\Delta/2}} \quad (3.19)$$

$$= -2 \sinh(\Delta/2) \frac{1}{\pi_1 e^{-\Delta/2} + \pi_2 e^{\Delta/2}} \quad (3.20)$$

$$= -2 \sinh(\Delta/2) \frac{2 \cosh(\beta \Delta/2)}{e^{(\beta-1)\Delta/2} + e^{(1-\beta)\Delta/2}} \quad (3.21)$$

$$= -2 \sinh(\Delta/2) \frac{\cosh(\beta \Delta/2)}{\cosh((1-\beta)\Delta/2)}, \quad (3.22)$$

which shows that the mean field has improved to pull Δ back to its steady state exponentially fast.

3.2 Improved decrease of Lyapunov function

In this section, we prove Proposition 2 of the main text.

In Section 2.2.1 of the main text we introduced the visit control mechanism, which proposes the class of controls

$$\pi_k^{\text{TSS}}(\theta) = \frac{\gamma_k (Z_k / Z_k^\circ)^{\eta/(\eta+1)}}{\sum_{\ell \in [K]} \gamma_\ell (Z_\ell / Z_\ell^\circ)^{\eta/(\eta+1)}}. \quad (3.23)$$

for some parameter $\eta \geq 0$. The mean field for (Z, o) is

$$\dot{Z}_k = \frac{Z_k^*}{\sum_{\ell \in [K]} \pi_\ell^{\text{TSS}}(\theta(t)) Z_\ell^* / Z_\ell(t)} - Z_k(t) \quad (3.24)$$

$$\dot{o}_k = \frac{\pi_k^{\text{TSS}}(\theta(t))}{\gamma_k(\mu(t))} \frac{Z_k^* / Z_k(t)}{\sum_{\ell \in [K]} \pi_\ell^{\text{TSS}}(\theta(t)) Z_\ell^* / Z_\ell(t)} - o_k(t). \quad (3.25)$$

The law 3.23 has been adopted because if the process $o(t)$ can be kept close to its stationary regime, then the rate at which $Z(t)$ approaches Z^* increases in magnitude monotonically with increasing $\eta \geq 0$. More precisely, we measure the rate of decay of the relative entropy (or Kullback-Leibler divergence)

$$V_\gamma(Z) = D_{\text{KL}}(\gamma || r) = - \sum_{k \in [K]} \gamma_k \log(r_k / \gamma_k), \quad (3.26)$$

where

$$r_k = \gamma_k x_k / \left(\sum_{\ell \in [K]} \gamma_\ell x_\ell \right) \quad (3.27)$$

with $x_k = Z_k^* / Z_k$. The relative entropy serves as a Lyapunov function for the dynamics of Z (see Lemma 5.2). We will show that the rate of decay increases monotonically with increasing $\eta \geq 0$ if the process $o(t)$ is assumed to be stationary, thus proving Proposition 2.

To ease notation clutter, we will drop the dependence on μ in $\gamma_k(\mu)$ and write only γ_k , and we will also stop denoting explicit time dependence. At stationarity of the tilts o (that is, $\dot{o}_k = 0$ for all $k \in [K]$), we have $Z_k^o = Z_k^*$ for all $k \in [K]$ and

$$\pi_k^{\text{TSS}}(\theta) = \frac{\gamma_k (Z_k / Z_k^*)^{\eta / (\eta + 1)}}{\sum_{\ell \in [K]} \gamma_\ell (Z_\ell / Z_\ell^*)^{\eta / (\eta + 1)}} = \frac{\gamma_k x_k^{-\eta / (\eta + 1)}}{\sum_{\ell \in [K]} \gamma_\ell x_\ell^{-\eta / (\eta + 1)}}. \quad (3.28)$$

We compute

$$\dot{Z}_k = \frac{Z_k^*}{\sum_{\ell \in [K]} \pi_\ell Z_\ell^* / Z_\ell} - Z_k = \frac{\sum_{\ell \in [K]} \gamma_\ell x_\ell^{-\eta / (\eta + 1)}}{\sum_{\ell \in [K]} \gamma_\ell x_\ell^{1 / (\eta + 1)}} Z_k^* - Z_k \quad (3.29)$$

and

$$\frac{d}{dt} V_\gamma(Z) = - \sum_{k \in [K]} \gamma_k \left(\frac{Z_k^* / Z_k}{\sum_{\ell \in [K]} \gamma_\ell Z_\ell^* / Z_\ell} - 1 \right) \frac{\dot{Z}_k}{Z_k} \quad (3.30)$$

$$= - \sum_{k \in [K]} \gamma_k \left(\frac{x_k}{\sum_{\ell \in [K]} \gamma_\ell x_\ell} - 1 \right) \left(\frac{\sum_{\ell \in [K]} \gamma_\ell x_\ell^{-\eta / (\eta + 1)}}{\sum_{\ell \in [K]} \gamma_\ell x_\ell^{1 / (\eta + 1)}} x_k - 1 \right). \quad (3.31)$$

If $\eta = 0$ (no visit control), then

$$\frac{d}{dt} V_\gamma(Z) = - \sum_{k \in [K]} \gamma_k \left(\frac{x_k}{\sum_{\ell \in [K]} \gamma_\ell x_\ell} - 1 \right)^2 \quad (3.32)$$

is negative. We wish to show that the rate of decay becomes progressively more negative as η is increased to infinity. It suffices to show that

$$\frac{d}{d\eta} \frac{d}{dt} V_\gamma(Z) = \frac{1}{(\eta+1)^2} \left(\sum_{k \in [K]} \gamma_k x_k \left(\frac{x_k}{\sum_{\ell \in [K]} \gamma_\ell x_\ell} - 1 \right) \right) \quad (3.33)$$

$$\times \left(\frac{\sum_{\ell \in [K]} \gamma_\ell x_\ell^{1/(\eta+1)} x_\ell^{-1} \log(x_\ell)}{\sum_{\ell \in [K]} \gamma_\ell x_\ell^{1/(\eta+1)}} - \frac{\sum_{\ell \in [K]} \gamma_\ell x_\ell^{1/(\eta+1)} x_\ell^{-1} \sum_{\ell \in [K]} \gamma_\ell x_\ell^{1/(\eta+1)} \log(x_\ell)}{\sum_{\ell \in [K]} \gamma_\ell x_\ell^{1/(\eta+1)} \sum_{\ell \in [K]} \gamma_\ell x_\ell^{1/(\eta+1)}} \right) \quad (3.34)$$

is negative. The overall factor on the first row of the right-hand side is positive, as follows from the Cauchy-Schwarz inequality. The factor on the second row of the right-hand side is negative. This follows from the correlation inequality

$$\mathbb{E}[f(X)g(X)] \geq \mathbb{E}[f(X)]\mathbb{E}[g(X)] \quad (3.35)$$

applied to the strictly increasing functions $f(x) = -x^{-1}$ and $g(x) = \log(x)$, where $f, g : (0, \infty) \rightarrow \mathbb{R}$. The correlation inequality can be deduced from

$$\sum_{k \in [K]} \sum_{\ell \in [K]} w_k w_\ell (f(x_k) - f(x_\ell))(g(x_k) - g(x_\ell)) \geq 0, \quad (3.36)$$

which holds by monotonicity of f and g , because either both $f(x_k) - f(x_\ell)$ and $g(x_k) - g(x_\ell)$ are positive or both are negative. Here $w_k = \gamma_k x_k^{1/(\eta+1)} / (\sum_{\ell \in [K]} \gamma_\ell x_\ell^{1/(\eta+1)})$. Owing to $\gamma \in \mathcal{P}^e(K)$, the correlation inequality is strict unless x_k is a constant independent of $k \in [K]$.

4 Variance calculations for TSS and MBAR

Free energy difference estimators have been studied for at least the past 70 years. Following the introduction of the Zwanzig relation [32], which is a form of importance sampling, Bennett [5] developed an improved estimator known as the Bennett Acceptance Ratio (BAR), which is the two-rung version of MBAR. These estimators are known by different names in different fields; [31] is perhaps the earliest mathematical treatment of the maximum likelihood estimator (which we call MBAR); other relevant works include [8, 27, 14, 25, 9, 26, 22], and undoubtedly many others.

The variance of any estimator for the free energy differences is related to the amount of overlap between the distributions. In the examples where we compute variance, the rung distribution π is always taken to be fixed. For such fixed $\pi \in \mathcal{P}^e(K)$, we define the overlap matrix O by

$$O_{ij} = \mathbb{E} \left[\frac{e^{F_i^* - H_i(X)} e^{F_j^* - H_j(X)}}{\left(\sum_{k \in [K]} \pi_k e^{F_k^* - H_k(X)} \right)^2} \right], \quad (4.1)$$

where X is distributed according to the mixture distribution $\sum_{k \in [K]} \pi_k e^{F_k^* - H_k(x)} dx$.

Throughout, we make the assumption that O is irreducible. This is an assumption that goes back to Vardi [31]. The motivation for this assumption is that the differences $F_i - F_j$ can only be resolved if there is overlap between the densities $\rho_i(x)$ and $\rho_j(x)$, or alternatively, if for any $i, j \in [K]$ there is a sequence $(i_k)_{k=1}^n$ such that $i_1 = i$, $i_n = j$, and

$$\int_S 1\{\rho_{i_k}(x) > 0\} 1\{\rho_{i_{k+1}}(x) > 0\} dx > 0, \quad k = 1, \dots, n-1. \quad (4.2)$$

See, for instance, [9] (bottom of p. 1071).

4.1 Variance of stochastic approximation

Consider the stochastic approximation recursion

$$\theta^{t+1} = \theta^t + \frac{1}{t+1} \Gamma G(\mathcal{X}^{t+1}; \theta^t), \quad (4.3)$$

and let $g(\theta) = \mathbb{E}_\theta[G(\mathcal{X}, \theta)]$ denote the mean field, where the expectation is with respect to the stationary distribution $p_\theta(x)dx$. When the function G and the gain matrix Γ satisfy the appropriate conditions to guarantee almost sure convergence of the stochastic approximation algorithm, the asymptotic covariance of the iterates θ^t is the solution Σ to the Lyapunov equation (see eqn. 3.2.7, Part I in Benveniste et al., 1990)

$$\left(\frac{I}{2} + \Gamma Dg(\theta^*) \right) \Sigma + \Sigma \left(\frac{I}{2} + \Gamma Dg(\theta^*) \right)^\top + \Gamma R(\theta^*) \Gamma^\top = 0. \quad (4.4)$$

Here, θ^* is the unique solution to $g(\theta) = 0$, $Dg(\theta)$ is the Jacobian of g with respect to θ , I is the identity matrix and $R(\theta^*)$ is the integrated autocovariance, defined by

$$R(\theta) = \sum_{n=-\infty}^{\infty} \text{Cov}_\theta(G(\mathcal{X}_0; \theta), G(\mathcal{X}_n; \theta)), \quad (4.5)$$

where $\mathcal{X}_n$ is generated according to the Markov chain which has stationary distribution $p_\theta(x)dx$.

The optimal choice of gain matrix, in the sense of minimizing Σ in the rank-ordering on positive semidefinite matrices, is the gain

$$\Gamma = -Dg(\theta^*)^{-1}, \quad (4.6)$$

which leads to the asymptotic covariance matrix (eqn. 3.2.8 in Benveniste et al. 1990)

$$\Sigma = Dg(\theta^*)^{-1} R(\theta^*) Dg(\theta^*)^{-T}. \quad (4.7)$$

The optimal gain matrix for the F component was identified in (2.39) to be $\Gamma = -I_{K \times K}$, so the asymptotic covariance matrix of the TSS free energy estimates TSS is given by

$$\Sigma_{\text{TSS}} = R(\theta^*), \quad (4.8)$$

where the observable function G which determines the integrated covariance R in (4.5) is the function $G = G^F$ from Section 2.1 the main text, which we recall is defined componentwise by

$$G_k^F(x; \theta) = 1 - \frac{e^{F_k - H_k(x)}}{\sum_{\ell \in [K]} \pi_\ell(\theta) e^{F_\ell - H_\ell(x)}}. \quad (4.9)$$

Suppose $\pi(\theta)$ is fixed and equal to some $\pi^* \in \mathcal{P}^e(K)$, and that the TSS samples $(\mathcal{X}^{t+1}, \mathcal{K}^{t+1})$ are drawn independently from the density p_{F^t, π^*} , the expression for the asymptotic covariance matrix simplifies to

$$(\Sigma_{\text{TSS}})_{ij} = \text{Cov}_{\theta^*}(G_i^{F^*}(\mathcal{X}; \theta^*), G_j^{F^*}(\mathcal{X}; \theta^*)) \quad (4.10)$$

$$= \mathbb{E}_{\theta^*} \left[\left(1 - \frac{e^{F_i^* - H_i(\mathcal{X})}}{\sum_{\ell \in [K]} \pi_\ell^* e^{F_\ell^* - H_\ell(\mathcal{X})}} \right) \left(1 - \frac{e^{F_j^* - H_j(\mathcal{X})}}{\sum_{\ell \in [K]} \pi_\ell^* e^{F_\ell^* - H_\ell(\mathcal{X})}} \right) \right] \quad (4.11)$$

$$= -1 + \mathbb{E}_{\theta^*} \left[\frac{e^{F_i^* - H_i(\mathcal{X})}}{\sum_{\ell \in [K]} \pi_\ell^* e^{F_\ell^* - H_\ell(\mathcal{X})}} \frac{e^{F_j^* - H_j(\mathcal{X})}}{\sum_{\ell \in [K]} \pi_\ell^* e^{F_\ell^* - H_\ell(\mathcal{X})}} \right]. \quad (4.12)$$

Written in matrix form, this is

$$\Sigma_{\text{TSS}} = O - \mathbf{1}\mathbf{1}^\top, \quad (4.13)$$

where $\mathbf{1} = (1, \dots, 1) \in \mathbb{R}^K$. Since the vector $\mathbf{1}$ is orthogonal to any vector $v_{ij} = e_j - e_i$, where e_i, e_j are the i^{th} and j^{th} unit basis vectors respectively, for the purposes of determining the variance of the free energy difference estimates we can take

$$\Sigma_{\text{TSS}} = O. \quad (4.14)$$

Thus, O is the asymptotic covariance matrix for TSS when samples $(\mathcal{X}^{t+1}, \mathcal{K}^{t+1})$ are drawn independently from $p_{\theta^t}(x, k)dx d\mathcal{K}$. The expression for the asymptotic variance of the maximum likelihood estimator is presented in the next Section (Section 4.2); the proof that $\Sigma_{\text{TSS}} = O$ is smaller than the MLE covariance matrix is in Section 4.5.

4.2 Variance of maximum likelihood estimation

Given n independent samples $\mathcal{X}_1, \dots, \mathcal{X}_n$, n_k of which are from distribution $p_k(x)dx$, $n = n_1 + \dots + n_K$, the maximum likelihood estimator for the free energy differences (which we call MBAR) is a solution F of the system of equations

$$e^{-F_k} = \frac{1}{n} \sum_{i=1}^n \frac{e^{-H_k(\mathcal{X}_i)}}{\sum_{\ell=1}^K \frac{n_\ell}{n} e^{F_\ell - H_\ell(\mathcal{X}_i)}}. \quad (4.15)$$

The solution is unique up to a common additive constant that disappears when computing a free energy difference. The existence and uniqueness of solutions to this system of equations has been investigated in depth by Vardi [31]. The large sample variance has also been studied in [9], [8], and subtle aspects of the theory have been studied in [14]. The MBAR estimator, which achieves the Cramer-Rao lower bound, has a covariance matrix which can be expressed as (see eqn. 6.4 in [14]):

$$\Sigma_{\text{MBAR}} = (\Pi - \Pi O \Pi)^+ - \Pi^{-1}, \quad (4.16)$$

where $\Pi = \text{diag}(\pi_1, \dots, \pi_K)$ and the superscript $+$ denotes the Moore-Penrose pseudoinverse. In the specific case $K = 2$, one can simplify the expression for the asymptotic variance of $F_2 - F_1$, which works out to be:

$$\lim_{t \rightarrow \infty} t \text{Var}(F_2^t - F_1^t) = \frac{1}{\pi_1 \pi_2} [O_{12}^{-1} - 1]. \quad (4.17)$$

This is the same expression as the one for two-state MBAR on the last page of [26]. For further properties of the matrix Σ_{MBAR} , see Section 4.5.

4.3 Uniform distributions computations

In this section we derive the expressions for the variance used in Section 2.2.3 of the main text, where

$$H_1(x) = \infty \cdot 1_{[-1+\delta, \delta]^c}(x), \quad H_2(x) = \infty \cdot 1_{[-\delta, 1-\delta]^c}(x) \quad (4.18)$$

for some $0 < \delta < 1/2$. Let $(\mathcal{X}^t, \mathcal{K}^t)$ be the Markov process generated with the simulated tempering process while updating F^t with stochastic approximation. More precisely, the density of the transition kernel for the process is P_{F^t} , where for each fixed $F \in \mathbb{R}^2$, $P_F = P_F^2 \circ P_F^1$, and

$$P_F^1(x', k' | x, k) = \delta(x' - x) \frac{\pi_{k'} e^{F_{k'} - H_{k'}(x)}}{\sum_{\ell=1}^2 \pi_\ell e^{F_\ell - H_\ell(x)}}, \quad P_F^2(x'', k'' | x', k') = \delta(k'' - k') p_{k'}(x''). \quad (4.19)$$

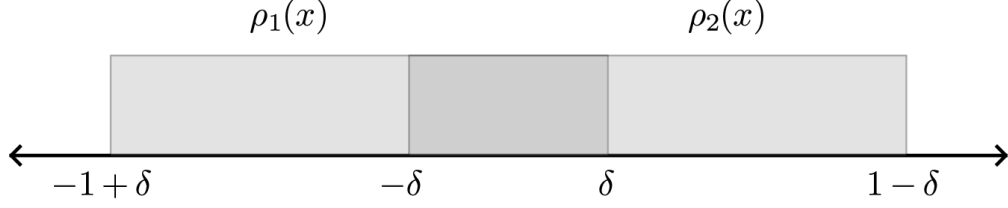


Figure 1: Densities of two uniform distributions, one on $[-1 + \delta, \delta]$ and the other on $[-\delta, 1 - \delta]$

The updates to F^t are given by

$$F_k^{t+1} = F_k^t + \frac{1}{t+1} G_k(\mathcal{X}^{t+1}; F^t) = F_k^t + \frac{1}{t+1} \left(1 - \frac{e^{F_k^t - H_k(\mathcal{X}^{t+1})}}{\sum_{\ell=1}^2 \pi_\ell e^{F_\ell^t - H_\ell(\mathcal{X}^{t+1})}} \right). \quad (4.20)$$

Since only free energy differences are of interest, we can define $\Delta^t = F_2^t - F_1^t$. We can also allow for a general gain matrix Γ , which in the present case will just be a scalar. The general update for Δ^t is

$$\Delta^{t+1} = \Delta^t + \frac{\Gamma}{t+1} \left(\frac{e^{-H_1(\mathcal{X}^{t+1})} - e^{\Delta^t - H_2(\mathcal{X}^{t+1})}}{\pi_1 e^{-H_1(\mathcal{X}^{t+1})} + \pi_2 e^{\Delta^t - H_2(\mathcal{X}^{t+1})}} \right). \quad (4.21)$$

By an abuse of notation, we will denote the term in parentheses in the above expression by $G(x; \Delta)$ (where $x = \mathcal{X}^{t+1}$ and $\Delta = \Delta^t$ above).

For fixed $\pi \in \mathcal{P}^{\mathcal{E}}(K)$, we will compute the asymptotic variance of the TSS free energy estimator Δ^t when $\mathcal{X}^t$ is sampled independently from $\rho_k(x)dx$ given $\mathcal{K}^t = k$, and where $\mathcal{K}^t$ is sampled with $v \geq 1$ iterations of the simulated tempering transition kernel $P_{F^t}^k$. To evaluate the expression for the asymptotic variance given in (4.7), we will require $R(\theta^*)$, the integrated covariance, and $Dg(\theta^*)$, the derivative of the mean field evaluated at θ^* . This last computation is particularly simple, because the Jacobian of the mean field for the free energies evaluated at θ^* was found in Section 2 to be $-I_{K \times K}$.

We will use the expression in (4.16) to compute the asymptotic variance of MBAR, with the assumption that a proportion π_k of samples is drawn independently from $\rho_k(x)dx$.

4.3.1 Computing the variance for TSS and MBAR

To compute the variance of the TSS free energy estimates, we need to compute the integrated covariance (4.5). For certain sampling schemes, it is possible to compute the integrated covariance explicitly by solving an associated Poisson equation. Tan has derived some of these expressions for the case of free energy estimation when independent samples are available from each distribution $\rho_k(x)dx$ (see the Supplementary Material in [28]), while a more general result on the covariance structure of Gibbs samplers with Rao-Blackwellized estimators is available in [18]. In this section we calculate the covariance matrices by explicit summation.

Recall the exact form of the Hamiltonians in the uniform distributions example,

$$H_1(x) = \infty \cdot 1_{[-1+\delta, \delta]^c}(x), \quad H_2(x) = \infty \cdot 1_{[-\delta, 1-\delta]^c}(x). \quad (4.22)$$

Observe that $G(x; \Delta)$ takes on three distinct values: $1/\pi_1$ when $x < -\delta$, $-1/\pi_2$ when $x > \delta$, and $(1 - e^\Delta)/(\pi_1 + \pi_2 e^\Delta)$ when $x \in [-\delta, \delta]$. Moreover, the likelihood of observing any one of these values at the

next iteration is determined by the value of G at the present iteration. In other words, $Y^t = G(\mathcal{X}^t; \Delta)$ is a three-state Markov chain for any $\Delta \in \mathbb{R}$, with transition matrix

$$P_\Delta = \begin{pmatrix} 1-2\delta & 2\delta & 0 \\ (1-2\delta) \cdot \frac{\pi_1}{\pi_1 + \pi_2 e^\Delta} & 2\delta & (1-2\delta) \cdot \frac{\pi_2 e^\Delta}{\pi_1 + \pi_2 e^\Delta} \\ 0 & 2\delta & 1-2\delta \end{pmatrix}, \quad (4.23)$$

for which the unique stationary distribution is

$$p_\Delta = \left((1-2\delta) \cdot \frac{\pi_1}{\pi_1 + \pi_2 e^\Delta}, 2\delta, (1-2\delta) \cdot \frac{\pi_2 e^\Delta}{\pi_1 + \pi_2 e^\Delta} \right). \quad (4.24)$$

The advantage of this perspective is that we can compute the integrated covariance (4.5) explicitly for this chain at $\Delta = \Delta^*$ with Y^0 distributed according to p_{Δ^*} (defined in (4.24)). To simplify the calculations we take $\pi_1 = \pi_2 = 1/2$, so that

$$P_{\Delta^*} = \begin{pmatrix} 1-2\delta & 2\delta & 0 \\ (1-2\delta)/2 & 2\delta & (1-2\delta)/2 \\ 0 & 2\delta & 1-2\delta \end{pmatrix}, \quad (4.25)$$

To compute the integrated covariance we need to compute the powers of P_{Δ^*} , so we diagonalize the matrix. We have $P_{\Delta^*} = ADA^{-1}$, where

$$A = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 0 & 1 - \frac{1}{2\delta} \\ 1 & -1 & 1 \end{pmatrix}, A^{-1} = \frac{1}{2} \begin{pmatrix} 1-2\delta & 4\delta & 1-2\delta \\ 1 & 0 & -1 \\ 2\delta & -4\delta & 2\delta \end{pmatrix}, D = \text{diag}(1, 1-2\delta, 0). \quad (4.26)$$

Thus, for integers $t \geq 0$ we have $P_{\Delta^*}^t = AD^t A^{-1}$, which is explicitly

$$P_{\Delta^*}^t = \frac{1}{2} \begin{pmatrix} 1-2\delta + p_\delta^t & 4\delta & 1-2\delta - p_\delta^t \\ 1-2\delta & 4\delta & 1-2\delta \\ 1-2\delta - p_\delta^t & 4\delta & 1-2\delta + p_\delta^t \end{pmatrix} = \frac{1}{2} \begin{pmatrix} p_\delta + p_\delta^t & 4\delta & p_\delta - p_\delta^t \\ p_\delta & 4\delta & p_\delta \\ p_\delta - p_\delta^t & 4\delta & p_\delta + p_\delta^t \end{pmatrix}, \quad (4.27)$$

where $p_\delta = (1-2\delta)$.

Now we assume Y^t is the chain generated with (4.23) and $\Delta = \Delta^*$. In this case Y^t takes on only the values ± 2 and 0, and has $\mathbb{E}[Y^t] = \mathbb{E}_{\Delta^*}[G(\mathcal{X}; \Delta^*)] = 0$. We use the values $\{-2, 0, 2\}$ to index the entries of the transition matrix, so that, for instance, $(P_{\Delta^*}^1)_{-2,-2} = (1-2\delta)$ and $(P_{\Delta^*}^1)_{0,2} = (1-2\delta)/2$. Noting that $(P_{\Delta^*}^t)_{2,2} = (P_{\Delta^*}^t)_{-2,-2}$ and $(P_{\Delta^*}^t)_{2,-2} = (P_{\Delta^*}^t)_{-2,2}$, we have

$$\begin{aligned} \mathbb{E}[Y^0 Y^t] &= \sum_{y^0 \in \{-2, 0, 2\}} \sum_{y^t \in \{-2, 0, 2\}} y^0 y^t \mathbb{P}(Y^t = y^t | Y^0 = y^0) p_{\Delta^*}(y^0) \\ &= \sum_{y^0 \in \{\pm 2\}} \sum_{y^t \in \{\pm 2\}} y^0 y^t \mathbb{P}(Y^t = y^t | Y^0 = y^0) p_{\Delta^*}(y^0) \\ &= 4 \left(\frac{1-2\delta}{2} [(P_{\Delta^*}^t)_{-2,-2} - (P_{\Delta^*}^t)_{-2,2}] + \frac{1-2\delta}{2} [(P_{\Delta^*}^t)_{2,2} - (P_{\Delta^*}^t)_{2,-2}] \right) \\ &= 4(1-2\delta) [(P_{\Delta^*}^t)_{-2,-2} - (P_{\Delta^*}^t)_{-2,2}], \\ &= 4(1-2\delta) p_\delta^t. \end{aligned} \quad (4.28)$$

Thus,

$$\sum_{t=1}^{\infty} \mathbb{E}[Y^0 Y^{v_t}] = 4(1-2\delta) \sum_{t=1}^{\infty} p_{\delta}^{v_t} = 4(1-2\delta) \frac{p_{\delta}^v}{1-p_{\delta}^v}. \quad (4.29)$$

The variance of TSS when using v moves per estimator update is

$$\Sigma_{\text{TSS}}^v = \text{Var}(Y^0) + 2 \sum_{t=1}^{\infty} \text{Cov}(Y^0, Y^{v_t}) = 4(1-2\delta) + 8(1-2\delta) \frac{p_{\delta}^v}{1-p_{\delta}^v}. \quad (4.30)$$

On the other hand, from (4.17) the variance of MBAR is

$$\frac{1}{\pi_1 \pi_2} [O_{12}^{-1} - 1] = 4 \left[\left(\int_{-1+\delta}^{1-\delta} \frac{e^{-H_1(x)-H_2(x)}}{\pi_1 e^{-H_1(x)} + \pi_2 e^{-H_2(x)}} dx \right)^{-1} - 1 \right] \quad (4.31)$$

$$= 4 \left[\frac{1}{2\delta} - 1 \right] = \frac{2p_{\delta}}{\delta}. \quad (4.32)$$

Recalling that $p_{\delta} = 1 - 2\delta$, we find $\Sigma_{\text{TSS}}^v \leq \Sigma_{\text{MBAR}}$ whenever $v \geq 1 - \log(1 + 2\delta)/\log(1 - 2\delta)$, and we note that the latter inequality is satisfied by all $v \geq 2$ if $\delta \in (0, 1/2)$.

4.4 General calculations for variance of TSS

Consider the stationary process (X^t, K^t) defined by the transition densities

$$P_K(x', k' | x, k) = \delta(x' - x) \frac{\pi_k \rho_k(x)}{\sum_{\ell \in [K]} \pi_{\ell} \rho_{\ell}(x)} \quad (4.33)$$

$$P_X(x'', k'' | x', k') = \rho_{k'}(x'') \delta(k'' - k'). \quad (4.34)$$

Here, $\rho_k(x) = e^{F_k^* - H_k(x)}$, $\lambda_k \in \Lambda$ for each $k \in [K]$, $\pi = (\pi_1, \dots, \pi_K) \in \mathcal{P}^{\varepsilon}(K)$. We will determine the asymptotic variance of the following empirical estimator:

$$S_T(w_j) = \frac{1}{T} \sum_{t=1}^T w_k(X^t) = \frac{1}{T} \sum_{t=1}^T \frac{\pi_k \rho_k(X^t)}{\sum_{\ell \in [K]} \pi_{\ell} \rho_{\ell}(X^t)}, \quad (4.35)$$

where

$$w_k(x) = \frac{\pi_k \rho_k(x)}{\sum_{\ell \in [K]} \pi_{\ell} \rho_{\ell}(x)} \quad (4.36)$$

Observe that $D_{\pi}G = \pi - w$, where $G = (G_1(x; F), \dots, G_K(x; F))$ with

$$G_k(x; F) = 1 - \frac{\rho_k(X^t)}{\sum_{\ell \in [K]} \pi_{\ell} \rho_{\ell}(X^t)}, \quad (4.37)$$

and $D_{\pi} = \text{diag}(\pi)$ is a diagonal matrix. With V denoting the asymptotic covariance matrix of (4.35), the asymptotic covariance of TSS will be given by

$$\Sigma_{\text{TSS}} = D_{\pi}^{-1} V D_{\pi}^{-1}. \quad (4.38)$$

In (4.14) we identified Σ_{TSS} in the case where the pair (X^t, K^t) is drawn independently from $p_{F^t-1}(x, k)$, because in this case the matrix V reduces to $\text{Cov}(G_k, G_j)$. In this case we find $\Sigma_{\text{TSS}} = O$, where O is the overlap matrix defined by

$$O_{jk} = \int_S \frac{\rho_j(x)\rho_k(x)}{\sum_{\ell \in [K]} \pi_\ell \rho_\ell(x)} dx.$$

When the sampling is not independent in either x , k , or both, computing V is more complicated. One way to identify V is to solve the system of Poisson equations with forcing functions w_j for the marginal chain X^t :

$$\hat{w}_j(x) - P\hat{w}_j(x) = w_j(x) - \pi_j, \quad j \in [K]. \quad (4.39)$$

The asymptotic variance V is then given by

$$E[\hat{w}\hat{w}^T - (P\hat{w})(P\hat{w})^T], \quad (4.40)$$

where E denotes an expectation with respect to the stationary density $p(x, k) = \pi_k \rho_k(x)$.

for the case of independent dynamics in X , with v steps of the transition kernel. More precisely, let P denote the transition kernel for the marginal chain, i.e.,

$$P(x'|x) = \sum_{k \in [K]} \rho_k(x') \frac{\pi_k \rho_k(x)}{\sum_j \pi_j \rho_j(x)} = \sum_{k \in [K]} \rho_k(x') w_k(x), \quad (4.41)$$

and for any function f define $Pf(x) = \int_S P(x'|x)f(x')dx'$. We will use independent samples generated from the chain with transition kernel P^v , which we now compute. We have

$$P^2(x''|x) = \int_S P(x''|x')P(x'|x)dx' \quad (4.42)$$

$$= \int_S \left(\sum_j w_j(x') \rho_j(x'') \right) \left(\sum_k w_k(x) \rho_k(x') \right) dx' \quad (4.43)$$

$$= \sum_{j,k} w_k(x) \rho_j(x'') \int_S w_j(x') \rho_k(x') dx' \quad (4.44)$$

$$= \sum_{j,k} w_k(x) \rho_j(x'') \pi_j O_{jk} \quad (4.45)$$

$$= \rho(x'')^T D_\pi O w(x). \quad (4.46)$$

In general, we have

$$P^v(x^v|x^0) = \rho(x^v)^T (D_\pi O)^{v-1} w(x^0). \quad (4.47)$$

The Poisson equation with forcing function w_j for the marginal chain X^t is

$$\hat{w}_j(x) - P^v \hat{w}_j(x) = w_j(x) - \pi_j. \quad (4.48)$$

Observe that, with $Q = D_\pi O$, given entrywise by $Q = (q_{k\ell})$

$$P^\nu w_j(x) = \int_S [\rho(x')^T (D_\pi O)^{\nu-1} w(x)] w_j(x') dx' \quad (4.49)$$

$$= \int_S \rho(x')^T Q^{\nu-1} w(x) w_j(x') dx' \quad (4.50)$$

$$= \sum_{k,\ell} \int_S \rho_k(x') q_{k\ell}^{\nu-1} w_\ell(x) w_j(x') dx' \quad (4.51)$$

$$= \sum_{k,\ell} w_\ell(x) q_{k\ell}^{\nu-1} \int_S \rho_k(x') w_j(x') dx' \quad (4.52)$$

$$= \sum_{k,\ell} \pi_{j O_{jk}} q_{k\ell}^{\nu-1} w_\ell(x) \quad (4.53)$$

$$= ((D_\pi O)^\nu w(x))_j. \quad (4.54)$$

The above calculation shows that $P^\nu w(x) = Q^\nu w(x)$. Note also that $P^\nu \pi = Q^\nu \pi = \pi$ and $(Q^\nu - \pi 1^T) = (Q - \pi 1^T)^\nu$ for any integer $\nu \geq 1$. With $\bar{w}_j(x) = w_j(x) - \pi_j$ we also have $\pi 1^T \bar{w} = 0$ because $1^T w = 1^T \pi = 1$. We deduce that

$$\hat{w}(x) = (I - (Q^\nu - \pi 1^T))^{-1} \bar{w}(x).$$

Indeed,

$$\hat{w}(x) - P^\nu \hat{w}(x) = (I - (Q^\nu - \pi 1^T))^{-1} \bar{w}(x) - P^\nu (I - (Q^\nu - \pi 1^T))^{-1} \bar{w}(x) \quad (4.55)$$

$$= (I - (Q^\nu - \pi 1^T))^{-1} (\bar{w}(x) - Q^\nu \bar{w}(x)) \quad (4.56)$$

$$= (I - (Q^\nu - \pi 1^T))^{-1} (I - (Q^\nu - \pi 1^T)) \bar{w}(x) \quad (4.57)$$

$$= \bar{w}(x). \quad (4.58)$$

The importance of the term $\pi 1^T$ is that it makes the matrix $I - (Q^\nu - \pi 1^T)$ invertible.

The asymptotic variance of the stationary sum is given by $V(\nu) = E[\hat{w} \hat{w}^T - (P^\nu \hat{w})(P^\nu \hat{w})^T]$, which we now compute. As before,

$$E[(\bar{w} \bar{w}^T)] = D_\pi O D_\pi - \pi \pi^T. \quad (4.59)$$

Let $B = (Q^\nu - \pi 1^T)$ and note that $Q^\nu \bar{w} = B \bar{w}$ because $\pi 1^T \bar{w} = 0$. The first term is

$$E[(\hat{w} \hat{w}^T)] = (I - B)^{-1} (D_\pi O D_\pi - \pi \pi^T) (I - B)^{-T},$$

The second term is

$$E[(P^\nu \hat{w})(P^\nu \hat{w})^T] = (I - B)^{-1} E[(Q^\nu \bar{w})(Q^\nu \bar{w})^T] (I - B)^{-T} \quad (4.60)$$

$$= (I - B)^{-1} B E[\bar{w} \bar{w}^T] B^T (I - B)^{-T} \quad (4.61)$$

$$= (I - B)^{-1} B (D_\pi O D_\pi - \pi \pi^T) B^T (I - B)^{-T} \quad (4.62)$$

$$= ((I - B)^{-1} - I) (D_\pi O D_\pi - \pi \pi^T) ((I - B)^{-1} - I)^T, \quad (4.63)$$

where in the last line we used the identity $(I - B)^{-1} B = (I - B)^{-1} - I$.

Observing that $D_\pi O D_\pi - \pi \pi^T = (D_\pi O - \pi 1^T) D_\pi = Q D_\pi = D_\pi Q^T$, the difference of the two terms is

$$E[(\hat{w} \hat{w}^T)] - E[(P \hat{w})(P \hat{w})^T] = (I - B)^{-1} Q D_\pi (I - B)^{-T} - ((I - B)^{-1} - I) Q D_\pi ((I - B)^{-1} - I)^T \quad (4.64)$$

$$= (I - B)^{-1} Q D_\pi + D_\pi Q^T (I - B)^{-T} - Q D_\pi \quad (4.65)$$

$$= A + A^T - Q D_\pi \quad (4.66)$$

$$= V(\mathbf{v}), \quad (4.67)$$

where $A = (I - B)^{-1} Q D_\pi$ and the last equality defines the symmetric positive semi-definite matrix $V(\mathbf{v})$. From (4.40) and (4.38), the asymptotic variance of the TSS free energy estimate is the same as the variance of the stationary sum in (4.35) multiplied by the matrix D_π^{-1} . That is,

$$\Sigma_{\text{TSS}}(\mathbf{v}) = D_\pi^{-1} V(\mathbf{v}) D_\pi^{-1}.$$

Here is a sanity check for the dependence of Σ_{TSS} on $\mathbf{v}$. We know the asymptotic covariance of TSS with perfectly self-adjusted samples is just the overlap matrix O , and so we must have $\lim_{\mathbf{v} \rightarrow \infty} \Sigma_{\text{TSS}}(\mathbf{v}) = O$. Since free energy differences can only be resolved up to a constant additive difference, we can neglect terms like $\pi 1^T$ and formally write $(I - B)^{-1} = (I - Q^v)^{-1}$, even though the matrix $I - Q^v$ is not itself invertible. Then:

$$\Sigma_{\text{TSS}}(\mathbf{v}) = D_\pi^{-1} (A + A^T - Q D_\pi) D_\pi^{-1} \quad (4.68)$$

$$= D_\pi^{-1} (I - Q^v)^{-1} Q + Q^T (I - Q^v)^{-1} D_\pi^{-1} - O \quad (4.69)$$

$$= O Q^{-1} (I - Q^v)^{-1} Q + Q^T (I - Q^v)^{-1} Q^{-T} O - O. \quad (4.70)$$

The first equality follows by definition of $V(\mathbf{v})$, the second and third because $Q = D_\pi O$. As $\mathbf{v} \rightarrow \infty$, $(Q - \pi 1^T)^v \rightarrow 0$, and formally we get $(I - Q^v)^{-1} \rightarrow I$. The terms reduce to $O + O - O = O$, as claimed.

4.5 General proof of variance reduction for TSS

In this section we provide the details of the proof of Proposition 3 in the main text. Let $\Pi = \text{diag}(\pi_1, \dots, \pi_K)$ and let O denote the overlap matrix defined in (4.1). Since we are considering asymptotic covariance matrices, we assume the rung distribution π has converged to some fixed vector with strictly positive entries. In particular, Π is positive definite.

The covariance matrix of the TSS free energy difference estimators was derived in Section 4.1; the MBAR estimator and its covariance matrix were defined in 4.2. The two covariance matrices are denoted Σ_{TSS} and Σ_{MBAR} respectively, and both are defined below for convenience. For either covariance matrix Σ , the variance of the free energy difference estimates $F_j - F_i$ is obtained through the quadratic form

$$(\mathbf{e}_j - \mathbf{e}_i)^\top \Sigma (\mathbf{e}_j - \mathbf{e}_i), \quad (4.71)$$

where $\mathbf{e}_i, \mathbf{e}_j \in \mathbb{R}^K$ are unit basis vectors with the i^{th} and j^{th} entries respectively equal to 1. The claim of Proposition 3 in the main text is proved if we can establish that

$$(\mathbf{e}_j - \mathbf{e}_i)^\top (\Sigma_{\text{MBAR}} - \Sigma_{\text{TSS}}) (\mathbf{e}_j - \mathbf{e}_i) \geq 0, \quad \forall i, j \in [K]. \quad (4.72)$$

Theorem 4.1 *Suppose that the distributions $\mathbf{p}_i, i \in [K]$ are such that the overlap matrix O is irreducible. Let $\pi \in \mathcal{P}^e(K)$ and suppose MBAR uses a proportion $\pi_k > 0$ of independent samples from $\mathbf{p}_k(x)dx$, whereas for each fixed $F \in \mathbb{R}^K$ and the same $\pi = (\pi_1, \dots, \pi_K)$, TSS uses independent samples from $p_{\pi, F}(x, k)dx C(dk)$. Then, for any $i, j \in [K]$, the asymptotic variance of the estimate $F_i - F_j$ for TSS is smaller than that of MBAR.*

Proof Recall that the asymptotic covariance matrix of the TSS free energy difference estimates was determined in Section 4.1 to be $\Sigma_{\text{TSS}} = O$ (see (4.8)). The asymptotic covariance matrix of MBAR is (see e.g. equation 6.4, [14], or [8, 27, 9])

$$\Sigma_{\text{MBAR}} = (\Pi - \Pi O \Pi)^+ - \Pi^{-1}, \quad (4.73)$$

where $\Pi = \text{diag}(\pi_1, \dots, \pi_K)$ and the superscript $+$ denotes the Moore-Penrose pseudoinverse.

The positive semidefinite symmetric matrix $\Pi^{1/2} O \Pi^{1/2}$ admits 1 as a positive eigenvalue with positive normalized eigenvector $\mathbf{e} = \Pi^{1/2} \mathbf{1}$. Indeed,

$$(\Pi^{1/2} O \Pi^{1/2} \mathbf{e})_i = \pi_i^{1/2} \sum_{j \in [K]} O_{ij} \pi_j = \pi_i^{1/2} \sum_{j \in [K]} \pi_j \int_S \frac{\rho_i(x) \rho_j(x)}{\sum_{k \in [K]} \pi_k \rho_k(x)} dx = \pi_i^{1/2} \int_S \rho_i(x) dx = \pi_i^{1/2}. \quad (4.74)$$

Since $\Pi^{1/2} O \Pi^{1/2}$ is also irreducible, the Perron-Frobenius theorem implies that $\Pi^{1/2} O \Pi^{1/2}$ admits 1 as a simple eigenvalue and all other eigenvalues (which are necessarily non-negative real numbers) have magnitude less than 1. Therefore, $B = I - \Pi^{1/2} O \Pi^{1/2}$ and $A = \Pi - \Pi O \Pi = \Pi^{1/2} B \Pi^{1/2}$ are positive semidefinite matrices with one-dimensional kernels. The kernels of B and A are spanned by the normalized eigenvectors $\mathbf{e} = \Pi^{1/2} \mathbf{1}$ and $\mathbf{e}' = \mathbf{1}/K^{1/2}$, respectively. The Moore-Penrose pseudoinverse A^+ has the property $AA^+ = I - \mathbf{e}' \mathbf{e}'^\top$. We compute

$$A(\Sigma_{\text{MBAR}} - \Sigma_{\text{TSS}}) = I - \mathbf{e}' \mathbf{e}'^\top + \Pi(I - O \Pi) \Pi^{-1} - \Pi(I - O \Pi) O = (\Pi O)^2 - \mathbf{e}' \mathbf{e}'^\top. \quad (4.75)$$

Using that $\mathbf{e}'^\top A = (A \mathbf{e}')^\top = \mathbf{0}^\top$, we further compute

$$A(\Sigma_{\text{MBAR}} - \Sigma_{\text{TSS}})A = (\Pi O)^2 A - \mathbf{e}' \mathbf{e}'^\top A = (\Pi O)^2 A = (\Pi O)A(O \Pi), \quad (4.76)$$

which is positive semidefinite. If $\mathbf{v} \in \langle \mathbf{1} \rangle^\perp$, then $\mathbf{u} = A^+ \mathbf{v} \in \langle \mathbf{1} \rangle^\perp$ has the property $\mathbf{v} = A \mathbf{u}$. Then

$$\mathbf{v}^\top (\Sigma_{\text{MBAR}} - \Sigma_{\text{TSS}}) \mathbf{v} = \mathbf{u}^\top A (\Sigma_{\text{MBAR}} - \Sigma_{\text{TSS}}) A \mathbf{u} = \mathbf{u}^\top (\Pi O) A (O \Pi) \mathbf{u} = \mathbf{w}^\top A \mathbf{w} \geq 0, \quad (4.77)$$

where $\mathbf{w} = (O \Pi) \mathbf{u}$. The theorem follows because $\mathbf{v} = \mathbf{e}_j - \mathbf{e}_i$ lie in the orthogonal complement $\langle \mathbf{1} \rangle^\perp$ for all $1 \leq i, j \leq K$. $\square$

5 Proof of convergence

Before proceeding to the technical results, let us sketch key steps of the proof of convergence. As is typical in stochastic approximation, one relates the stochastic system to the (deterministic) mean-field system of coupled ordinary differential equations (ODEs), and then establishes some form of convergence of the deterministic system (see [6, 15] for broad overviews). We rely on existing results that phrase the convergence result in terms of regularity properties of the Markov process and stability properties of the system of ODEs.

A particular property of the system at hand is that the quantity

$$\frac{e^{F_k - F_k^*}}{\sum_{j \in [K]} \pi_j e^{F_j - F_j^*}} \quad (5.1)$$

can be controlled as a function of F and $\varepsilon > 0$, uniformly in $\pi \in \mathcal{P}^\varepsilon(K)$, and consequently we can say a lot about the system without knowing anything about π specifically. The key step is to find a Lyapunov function

that is uniform in π ; see Lemma 5.2 and Proposition 5.1. We then observe that $F(t) \rightarrow F^*$ regardless of $\pi \in \mathcal{P}^e(K)$, and consequently the system of mean field equations can be approximated as

$$1 - \frac{e^{F_k(t) - F_k^*}}{\sum_{\ell \in [K]} \pi_\ell e^{F_\ell(t) - F_\ell^*}} \approx 0 \quad (5.2)$$

$$\frac{e^{F_k(t) - F_k^*}}{\sum_{\ell \in [K]} \pi_\ell e^{F_\ell(t) - F_\ell^*}} (\mu_{km}^* - \mu_{km}(t)) \approx \mu_{km}^* - \mu_{km}(t). \quad (5.3)$$

The ODE dynamics for $\mu_{km}(t)$ imply $\mu_{km}(t) \rightarrow \mu_{km}^*$, also uniformly in $\pi \in \mathcal{P}^e(K)$. This uniformity allows us to establish convergence of $(F(t), \mu(t))$ for any trajectory $a(t)$, where

$$\dot{a}_j(t) = g_j^a(F(t), \mu(t), a(t)) \quad (5.4)$$

is the mean field associated to the stochastic process $\{a^t\}$ defined by a general function $G_j^a(x, k; F, \mu, a)$,

$$a_j^{t+1} = a_j^t + \frac{1}{t+1} G_j^a(x, k; F(t), \mu(t), a(t)), \quad j = 1, \dots, J \quad (5.5)$$

so that

$$g_j^a(F, \mu, a) = \mathbb{E}_{\pi, F} [G_j^a(\mathcal{X}, \mathcal{K}; F, \mu, a)], \quad j = 1, \dots, J. \quad (5.6)$$

As a special case, we will apply the results to the choice $a = o$, where $o = (o_1, \dots, o_K)$ denote the tilts, for which

$$G_j^o(x, k; \theta) = \frac{1}{\gamma_j(\mu)} 1_{\{j\}}(k) - o_j, \quad j = 1, \dots, K. \quad (5.7)$$

The rest of the work involves establishing that under a stability assumption on the dynamics for a , the whole system is globally asymptotically stable. We then use a converse Lyapunov theorem to produce a Lyapunov function necessary for the proof of convergence of the stochastic approximation procedure.

The general results we use are Theorem 5.5 and Proposition 6.1 in [2]. To apply this theorem, we need to check three conditions: a stability condition (Condition (A1) in [2]), a drift condition (DRI), and a condition on the gain sequences (A4). The stability condition requires us to find a global Lyapunov function, which is done in Section 5.2, while the drift condition requires us to check various regularity properties, which is done in Section 5.4.

We will make further assumptions on the compactness of Θ , but we note that none of these assumptions are used in establishing the existence of a global Lyapunov function for the free energy estimates F . When there exists a global Lyapunov function, [2] outlines a reprojection technique that offers stronger convergence results (see also [15] for discussion on reprojection techniques). We have not implemented such a technique, because in practice we observe boundedness and eventual convergence of the estimates. It is of theoretical interest to investigate the removal of the compactness assumption, but we do not pursue this question in the present work.

5.1 Setup

We recall the definitions from the main text, with the modification that we use a general stochastic approximation variable a instead of our particular choice of tilts o . The parameter of interest is $\theta = (F, \mu, a)$, which takes values in a compact subset $\Theta \subset \mathbb{R}^K \times \mathbb{R}^{KM} \times A$, where $A \subset \mathbb{R}^I$ is compact, and the rung distributions $\pi(\theta)$ and $\gamma(\mu)$ are user-defined deterministic functions $\pi: \Theta \rightarrow \mathcal{P}^e(K)$ and $\gamma: \mathbb{R}^{KM} \rightarrow \mathcal{P}^e(K)$, where we recall

that $\mathcal{P}^\varepsilon(K)$ is the space of probability distributions ω on $[K] = \{1, \dots, K\}$ with $\omega_k \geq \varepsilon$ for all $k \in [K]$. We assume that π is continuously differentiable with respect to θ , and to avoid degeneracies, we assume that it is invariant under additive shifts of the free energies, i.e.

$$\pi(F, \mu, a) = \pi(F + \alpha \vec{1}, \mu, a) \quad (5.8)$$

for all $\alpha \in \mathbb{R}$, where $\vec{1} = (1, \dots, 1) \in \mathbb{R}^K$.

The functions $\psi_1(x), \dots, \psi_m(x)$ can be evaluated for every $x \in \mathcal{S}$, and their averages with respect to $\rho_1(x)dx, \dots, \rho_K(x)dx$ define the quantities μ_{km}^* , which are assumed finite. In a standard TSS simulation, the μ_{km}^* are used to determine the coordinate invariant distribution defined in Section 2.2.2 the main text and discussed in Section 7.1.3 of the present supplementary material.

We will demonstrate the convergence of $\theta^t = (F^t, \mu^t, a^t)$ as defined in the stochastic algorithm algorithm

$$F_k^{t+1} = F_k^t + \frac{1}{t+1} (1 - R_k(\mathcal{X}^{t+1}; \theta^t)) \quad (5.9)$$

$$\mu_{km}^{t+1} = \mu_{km}^t + \frac{1}{t+1} R_k(\mathcal{X}^{t+1}; \theta^t) (\psi_m(\mathcal{X}^{t+1}) - \mu_{km}^t) \quad (5.10)$$

$$a_i^{t+1} = a_i^t + \frac{1}{t+1} G_i^a(\mathcal{X}^{t+1}, \mathcal{K}^{t+1}; \theta^t), \quad (5.11)$$

where

$$R(x, \theta) = \frac{e^{F_k - H_k(x)}}{\sum_{\ell \in [K]} \pi_\ell(\theta) e^{F_\ell - H_\ell(x)}}, \quad \theta = (F, \mu, a). \quad (5.12)$$

We assume that the observable G^a is continuously differentiable in $(x, k; \theta)$, and moreover that it is invariant under additive shifts of the free energies, that is,

$$G_j^a(x, k; F, \mu, a) = G_j^a(x, k; F + \alpha \vec{1}, \mu, a) \quad (5.13)$$

for all $j = 1, \dots, J$, $\alpha \in \mathbb{R}$, $F \in \mathbb{R}^K$, $(x, k) \in \mathcal{S} \times [K]$, $\mu \in \mathbb{R}^{KM}$ and $a \in \mathbb{R}^I$. Moreover, we assume that a^* is a globally asymptotically stable equilibrium for the system

$$g_i^a(F^*, \mu^*, a) = \mathbb{E}_{\pi(F^*, \mu^*, a), F^*} [G_i^a(\mathcal{X}, \mathcal{K}; F^*, \mu^*, a)] = 0. \quad (5.14)$$

In the special case $a = o$, the mean field is

$$\dot{o}_k(t) = \frac{\pi_k(F^*, \mu^*, o(t))}{\gamma_k(\mu^*)} - o_k(t), \quad k = 1, \dots, K. \quad (5.15)$$

In particular, this implies $\pi_k(\theta(t)) \rightarrow \gamma_k(\mu^*)$ for general functions π that have $o^* = (1, \dots, 1) \in \mathbb{R}^K$ as their global asymptotically stable equilibrium, and consequently the distribution of the rung process $\mathcal{K}^t$ will converge to $\gamma(\mu^*)$. With regard to the compactness of the domain of $o = (o_1, \dots, o_K)$, see (5.24).

5.2 Stability assumption (A1)

The goal of this section is to find a Lyapunov function which satisfies Assumption (A1) of [2] for the mean field equations (2.13), (2.14) and (2.7). This is done in Proposition 5.3; the conditions (A1) are listed immediately preceding the proposition, while the relevant definitions from ODE stability theory are provided in Section 5.2.2.

5.2.1 Change of coordinates

The stochastic approximation recursions (5.9) depend on F only up to a constant shift $F_k \rightarrow F_k + \alpha$ common to all $k \in [K]$. It will be simpler to apply results from stability theory if we identify a unique equilibrium point for the mean field of F . To this end, we define the vector subspace $\mathbb{L} = \text{span}(\vec{v}) = \{\alpha \vec{1} : \alpha \in \mathbb{R}\}$ and the norm

$$\|F\|_{\mathbb{L}}^2 = \inf_{\alpha \in \mathbb{R}} \|F - \alpha \vec{1}\|^2. \quad (5.16)$$

The space $(\mathbb{R}^K/\mathbb{L}, \|\cdot\|_{\mathbb{L}})$ is isometric to $(\mathbb{R}^{K-1}, \|\cdot\|)$, where $\|\cdot\|$ denotes the standard Euclidean 2-norm. Observe that

$$\|F\|_{\mathbb{L}}^2 = \|F - \text{proj}_{\mathbb{L}}(F)\|^2 = \sum_{k \in [K]} (F_k - \langle F \rangle)^2 = \|F - \langle F \rangle\|^2, \quad (5.17)$$

where

$$\langle F \rangle = \frac{1}{K} \sum_{k \in [K]} F_k. \quad (5.18)$$

In this norm, we wish to establish $\|F(t) - F^*\|_{\mathbb{L}} \rightarrow 0$ as $t \rightarrow \infty$ in a precise sense defined below as global asymptotic stability. To further simplify notation, we also assume without loss of generality that $F_k^* = 0$ for all $k \in [K]$.

Now, let $\bar{F}_k(t) = F_k(t) - \langle F(t) \rangle$, and note that $\|\bar{F}(t)\| = \|\bar{F}(t)\|_{\mathbb{L}}$. The dynamics for $\bar{\theta} = (\bar{F}, \mu, a)$ are

$$\dot{\bar{F}}_k(t) = \left[\frac{1}{K} \sum_{j \in [K]} \frac{e^{\bar{F}_j(t)}}{\sum_{\ell \in [K]} \pi_{\ell}(\bar{\theta}(t)) e^{\bar{F}_{\ell}(t)}} \right] - \frac{e^{\bar{F}_k(t)}}{\sum_{\ell \in [K]} \pi_{\ell}(\bar{\theta}(t)) e^{\bar{F}_{\ell}(t)}} \quad (5.19)$$

$$\dot{\mu}_{km}(t) = \frac{e^{\bar{F}_k(t)}}{\sum_{\ell \in [K]} \pi_{\ell}(\bar{\theta}(t)) e^{\bar{F}_{\ell}(t)}} (\mu_{km}^* - \mu_{km}(t)) \quad (5.20)$$

$$\dot{a}_i(t) = g_i^a(\bar{F}(t), \mu(t), a(t)), \quad (5.21)$$

We concisely write this system as $\dot{\bar{\theta}} = \bar{g}(\bar{\theta})$, where $\bar{g} = (\bar{g}^F, \bar{g}^{\mu}, \bar{g}^a)$ is defined by (5.19), (5.20) and (5.21). Note that the stability properties of $(\bar{F}, \mu, a)$ are exactly the same as those of (F, μ, a) , owing to the equality

$$\frac{e^{F_k(t)}}{\sum_{\ell \in [K]} \pi_{\ell}(\theta(t)) e^{F_{\ell}(t)}} = \frac{e^{F_k(t) - \langle F(t) \rangle}}{\sum_{\ell \in [K]} \pi_{\ell}(\theta(t)) e^{F_{\ell}(t) - \langle F(t) \rangle}}. \quad (5.22)$$

as well as the shift invariance of G_i^a and π assumed in (5.13) and (5.8).

To summarize, we seek to show that $\|\bar{F}(t)\|_{\mathbb{L}} \rightarrow 0$, $\|\mu(t) - \mu^*\| \rightarrow 0$ for any choice of G^a , and additionally if we assume the system (5.14) has a^* as its global asymptotically stable steady state, then we also show $\|a(t) - a^*\| \rightarrow 0$.

Finally, for the specific case $a = o$, we have

$$G_j^o(x, k; \theta) = \frac{1}{\gamma_j(\mu)} 1_{\{j\}}(k) - o_j. \quad (5.23)$$

In this case, the state space for $o(t)$ can be assumed compact. Since $\gamma \in \mathcal{P}^e(K)$, from the integrated form of the recurrence (5.11),

$$o_k^T = \frac{1}{T} \sum_{t=1}^T \frac{1_{\{k\}}(\mathcal{K}^t)}{\gamma_k(\mu^{t-1})}, \quad (5.24)$$

we infer o^t , hence $o(t)$, lie in the convex hull $O^e \subset \mathbb{R}_{\geq 0}^K$ defined by $o_k^t \geq 0$ for all $k \in [K]$ and $\sum_{k \in [K]} o_k^t \leq \varepsilon^{-1}$. In this case, we can take $A = O^e$.

5.2.2 Producing a Lyapunov function

We provide some definitions from stability theory. These will be used to establish convergence of $F(t)$ and $\mu(t)$, and if needed, convergence of $a(t)$. We use an ε - δ formulation of asymptotic stability, which is equivalent to modern formulations in terms of classes functions (see, for instance, [20, 13, 29]). Consider a general non-autonomous system

$$\dot{z}(t) = h(z(t), t). \quad (5.25)$$

It is always assumed that h is everywhere continuously differentiable and therefore satisfies the mild regularity properties required for existence and uniqueness of solutions to ordinary differential equations (see for instance [11]). The variable z takes values in $Z \subset \mathbb{R}^{n_z}$. We adopt the convention that $z(t; z_0, t_0)$ denotes a solution of (5.25) with initial condition $z(t_0; z_0, t_0) = z_0$. If t_0 is not specified and the system is autonomous, we sometimes use $z(t; z_0)$ to denote the solution with $z(0; z_0) = z_0$.

The following definition is taken from properties (ii), (v) in Section 3 of [20].

Definition 5.1 *The equilibrium z^* of the non-autonomous system is called uniformly asymptotically stable in the large if the following two properties are satisfied:*

1. (uniform stability) *for any $\eta > 0$, there is $\delta = \delta(\eta) > 0$, independent of t_0 , such that for any $|z_0 - z^*| < \delta$ and $t_0 \geq 0$, we have $|z(t; z_0, t_0) - z^*| < \eta$ for all $t \geq t_0$.*
2. (uniform asymptotic stability) *for any $\eta > 0$ and $\rho > 0$ there is $T = T(\rho, \eta) < \infty$, independent of t_0 , such that $|z(t; z_0, t_0) - z^*| < \eta$ whenever $t \geq t_0 + T$, $t_0 \geq 0$, and $|z_0 - z^*| < \rho$.*

We will also refer to uniform asymptotic stability in the large by the name **global asymptotic stability**.

A standard way of establishing global asymptotic stability is through the use of Lyapunov functions.

Definition 5.2 *A function $\alpha : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$ is said to be of class $\mathcal{K}_\infty$ if it is continuous, zero at zero, strictly increasing, and satisfies $\lim_{c \rightarrow \infty} \alpha(c) = \infty$. A function $W : D \subset \mathbb{R}^n \rightarrow \mathbb{R}_{\geq 0}$ is said to be positive definite if it is continuous, $W(0) = 0$, and $W(x) > 0$ for all $x \neq 0$.*

Let $V(t, z) : [0, \infty) \times Z \rightarrow [0, \infty)$ be a function defined on some subset $Z \subset \mathbb{R}^n$. Suppose that there exist functions $\alpha_1, \alpha_2 \in \mathcal{K}_\infty$ such that

$$\alpha_1(\|z\|) \leq V(t, z) \leq \alpha_2(\|z\|), \quad t \geq 0, z \in Z. \quad (5.26)$$

Suppose also that there is a positive definite function $W : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$ such that

$$V(t, z) \geq W(\|z\|), \quad t \geq 0, z \in Z. \quad (5.27)$$

*We call such a function V a **candidate Lyapunov function**. A function V is said to be a **Lyapunov function** for the system (5.25) if, in addition, there is a positive definite function $\sigma : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$ such that, for all $t \geq 0$*

$$\frac{d}{dt} V(t, z(t)) \leq -\sigma(\|z(t)\|), \quad t \geq 0. \quad (5.28)$$

We remark that given a positive-definite W defined on $D = \mathbb{R}_{\geq 0}$, it is straightforward to define a positive definite $\tilde{W}$ on $\mathbb{R}^n$ by $\tilde{W}(x) = W(\|x\|)$. We also note that the functions α_1, α_2 are said to be of class $\mathcal{K}_\infty$ (see e.g. [13]).

Let $z = (x, y) \in X \times Y$ for some $X, Y \subset \mathbb{R}^{n_x} \times \mathbb{R}^{n_y}$ and consider the system

$$\begin{pmatrix} \dot{x}(t) \\ \dot{y}(t) \end{pmatrix} = \begin{pmatrix} f(x(t), y(t)) \\ g(x(t), y(t)) \end{pmatrix}, \quad (5.29)$$

which we write as $\dot{z} = h(z)$ with $h(z(x, y)) = (f(x, y), g(x, y))$. We use the norm $|z| = |x| + |y|$, so in particular $|z - z^*| \geq |x - x^*| + |y - y^*|$. Let $z_0 = (x_0, y_0)$ and let $y(t; z_0)$ denote the solution of (5.29) with initial condition (x_0, y_0) . We can view $\dot{x} = f(x, y(t; z_0))$ as a time-dependent system for x with initial condition x_0 . Below, we study the stability properties of this time-dependent system for any trajectory $y(t), y : [0, \infty) \rightarrow Y$.

Lemma 5.1 *Let $U(x)$ be a candidate Lyapunov function in the sense of Definition 5.2, and suppose that there is a positive definite function σ such that*

$$\sigma(|x - x^*|) \leq \inf_{y \in Y} -\langle \nabla U(x), f(x, y) \rangle. \quad (5.30)$$

Then the state x^ is a globally asymptotically stable equilibrium for the time-dependent system*

$$\dot{x} = f(x, y(t)) \quad (5.31)$$

uniformly in the smooth map $y(t), y : [0, \infty) \rightarrow Y$.

Proof Let $y(t)$ be any smooth trajectory $y : [0, \infty) \rightarrow Y$, let $x_0 \in X$, and denote by $x(t; x_0)$ the solution of (5.31). Using the chain rule and the inequality (5.30), we find

$$\frac{d}{dt} U(x(t; x_0)) \leq \sup_y \langle \nabla U(x(t; x_0)), f(x(t; x_0), y) \rangle \leq -\sigma(|x(t; x_0) - x^*|).$$

By Theorem 22 in [20], the equilibrium x^* is uniform-asymptotically stable “in the large” (i.e., globally asymptotically stable). It follows from the uniformity of the bound (5.30) that the stability is uniform in the trajectory $y(\cdot)$, in the sense that in the second condition of Definition 5.1, $T(\rho, \eta)$ can be chosen independently of $y(\cdot)$ (see the proof of Theorem 13 in [20], or the proof of Theorem 3.1 in section X.3 of [11]). $\square$

We first identify a candidate Lyapunov function for the system $\bar{F}$ which will be suitably uniform with respect to a . We then use Lemma 5.1 to establish the global asymptotic stability of the equilibrium (F^*, μ^*) .

Lemma 5.2 *For any $\gamma \in \mathcal{P}^e(K)$, the function*

$$V_\gamma(F) = \log \left(\sum_{k \in [K]} \gamma_k e^{F_k} \right) - \sum_{k \in [K]} \gamma_k F_k$$

is a candidate Lyapunov function for the equilibrium $F^ = 0$, in the sense that V_γ is positive definite, and there exist α_1, α_2 as in Definition 5.2 such that $\alpha_1(\|F\|_{\mathbb{L}}) \leq V_\gamma(F) \leq \alpha_2(\|F\|_{\mathbb{L}})$.*

Proof Let $r(F) = (r_1(F), \dots, r_K(F))$ be defined by

$$r_k(F) = \frac{\gamma_k e^{F_k}}{\sum_{\ell \in [K]} \gamma_\ell e^{F_\ell}}$$

and note that γ and $r(F)$ are mutually absolutely continuous for all $F \in \mathbb{R}^K$ and that $V_\gamma(F) = D_{\text{KL}}(\gamma \| r(F))$, where $D_{\text{KL}}(\cdot \| \cdot)$ is the relative entropy (also known as the Kullback-Leibler divergence):

$$V_\gamma(F) = \log \left(\sum_{k \in [K]} \gamma_k e^{F_k} \right) - \sum_{k \in [K]} \gamma_k F_k = - \sum_{k \in [K]} \gamma_k \log \left(\frac{r_k(F)}{\gamma_k} \right) = D_{\text{KL}}(\gamma \| r(F)). \quad (5.32)$$

Relative entropy is known to be positive definite (apply Jensen's inequality). For any two probability distributions γ', γ'' , $D_{\text{KL}}(\gamma' \| \gamma'') = 0$ if and only if $\gamma' = \gamma''$; the condition $r_k(F) = \gamma_k$ for all $k \in [K]$ forces $F_k = \text{const}$, which is equivalent to $\|F\|_{\mathbb{L}} = 0$. It is also easy to check that $V_\gamma(F) \rightarrow \infty$ linearly with $\|F\|_{\mathbb{L}} \rightarrow \infty$. Together with the fact that $\varepsilon \leq \gamma_k \leq 1 - \varepsilon$ for all $k \in [K]$, we find that there exist $\alpha_1, \alpha_2 \in \mathcal{K}_\infty$ such that $\alpha_1(\|F\|_{\mathbb{L}}) \leq V_\gamma(F) \leq \alpha_2(\|F\|_{\mathbb{L}})$ for all $F \in \mathbb{R}^K$. $\square$

We record the following simple estimate for use in the subsequent proposition.

Lemma 5.3 *For any $\gamma \in \mathcal{P}^\varepsilon(K)$,*

$$\frac{e^{F_k}}{\sum_{\ell \in [K]} \gamma_\ell e^{F_\ell}} \geq e^{-2\|F\|_{\mathbb{L}}}. \quad (5.33)$$

Proof Recall that $\|F\|_{\mathbb{L}}^2 = \sum_{k \in [K]} (F_k - \langle F \rangle)^2$, so for any $\ell, k \in [K]$ we have $|F_\ell - F_k| \leq |F_\ell - \langle F \rangle| + |\langle F \rangle - F_k| \leq 2\|F\|_{\mathbb{L}}$. Assume without loss of generality that $F_K = \max_{k \in [K]} F_k$. Then $e^{F_k - F_K} \leq 1$ for all $k \in [K]$, hence $\sum_{k \in [K]} \gamma_k e^{F_k - F_K} \leq 1$ and

$$\frac{e^{F_k}}{\sum_{\ell \in [K]} \gamma_\ell e^{F_\ell}} = \frac{e^{F_k - F_K}}{\sum_{\ell \in [K]} \gamma_\ell e^{F_\ell - F_K}} \geq e^{F_k - F_K} \geq e^{-2\|F\|_{\mathbb{L}}}. \quad (5.34)$$

$\square$

Proposition 5.1 *Let $\bar{\theta}^0 = (\bar{F}^0, \mu^0, a^0) \in \Theta$ be any initial condition, and let $\bar{\theta}(t) = \bar{\theta}(t; \bar{\theta}^0)$ denote the solution of the time-independent system $\dot{\bar{\theta}} = \bar{g}(\bar{\theta})$ defined by (5.19), (5.20), (5.21). The state (F^*, μ^*) is globally asymptotically stable for the time-dependent system*

$$\dot{\bar{F}}_k(t) = \left[\frac{1}{K} \sum_{j \in [K]} \frac{e^{\bar{F}_j(t)}}{\sum_{\ell \in [K]} \pi_\ell(\bar{\theta}(t)) e^{\bar{F}_\ell(t)}} \right] - \frac{e^{\bar{F}_k(t)}}{\sum_{\ell \in [K]} \pi_\ell(\bar{\theta}(t)) e^{\bar{F}_\ell(t)}} \quad (5.35)$$

$$\dot{\mu}_{km}(t) = \frac{e^{\bar{F}_k(t)}}{\sum_{\ell \in [K]} \pi_\ell(\bar{\theta}(t)) e^{\bar{F}_\ell(t)}} (\mu_{km}^* - \mu_{km}) \quad (5.36)$$

uniformly in the trajectory $a(t) = a(t; \bar{\theta}^0)$.

Proof For any fixed $\gamma \in \mathcal{P}^\varepsilon(K)$, define the function

$$U(\bar{F}, \mu) = V_\gamma(\bar{F}) + \frac{1}{2} \sum_{k \in [K]} \sum_{m=1}^M (\mu_{km} - \mu_{km}^*)^2. \quad (5.37)$$

We have shown in Lemma 5.2 that V_γ is a candidate Lyapunov function, and it is clear that any sum of quadratic terms is as well; U is therefore a candidate Lyapunov function. If we differentiate $U(\bar{F}(t), \mu(t))$ with respect to t and apply the chain rule, we get

$$\frac{d}{dt} U(\bar{F}, \mu) = \langle \nabla_F U(\bar{F}, \mu), \bar{g}^F(\bar{F}, \mu, a) \rangle + \langle \nabla_\mu U(\bar{F}, \mu), g^\mu(\bar{F}, \mu, a) \rangle, \quad (5.38)$$

where in the above display ∇_F and ∇_μ are derivatives with respect to F and μ respectively, and the t dependence of $\bar{F}, \mu$ and o has been suppressed to avoid excessive parentheses. To apply Lemma 5.1 with $x = (\bar{F}, \mu)$ and $y = o$, we need to bound the terms $\langle \nabla_F U, \bar{g}^F(\bar{F}, \mu, a) \rangle$ and $\langle \nabla_\mu U, g^\mu(\bar{F}, \mu, a) \rangle$ (defined in (5.19), (5.20)) in terms of $\|\bar{F}\| + \|\mu - \mu^*\|$, uniformly in a .

We first handle the inner product involving $\bar{g}^F$. Writing $\bar{\theta} = (\bar{F}, \mu, a)$, we have

$$\langle \nabla V_\gamma(F), \bar{g}^F(\bar{\theta}) \rangle = \sum_{k \in [K]} \gamma_k \left(\frac{e^{\bar{F}_k}}{\sum_{\ell \in [K]} \gamma_\ell e^{\bar{F}_\ell}} - 1 \right) \left(\left[\frac{1}{K} \sum_{j \in [K]} \frac{e^{\bar{F}_j}}{\sum_{\ell \in [K]} \pi_\ell(\bar{\theta}) e^{\bar{F}_\ell}} \right] - \frac{e^{\bar{F}_k}}{\sum_{\ell \in [K]} \pi_\ell(\bar{\theta}) e^{\bar{F}_\ell}} \right) \quad (5.39)$$

$$= - \sum_{k \in [K]} \gamma_k \left(\frac{e^{\bar{F}_k}}{\sum_{\ell \in [K]} \gamma_\ell e^{\bar{F}_\ell}} - 1 \right) \left(\frac{e^{\bar{F}_k}}{\sum_{\ell \in [K]} \pi_\ell(\bar{\theta}) e^{\bar{F}_\ell}} - 1 \right) \quad (5.40)$$

To establish the uniform descrescence property (5.30), it suffices to show that the sum above can be bounded below uniformly in $\pi \in \mathcal{P}^e(K)$. For $c \geq 0$, define

$$\sigma^F(c) = \inf_{\gamma' \in \mathcal{P}^e(K)} \inf_{F' : \|F'\|_{\mathbb{L}} = c} \sum_{k=1}^K \gamma_k \left(\frac{e^{F'_k}}{\sum_{\ell \in [K]} \gamma_\ell e^{F'_\ell}} - 1 \right) \left(\frac{e^{F'_k}}{\sum_{\ell \in [K]} \gamma'_\ell e^{F'_\ell}} - 1 \right). \quad (5.41)$$

We claim that σ^F is positive definite: $\sigma^F(c) = 0$ if and only if $c = 0$, and $\sigma^F > 0$ otherwise. Clearly $\sigma^F(0) = 0$, with the infimum being attained by $F' = (0, \dots, 0)$. To see that $\sigma^F(c) > 0$ for $c > 0$, let $F' \in \mathbb{R}^K, \gamma' \in \mathcal{P}^e(K)$ be arbitrary and define the two functions $f, g : \mathbb{R} \rightarrow \mathbb{R}$ by

$$f(w) = \frac{e^w}{\sum_{\ell \in [K]} \gamma_\ell e^{F'_\ell}} - 1, \quad g(w) = \frac{e^w}{\sum_{\ell \in [K]} \gamma'_\ell e^{F'_\ell}} - 1. \quad (5.42)$$

Both f and g are monotonically increasing, so if $f(F'_k) - f(F'_\ell) \geq 0$ we must have $g(F'_k) - g(F'_\ell) \geq 0$, and if $f(F'_k) - f(F'_\ell) \leq 0$ we must have $g(F'_k) - g(F'_\ell) \leq 0$. Therefore,

$$\sum_{k \in [K]} \sum_{\ell \in [K]} \gamma_k \gamma_\ell (f(F'_k) - f(F'_\ell)) (g(F'_k) - g(F'_\ell)) = \frac{\sum_{\ell \in [K]} \gamma_\ell e^{F'_\ell}}{\sum_{\ell \in [K]} \gamma'_\ell e^{F'_\ell}} \sum_{k \in [K]} \sum_{\ell \in [K]} \gamma_k \gamma_\ell |f(F'_k) - f(F'_\ell)|^2 \geq 0. \quad (5.43)$$

Because all terms in the above sum are positive, equality holds if and only if $f(F'_1) = \dots = f(F'_K)$, which forces $F'_1 = \dots = F'_K$ or, equivalently, $\|F'\|_{\mathbb{L}} = 0$. Observe that

$$\sum_{k \in [K]} \gamma_k f(F'_k) g(F'_k) \geq \left(\sum_{k \in [K]} \gamma_k f(F'_k) \right) \left(\sum_{k \in [K]} \gamma_k g(F'_k) \right) = 0, \quad (5.44)$$

where the inequality follows by re-arranging (5.43) and the equality follows from the specific form of f , which satisfies $\sum_{k \in [K]} \gamma_k f(F'_k) = 0$ for any $F' \in \mathbb{R}^K$. Since $\mathcal{P}^e(K)$ is compact and the set $\{F : \|F\|_{\mathbb{L}} = c\}$ is compact in $\mathbb{R}^{K-1}$, the infima in (5.41) are attained. It follows that σ^F is positive definite and that

$$\sup_{\mu \in \mathbb{R}^{KM}, a \in A} \langle \nabla_F V_\gamma(\bar{F}), \bar{g}^F(\bar{F}, \mu, a) \rangle \leq -\sigma^F(\|\bar{F}\|_{\mathbb{L}}). \quad (5.45)$$

Now we return to the second term in (5.38). Using the bound in Lemma 5.3, we have

$$\sum_{k \in [K]} \sum_{m=1}^M \frac{e^{\bar{F}_k}}{\sum_{j \in [K]} \pi(\bar{\theta}) e^{\bar{F}_j}} (\mu_{km} - \mu_{km}^*)^2 \geq e^{-2\|\bar{F}\|_{\mathbb{L}}} \sum_{k \in [K]} \sum_{m=1}^M (\mu_{km} - \mu_{km}^*)^2 = e^{-2\|\bar{F}\|_{\mathbb{L}}} \|\mu - \mu^*\|^2. \quad (5.46)$$

Finally, let $\sigma^\mu(c, d) = e^{-2c}d$. It is straightforward to find a positive definite function σ that $\sigma(c + d) \leq \sigma^F(c) + \sigma^\mu(c, d)$ (though σ will satisfy $\lim_{c \rightarrow \infty} \sigma(c) = 0$; see [12] for this style of argument). From equations (5.45) and (5.46) we find

$$\sup_{o \in O^e} \{ \langle \nabla_F U(\bar{F}, \mu), \bar{g}^F(\bar{F}, \mu, a) \rangle + \langle \nabla_\mu U(\bar{F}, \mu), g^\mu(\bar{F}, \mu, a) \rangle \} \leq -\sigma(\|\bar{F}\|_{\mathbb{L}} + \|\mu - \mu^*\|). \quad (5.47)$$

The conditions of Lemma 5.1 are thus satisfied, with $x^* = (F^*, \mu^*)$ ($F^* = 0$) and an arbitrary trajectory $y(t) = a(t)$. It follows that (F^*, μ^*) is a globally asymptotically stable equilibrium for the system $\dot{\bar{\theta}} = \bar{g}(\bar{\theta})$. $\square$

Proposition 5.1 establishes the global asymptotic stability of (F^*, μ^*) independently of the trajectory $a(t) = a(t; \bar{\theta}^0)$. We remark that this is enough to guarantee convergence of the stochastic iterates (F^t, μ^t) if $\pi(\bar{\theta}) = \gamma(\mu)$ (see 5.1).

We now proceed to establish the global asymptotic stability of (F^*, μ^*, a^*) using only the assumptions that π is continuously differentiable and that a^* is a globally asymptotically stable equilibrium for the system (5.15). We require a definition (see Definition 56.1 in [29]) and a Lemma.

Definition 5.3 Suppose that the system

$$\dot{z}(t) = h(z(t), t) \quad (5.48)$$

has a unique solution and unique equilibrium z^* . Consider any function s for which the system

$$\dot{z}(t) = h(z(t), t) + s(z(t), t) \quad (5.49)$$

has a unique solution, denoted $\tilde{z}(t; z_0, t_0)$. The equilibrium z^* for the system (5.48) is called **totally stable** if for any $\eta > 0$, there are $\delta_1(\eta) > 0$ and $\delta_2(\eta) > 0$ such that

$$|z_0 - z^*| < \delta_1, \quad |s(z(t), t)| < \delta_2 \quad (5.50)$$

together imply $|\tilde{z}(t; z_0, t_0)| < \eta$.

Lemma 5.4 Suppose that h is continuously differentiable. If z^* is a globally asymptotically stable equilibrium for the system (5.48), it is totally stable.

Moreover, there is an $r_1 > 0$ such that, for any $0 < \delta < r_1$ and any $r_2 \in (\delta, r_1]$, there is a $\Delta > 0$ and a $T > 0$ such that, for any $t_0 \geq 0$ and any $z_0 \in \mathbb{R}^{n_z}$, $\delta \leq |z_0 - z^*| \leq r_2$, the solution $z(t; t_0, z_0)$ of (5.48) satisfies $|z(t; t_0, z_0) - z^*| < \delta$ for all $t \geq t_0 + T$, provided that $|s(t, z)| < \Delta$ for all $t \geq t_0, |z - z^*| \leq r_1$.

The proof of the first statement in Lemma 5.4 can be found in [29], Theorem 56.4; the proof of the second statement can be found in [11], Theorem 5.2, Section X.5.

Lemma 5.5 Let Y be compact, and suppose that y^* is a globally asymptotically stable equilibrium for the system

$$\dot{y}(t) = g(x^*, y(t)) \quad (5.51)$$

and that $\dot{x}(t) = f(x(t), y(t))$ satisfies the conditions of Lemma 5.1. Then (x^*, y^*) is a globally asymptotically stable equilibrium of the system $(\dot{x}, \dot{y}) = (f(x(t), y(t)))$.

Proof First we prove uniform stability. Let $\eta > 0$ be given. Since y^* is a globally asymptotically stable equilibrium of $\dot{y} = g(x^*, y)$, it is totally stable by Lemma 5.4. Therefore, for the given $\eta > 0$, there exist $\delta_1 = \delta_1(\eta) > 0$ and $\delta_2 = \delta_2(\eta) > 0$ such that the unique solution $y(t; z_0)$ of the system

$$\dot{y}(t) = g(x(t), y(t)) = g(x^*, y(t)) + s(t, y(t)), \quad y(0) = y_0 \quad (5.52)$$

satisfies

$$|y(t; z_0) - y^*| < \eta/2 \quad (5.53)$$

for all $t \geq 0$ whenever $|y_0 - y^*| < \delta_1$ and $|s(t, y)| < \delta_2$ for all $t \geq 0$ and $y \in B(y^*, \eta)$.

Let $r(x, y) = g(x, y) - g(x^*, y)$, and observe from (5.52) that $s(t, y) = r(x(t; z_0), y)$. Since g is continuous, r is continuous. By continuity of r , for each fixed y there is $\beta_y = \beta_y(\delta_2) > 0$ such that

$$|r(x, y)| < \delta_2 \quad (5.54)$$

whenever $|x - x^*| < \beta_y$. Without loss of generality, we can assume the map $y \rightarrow \beta_y$ is continuous, and hence it achieves its minimum on the compact set $\overline{B(y^*, \eta)}$; we denote $\beta = \min_{y \in \overline{B(y^*, \eta)}} \beta_y > 0$.

Next, we use the uniform stability of x^* for $\dot{x} = f(x, y(t))$, which holds uniformly in the trajectory $y(\cdot)$, $y : [0, \infty) \rightarrow Y$. In particular, it holds for $y(t) = y(t; z_0)$, where $y(t; z_0)$ denotes the solution of (5.52) with $y(0; z_0) = y_0$. For the given $\beta > 0$, there is $\delta_x > 0$ such that

$$|x(t; z_0) - x^*| < \min(\beta, \eta/2) \quad (5.55)$$

for all $t \geq 0$ whenever $x_0 \in B(x^*, \delta_x)$.

We claim that $\delta = \min(\delta_1, \delta_x) > 0$ satisfies the conditions of uniform stability. Let z_0 satisfy $|z_0 - z^*| < \delta$, so by the triangle inequality $|x_0 - x^*| < \delta$ and $|y_0 - y^*| < \delta$. By the choice of $\delta \leq \delta_x$, (5.55) holds for all $t \geq 0$, so $x(t; z_0)$ stays within $\eta/2$ of x^* . Since $x(t; z_0)$ also stays within β of x^* for all $t \geq 0$, (5.54) holds, and therefore $|s(t, y)| = |r(x(t; z_0), y)| < \delta_2$ for all $t \geq 0$ and $y \in B(y^*, \eta)$. Note also that $|y_0 - y^*| < \delta \leq \delta_1$ by the assumption $|z_0 - z^*| < \delta$. By the total stability of (5.52), we have $|y(t; z_0) - y^*| < \eta/2$ for all $t \geq 0$. Thus we have $|z(t; z_0) - z^*| \leq \eta/2 + \eta/2 = \eta$ for all $t \geq 0$, which establishes the uniform stability of z^* for the system $\dot{z} = g(z)$.

Next we establish equiasymptotic stability of z^* . Let $\eta > 0$ and $\rho > 0$ be given. Using the total stability of (5.51), there are once again $\delta_1 = \delta_1(\eta) > 0$, $\delta_2 = \delta_2(\eta) > 0$ such that

$$|y(t; t_0, y_0) - y^*| < \eta/2 \quad (5.56)$$

for all $t \geq t_0$ whenever $|y_0 - y^*| < \delta_1$ and $|s(t, y)| < \delta_2$. Here, we write $y(t; t_0, y_0)$ to distinguish the solution starting from y_0 at t_0 (rather than $t_0 = 0$), and we note that $y(t; z_0) = y(t; t_0, y(t_0; z_0))$. Notice also that we may assume without loss of generality that $\delta_1 < \eta/2$, as otherwise the statement is vacuous.

To show that the perturbed system (5.52) still brings trajectories close to the equilibrium y^* of the unperturbed system, we apply the second part of Lemma 5.4 to the system (5.52) with $r_1 = \sup_{y, y' \in Y} |y - y'|$, $r_2 = r_1$ and the δ_1 we have just obtained above. The theorem yields $\Delta = \Delta(\delta_1, r_2) > 0$ and $T_Y = T_Y(\eta, r_2)$ such that, the solution $y(t; t_0, y_0)$ of (5.52) with $y(t_0) = y_0$ satisfies

$$|y(t; z_0) - y^*| < \delta_1 \quad (5.57)$$

for $t \geq T_Y + t_0$, provided that $|s(t, y)| < \Delta$ for all $t \geq t_0$ and $r_2 \geq |y_0 - y^*| \geq \delta_1$.

Now we use the fact that $\dot{x} = f(x, y(t))$ is globally asymptotically stable uniformly in the trajectory $y(\cdot)$ to satisfy the bound $|s(t, y)| < \min(\delta_2, \Delta)$ for sufficiently large t . To this end, we again use the continuity of g to find a $\delta_x = \delta_x(\delta_2, \Delta) > 0$ such that

$$|s(t, y)| < \min(\delta_2, \Delta) \quad (5.58)$$

holds whenever $x \in B(x^*, \delta_x)$. This time the δ_x must work for all $y \in Y$, which is possible by compactness of Y . By the global asymptotic stability, for the given $\rho > 0$ there is $T_X = T_X(\rho, \eta, \delta_x) < \infty$ such that

$$|x(t; z_0) - x^*| < \min(\eta/2, \delta_x) \quad (5.59)$$

whenever $|x_0 - x^*| < \rho$ and $t \geq T_X$.

We now compile the estimates. Set $T = T(\rho, \eta) = T_X + T_Y$. We claim that for any $z_0 \in B(z^*, \rho)$, we have $|z(t; z_0) - z^*| < \eta$ for all $t \geq T$. First, it is straightforward that $x(t; z_0)$ arrives within $\eta/2$ of x^* by time T_X , owing to (5.59). By the same estimate, $x(t; z_0)$ is within δ_x of x^* at all times $t \geq T_X$ for any $z_0 \in B(z^*, \rho)$, so (5.58) holds for all $t \geq T_X$ and $y \in Y$.

Now, at time $t_0 = T_X$, either $y(t; z_0)$ is in $B(y^*, \delta_1)$ or it is not. If it is, then together with (5.58) we have that (5.56) holds by the total stability estimate, and hence $|y(t; z_0) - y^*| < \eta/2$ for all $t \geq T_X$. If not, then by estimates (5.57) and (5.58) we have $|y(t; z_0) - y^*| < \delta_1 < \eta/2$ for all $t \geq T_X + T_Y$. Either way, $|y(t; z_0) - y^*| < \eta/2$ for all $t \geq T$. Together with estimate (5.59) and the triangle inequality, we obtain the desired result. $\square$

Proposition 5.2 *Suppose that (5.14) is uniformly globally asymptotically stable. Then the time-independent system $\dot{\bar{\theta}}(t) = \bar{g}(\bar{\theta}(t))$ is (uniformly) globally asymptotically stable.*

Proof We know from Proposition 5.1 that the equilibrium $(0, \mu^*)$ is globally asymptotically stable for the system $(\dot{F}, \dot{\mu}) = (\bar{g}^F, g^\mu)$. The proof of the present proposition follows by applying Lemma 5.5 with $x = (\bar{F}, \mu)$ and $y = a$ to the system $f = (\bar{g}^F, g^\mu)$ and $g = g^a$. $\square$

Next, we establish the global asymptotic stability of the system (5.21) when π is taken to be TSS's visit control.

Lemma 5.6 *For any $\varepsilon_\pi \in [0, 1]$, let*

$$\pi_k^{\varepsilon_\pi}(o) = (1 - \varepsilon_\pi)\pi_k(o) + \varepsilon_\pi\gamma_k(\mu^*), \quad (5.60)$$

where π_k is the unregularized TSS visit control,

$$\pi_k(o) = \frac{\gamma_k(\mu^*)o_k^{-\eta}}{\sum_{\ell \in [K]} \gamma_\ell(\mu^*)o_\ell^{-\eta}}. \quad (5.61)$$

Then, for any $\varepsilon_\pi \in [0, 1]$, the system (5.15) with $\pi = \pi^{\varepsilon_\pi}$ has **1** as its globally asymptotically stable equilibrium.

Proof When $\eta = 0$, we have $\pi_\pi^e = \gamma$ for all $\varepsilon_\pi \in [0, 1]$ and the differential equation (5.15) for o becomes $\dot{o}_k = 1 - o_k$, which is trivially globally asymptotically stable. For $\eta > 0$ it suffices to exhibit a global

Lyapunov function. Let $U(o) = \frac{1}{2} \sum_{k \in [K]} \gamma_k (1 - o_k)^2$, and note that U is a candidate Lyapunov function (in the sense of Definition 5.2 in Section 5.2). We have, for any $\varepsilon_\pi \in [0, 1]$,

$$\langle \nabla U(o), \dot{o} \rangle = - \sum_{k \in [K]} \gamma_k (1 - o_k) \left(\frac{\pi_k^{\varepsilon_\pi}}{\gamma_k(\mu^*)} - o_k \right) \quad (5.62)$$

$$= - \sum_{k \in [K]} \gamma_k (1 - o_k) \left((1 - \varepsilon_\pi) \frac{o_k^{-\eta}}{\sum_{\ell \in [K]} \gamma_\ell o_\ell^{-\eta}} - (1 - \varepsilon_\pi) o_k + \varepsilon_\pi - \varepsilon_\pi o_k \right) \quad (5.63)$$

$$= -(1 - \varepsilon_\pi) \left[1 - E_\gamma[O] - \frac{E_\gamma[O^{1-\eta}]}{E_\gamma[O^{-\eta}]} + E_\gamma[O^2] \right] - \varepsilon_\pi E_\gamma[(1 - O)^2], \quad (5.64)$$

where $E_\gamma[f(O)] = \sum_{k \in [K]} \gamma_k f(o_k)$ for any function $f : [K] \rightarrow \mathbb{R}$. By the correlation inequality (see (3.35) and (3.36)) applied to the two increasing functions $f(o) = o$ and $g(o) = -o^{-\eta}$ (note $\eta > 0$ and $o_k > 0$), we have

$$-E_\gamma[O^{1-\eta}] = E_\gamma[f(O)g(O)] \geq E_\gamma[f(O)]E_\gamma[g(O)] = -E_\gamma[O]E_\gamma[O^{-\eta}], \quad (5.65)$$

and so $E_\gamma[O^{1-\eta}]/E_\gamma[O^{-\eta}] \leq E_\gamma[O]$. Therefore,

$$1 - E_\gamma[O] - \frac{E_\gamma[O^{1-\eta}]}{E_\gamma[O^{-\eta}]} + E_\gamma[O^2] \geq 1 - 2E_\gamma[O] + E_\gamma[O^2] = E_\gamma[(1 - O)^2]. \quad (5.66)$$

It follows from (5.64) that $\langle \nabla U(o), \dot{o} \rangle \leq -2U(o)$, which concludes the proof. $\square$

Finally, we conclude the existence of an appropriate Lyapunov function which satisfies the assumptions below. We will invoke a classical converse Lyapunov theorem; see [13, 11, 20, 29].

Assumption (A1). Assume Θ is an open subset of $\mathbb{R}^q$, $g : \Theta \rightarrow \mathbb{R}^q$ is continuous and there exists a continuously differentiable function $w : \Theta \rightarrow [0, \infty)$ such that

1. There exists $M_0 > 0$ such that

$$\mathcal{L} = \{\theta \in \Theta : \langle \nabla w(\theta), g(\theta) \rangle = 0\} \subset \{\theta \in \Theta : w(\theta) < M_0\}, \quad (5.67)$$

2. There exists $M_1 \in (M_0, \infty]$ such that $\{\theta \in \Theta : w(\theta) \leq M_1\}$ is a compact set,
3. For any $\theta \in \Theta \setminus \mathcal{L}$, $\langle \nabla w(\theta), g(\theta) \rangle < 0$,
4. The closure of $w(\mathcal{L})$ has an empty interior.

Proposition 5.3 *There exists a global Lyapunov function for the system $\dot{\bar{\theta}} = \bar{g}(\bar{\theta}(t))$ which satisfies Assumption (A1).*

Proof By Theorem 23 in [20], there is a function $w(\bar{\theta})$ and $\alpha_1, \alpha_2 \in \mathcal{K}_\infty$ such that $\alpha_1(\|\bar{\theta} - \bar{\theta}^*\|) \leq w(\bar{\theta}) \leq \alpha_2(\|\bar{\theta} - \bar{\theta}^*\|)$. The function w vanishes only at the point $\bar{\theta}^* = (0, \mu^*, a^*)$, is positive otherwise, and smooth ($w \in C^\infty$). Moreover, $\langle \nabla w(\bar{\theta}), \bar{g}(\bar{\theta}) \rangle \leq -\sigma(\|\bar{\theta} - \bar{\theta}^*\|)$ for some positive definite function σ .

Now we verify that this function w satisfies Assumption (A1). In our case $\mathcal{L} = \{\bar{\theta}^*\} = \{(0, \mu^*, a^*)\}$ and $w(\bar{\theta}) = 0$ if and only if $\bar{\theta} = \bar{\theta}^* \in \mathcal{L}$. Thus 1. is satisfied with any $M_0 > 0$.

Since $\alpha_1 \in \mathcal{K}_\infty$ and $w(\bar{\theta}) \geq \alpha_1(\|\bar{\theta} - \bar{\theta}^*\|)$, the function $w(\bar{\theta})$ is coercive ($w(\bar{\theta}) \rightarrow \infty$ as $\|\bar{\theta}\| \rightarrow \infty$) and therefore has compact level sets, so there is $M_1 > 0$ such that $\{\bar{\theta} \in \Theta : w(\bar{\theta}) \leq M_1\}$ is compact. Thus condition 2. can be satisfied, and we can choose any $M_0 \in (0, M_1)$ to satisfy condition 1.

Condition 3. follows immediately from the positive definiteness of the function σ .

Finally, $w(\mathcal{L}) = \{0\}$, and so the closure has empty interior. $\square$

5.3 Assumption (A4)

Assumption (A4) in [2] places conditions on the step sizes. The stochastic approximation algorithm with step size $\beta_k = 1/k$ satisfies

$$\sum_{k=1}^{\infty} \gamma_k = \infty, \quad \sum_{k=1}^{\infty} \gamma_k^{\delta} < \infty \quad (5.68)$$

for any $\delta > 1$. In the notation of [2], we need $\delta \in (1, p(1+\alpha)/(p+\alpha))$ for some $p \geq 2$ and $\alpha \in (0, 1]$. In our case we take $\alpha = 1$, $p \geq 2$, and we set $\varepsilon_k = C\gamma_k^{\eta}$ for some constant C and η such that

$$\frac{\delta-1}{\alpha} \leq \eta \leq \frac{p-\delta}{p}, \quad (5.69)$$

taking p as large as necessary. The sequence $\{\varepsilon_k\}_{k \geq 1}$ then satisfies Assumption (A4) of [2].

5.4 Drift conditions (DRI)

The drift conditions require us to verify basic properties to ensure ergodicity of the transition kernels for each fixed $\theta \in \Theta$. We reproduce the drift condition (DRI) in Section 6. of [2], which guarantees their Assumptions (A2) and (A3). Let $\Omega = \mathcal{S} \times [K]$. For any function $U : \Omega \rightarrow [1, \infty)$, define the norm

$$\|g\|_U = \sup_y \frac{|g(y)|}{U(y)}, \quad (5.70)$$

and let $\mathcal{L}_U = \{g : \Omega \rightarrow \mathbb{R}^{n_{\theta}} \mid \|g\|_U < \infty\}$.

(DRI). For any $\theta \in \Theta$, P_{θ} is ψ -irreducible and aperiodic. In addition, there exist a function $U : \Omega \rightarrow [1, \infty)$ and constants $p \geq 2$, $\beta \in [0, 1]$ such that for any compact set $\mathbb{K} \subset \Theta$,

(DRI1) there exist an integer m , constants $\lambda \in (0, 1)$, $b, \kappa, \delta > 0$ and a set $C \subset \Theta$ with a probability measure probability measure q such that

1. $\sup_{\theta \in \mathbb{K}} P_{\theta}^m U^p(y) \leq \lambda U^p(y) + b 1_C(y)$,
2. $\sup_{\theta \in \mathbb{K}} P_{\theta} U^p(y) \leq \kappa U^p(y)$, for all $y \in \Omega$,
3. $\inf_{\theta \in \mathbb{K}} P_{\theta}^m(y, A) \geq \delta q(A)$, for all $y \in C$ and Borel-measurable $A \subset \Omega$

(DRI2) there exists a constant $c > 0$ such that, for all $y \in \Omega$,

1. $\sup_{\theta \in \mathbb{K}} |G_{\theta}(y)| \leq c U(y)$
2. $\sup_{(\theta, \theta') \in \mathbb{K}} |\theta - \theta'|^{-\beta} |G_{\theta}(y) - G_{\theta'}(y)| \leq c U(y)$

(DRI3) there exists a constant $c > 0$ such that, for all $(\theta, \theta') \in \mathbb{K} \times \mathbb{K}$,

1. $\|P_{\theta}g - P_{\theta'}g\|_U \leq c \|g\|_U |\theta - \theta'|^{\beta}$, for all $g \in \mathcal{L}_U$
2. $\|P_{\theta}g - P_{\theta'}g\|_{U^p} \leq c \|g\|_{U^p} |\theta - \theta'|^{\beta}$, for all $g \in \mathcal{L}_{U^p}$

5.4.1 Assumptions for uniform ergodicity

Uniform ergodicity is achieved on compact spaces under mild conditions on the kernels $\mathcal{T}_k$ (see Theorem 16.2.5 in [21]) and guarantees existence and regularity of solutions to the Poisson equation (see Chapter 17 in [21]). The uniform ergodicity allows us to immediately fulfill Assumptions (A2) and (A3) in [2]. In this section, we argue that if $\mathcal{T}_k$ is weak Feller for each $k \in [K]$, and has $\rho_k(x)$ as its unique stationary density then the chain $P_\theta = P_\theta^x \circ P_\theta^k$ is uniformly ergodic with $\pi_k \rho_k(x)$ as its unique stationary density. In the statement of Theorem 5.1 we simply assume the uniform ergodicity.

It is well known that the simulated tempering sampler $P_\theta = P_\theta^x \circ P_\theta^k$ produces a Markov process which is ψ -irreducible and aperiodic, so we only sketch a proof below.

Lemma 5.7 *For each fixed $\theta \in \Theta$, the chain defined by P_θ is ψ -irreducible and aperiodic.*

Proof We note that $\varphi = p_{\pi,F}(x,k)dx C(dk)$ defines a positive measure on $S \times [K]$, and we assume that the distributions are chosen such that $S \times [K]$ is φ -communicating, hence the chain is φ -irreducible in the sense of Meyn and Tweedie [21] (see Proposition 4.2.2). It is also aperiodic, since the probability of staying at any $k \in [K]$ is nonzero and the dynamics in x are assumed aperiodic. Thus the chain is ψ -irreducible and aperiodic. $\square$

A weak Feller chain is one for which $x \rightarrow P(x, O)$ is lower semicontinuous for any open set O , where P denotes the transition kernel of the chain (see p. 131 of [21]). An alternative definition of weak Feller is to say that the transition kernel maps bounded continuous functions to bounded continuous functions. If we assume that for each $k \in [K]$, $\mathcal{T}_k$ is weak Feller, then for any fixed $\pi \in \mathcal{P}^e(K)$, the composed kernel $P_\theta(x', k'|x, k) = \mathcal{T}_{k'}(x'|x) R_\theta(k'|x)$ maps bounded continuous functions to bounded continuous functions, where

$$R_\theta(k|x) = \frac{\pi_k(\theta) e^{F_k - H_k(x)}}{\sum_{\ell \in [K]} \pi_\ell(\theta) e^{F_\ell - H_\ell(x)}}. \quad (5.71)$$

Thus the composed kernel is also weak Feller.

We use the following results.

Lemma 5.8 *(Theorem 6.2.9 in [21]) If a ψ -irreducible Markov chain with transition kernel $P(x, \cdot)$, $x \in X$ is weak Feller and if $\text{supp}(\psi)$ has nonempty interior, then the chain is a T-chain.*

Our chain is therefore a T-chain.

Proposition 5.4 *(Theorem 6.2.5, (ii) in [21]) If $\{X^n\}$ is a ψ -irreducible T-chain then every compact set is petite.*

Note that for aperiodic and irreducible chains, as is the case here, every petite set is small (see [21], Theorem 5.5.7); we will use this below. Finally, note that on a compact space, a ψ -irreducible and aperiodic T-chain is uniformly ergodic (as mentioned above; see Theorem 16.2.5 in [21]). It follows that the kernel is uniformly ergodic, with unique stationary density $\pi_k(\theta) \rho_k(x)$.

5.4.2 Verification of (DRI)

The results in [2] also provide simple verifiable conditions under which their Assumptions (A2) and (A3) hold. In this section, we verify these alternative conditions.

We take $\beta = 1$ and any $p \geq 2$ for which Assumption (A4) is satisfied. We also take $U(y) = 1$. Let the compact set $\mathbb{K} \subset \Theta$ be given. Below, we verify the drift conditions.

(DRI1) As in [7], we take any $0 < \lambda < 1$, set $b = 1 - \lambda < 1$. Since the chain is an aperiodic T-chain, every compact set is small. Since $C = \mathcal{S} \times [K]$ is compact, C is a small set, and

$$\sup_{\theta \in \mathbb{K}} (P_\theta U^p)(y) \leq 1 \equiv \lambda U^p(y) + b 1_C(y). \quad (5.72)$$

With $\kappa > 1$, we also have

$$\sup_{\theta \in \mathbb{K}} (P_\theta U^p)(y) \leq \kappa \equiv \kappa U^p(y). \quad (5.73)$$

Finally, recall that $P_\theta = P_\theta^x \circ P_\theta^k$, and moreover that $P_\theta^x((x', k') | (x, k)) = T_k(x' | x) \delta(k' - k)$ is independent of θ . For each $k \in [K]$, $\mathcal{S}$ is a small set for T_k , hence there exists an integer $s_k \geq 1$, a $\delta_k \in (0, 1)$ and a probability measure $q_k(dx)$ such that

$$T_k^{s_k}(A | x) \geq \delta_k q_k(A) \quad (5.74)$$

for all Borel-measurable $A \subset \mathcal{S}$. Since $[K]$ is finite, one can find uniform $\delta > 0$ and $s \in \mathbb{N}$ and probability measure q such that

$$T_k^s(A | x) \geq \delta q(A) \quad (5.75)$$

for all $k \in [K]$, $x \in \mathcal{S}$ and Borel subsets $A \subset \mathcal{S}$. Moreover, owing to the finiteness of $[K]$ and the ergodicity of the simulated tempering sampler, it is straightforward to show that there is a probability measure $r(k)C(dk)$ such that

$$P_\theta^k((x', B) | (x, k)) \geq \varepsilon \delta(x' - x) r(B) \quad (5.76)$$

for all $(x, k) \in \mathcal{S} \times [K]$. It follows that there is a uniform $\delta' > 0$, integer $s' \geq 1$ and probability measure $q' \in \mathcal{P}(\mathcal{S} \times [K])$ such that for any Borel $A \times B \subset \mathcal{S} \times [K]$,

$$\inf_{\theta \in \mathbb{K}} P_\theta^{s'}(y, A) \geq \delta' q(A), \quad (5.77)$$

so the minorization condition is satisfied.

(DRI2) From (2.2), (2.3), (2.4) with $\theta = (Z, \xi, o)$ and $y = (x, \ell)$, the functions $G(y; \theta) = (G^Z(y; \theta), G^\xi(y; \theta), G^o(y; \theta))$ are defined by

$$G_k^Z(y; \theta) = \frac{e^{-H_k(x)}}{\sum_{\ell \in [K]} \pi_\ell(\theta) e^{-H_\ell(x)} / Z_\ell} - Z_k \quad (5.78)$$

$$G_{km}^\xi(y; \theta) = \frac{e^{-H_k(x)}}{\sum_{\ell \in [K]} \pi_\ell(\theta) e^{-H_\ell(x)} / Z_\ell} \Psi_m(x) - \xi_{km} \quad (5.79)$$

and where $G^a(y; \theta)$ is an arbitrary continuously differentiable function in (y, θ) . Note that each component is continuous in (y, θ) . Moreover, $\Omega = \mathcal{S} \times [K]$ is compact, and the drift conditions allow us to restrict to a supremum over $\theta \in \mathbb{K}$ where $\mathbb{K}$ is a compact set. Since $(y, \theta) \rightarrow G(y; \theta)$ is continuous, it achieves its supremum on the compact set $\Omega \times \mathbb{K}$. Thus part 1. of condition (DRI2) is satisfied with $U(y) = 1$.

To see that

$$\sup_{(\theta, \theta') \in \mathbb{K} \times \mathbb{K}} \frac{|G(y, \theta) - G(y, \theta')|}{|\theta - \theta'|} \leq C, \quad (5.80)$$

note that $G^Z(y; \theta)$, $G^\xi(y, \theta)$ and $G^o(y, \theta)$ are all continuously differentiable in θ for each fixed $y = (x, k)$, so (5.80) follows from the mean value theorem. Thus part 2. of (DRI2) is satisfied.

(DRI3) To establish the Lipschitz-style property of $P_\theta = P_\theta^x \circ P_\theta^k$ with respect to θ , it suffices to establish it separately for both kernels, i.e. with $P^1 = P^x$ and $P^2 = P^k$, we want

$$\|P_\theta^i h\|_U \leq C_i \|h\|_U, \quad \|P_\theta^i h - P_{\theta'}^i h\|_U \leq C_i \|h\|_U |\theta - \theta'|^\beta. \quad (5.81)$$

Indeed, by the triangle inequality,

$$\|P_\theta h - P_{\theta'} h\|_U \leq \|P_\theta^2(P_\theta^1 h - P_{\theta'}^1 h)\|_U + \|P_\theta^2 P_{\theta'}^1 h - P_{\theta'}^2 P_{\theta'}^1 h\|_U \quad (5.82)$$

$$\leq C_2 \|P_\theta^1 h - P_{\theta'}^1 h\|_U + C_2 \|P_{\theta'}^1 h\|_U |\theta - \theta'|^\beta \quad (5.83)$$

$$\leq 2C_1 C_2 |\theta - \theta'|^\beta \|h\|_U. \quad (5.84)$$

For the first kernel, recall from Section 2.1 of the main text that $P_\theta^x(x', k' | x, k) = \mathcal{T}_k^{(n)}(x' | x) dx'$, which does not depend on θ . Thus (5.81) holds trivially for $i = 1$ and both U and U^P . For the second kernel, note that for any $g \in \mathcal{L}_U$,

$$P_\theta^k g(x, k) - P_{\theta'}^k g(x, k) \quad (5.85)$$

$$\sum_{k' \in [K]} \int_{\mathcal{S}} g(x', k') (P_\theta^k(x', k' | x, k) - P_{\theta'}^k(x', k' | x, k)) dx' \quad (5.86)$$

$$\sum_{k' \in [K]} \int_{\mathcal{S}} g(x', k') \delta(x - x') \left(\frac{\pi_{k'}(\theta) e^{-H_{k'}(x')/Z_{k'}}}{\sum_{\ell \in [K]} \pi_\ell(\theta) e^{-H_\ell(x')/Z_\ell}} - \frac{\pi_{k'}(\theta') e^{-H_{k'}(x')/Z_{k'}}}{\sum_{\ell \in [K]} \pi_\ell(\theta') e^{-H_\ell(x')/Z_\ell}} \right) dx' \quad (5.87)$$

$$= \sum_{k' \in [K]} g(x, k') \left(\frac{\pi_{k'}(\theta) e^{-H_{k'}(x)/Z_{k'}}}{\sum_{\ell \in [K]} \pi_\ell(\theta) e^{-H_\ell(x)/Z_\ell}} - \frac{\pi_{k'}(\theta') e^{-H_{k'}(x)/Z_{k'}}}{\sum_{\ell \in [K]} \pi_\ell(\theta') e^{-H_\ell(x)/Z_\ell}} \right) \quad (5.88)$$

The term in parentheses above is continuously differentiable w.r.t. θ , and by the mean value theorem its absolute value can be bounded above $c \|\theta - \theta'\|$, where $c > 0$ is a constant. It follows that

$$|P_\theta^k g(x, k) - P_{\theta'}^k g(x, k)| \leq cK \|g\|_U \|\theta - \theta'\|. \quad (5.89)$$

Since $U \equiv 1$ this demonstrates conditions 1. and 2. of (DRI3) and concludes the verification of assumptions.

5.5 Statement of theorem

Having verified all the assumptions necessary to apply Theorem 5.5 in [2], we can now state our convergence theorem. As remarked at the beginning of Section 5, we have restricted Θ to be a (arbitrarily large) compact set, and in Section 5.2 we have shown that θ^* is attracting on all of Θ . We do not use the full strength of the result in [2], which allows the iterates θ' to escape arbitrarily large compact sets. Instead, we state the result in a way that assumes boundedness of the iterates and deduces convergence (see, for instance, [6], Corollary 16 in Section 1.8, or [15]). For assumptions on the transition kernel $\mathcal{T}_k$ that lead to uniform ergodicity of P_θ , see 5.4.1.

Theorem 5.1 *Suppose $P_\theta = P_\theta^x \circ P_\theta^k$ is a uniformly ergodic transition kernel with unique stationary density $\pi_k(\theta) e^{F_k - H_k(x)}$ for each $\theta \in \Theta$. Let $\pi : \Theta \rightarrow \mathcal{P}^e(K)$, $\gamma : \mathbb{R}^{KM} \rightarrow \mathcal{P}^e(K)$ and $G_j^a : \Omega \times \Theta \rightarrow \mathbb{R}$ be continuously differentiable deterministic functions, and assume π and G^a are invariant to additive shifts of F , as in (5.8) and (5.13). Furthermore, assume that a^* is a globally asymptotically stable equilibrium of the system*

(5.14). If the iterates (F^t, μ^t, a^t) as defined by (5.9), (5.10) and (5.11) are almost surely bounded, then, with probability 1, $\theta^t = (F^t, \mu^t, a^t)$ satisfies $\lim_{t \rightarrow \infty} \theta^t \rightarrow \theta^* = (F^*, \mu^*, a^*)$. Moreover, the corresponding process $(X^t, \mathcal{K}^t)$ with transition kernel $P_{\theta^t} = P_{\theta^t}^k \circ P_{\theta^t}^x$ converges in distribution to $\pi_k(\theta^*) \rho_k(x) dx C(dk)$, where $\theta^* = (F^*, \mu^*, a^*)$.

Proof Proposition 6.1 in [2] shows that the drift conditions (DRI) verified above ensure that assumptions (A2) and (A3) in the same paper are satisfied, along with assumption (A4) in Section 5.3. Furthermore, for π taking values in $\mathcal{P}^e(K)$, we have verified in Section 5.2, Proposition 5.3 condition (A1) is satisfied. We thus fulfill the conditions of Theorem 5.5 in [2] on the compact sets $\mathcal{X} \times [K]$ and Θ , which establishes that $\theta^t \rightarrow \theta^*$ for a suitably re-projected algorithm, which is distinct from our algorithm only in that it may modify the process once it exits Θ . By assumption, $\theta^t \in \Theta$ almost surely, so $\theta \rightarrow \theta^*$ w.p.1. $\square$

Theorem 5.1 can be applied to the TSS algorithm. Let π^{TSS} be the ε_π -regularized visit controlled rung distribution,

$$\pi^{\text{TSS}}(F, \mu, o) = (1 - \varepsilon_\pi) \pi_k(\mu, o) + \varepsilon_\pi \gamma_k(\mu), \quad (5.90)$$

where

$$\pi_k(F, \mu, o) = \frac{\gamma_k(\mu) o_k^{-\eta}}{\sum_{\ell \in [K]} \gamma_\ell(\mu) o_\ell^{-\eta}}. \quad (5.91)$$

From Lemma 5.6, when $(F, \mu) = (F^*, \mu^*)$, the system (5.15) with $\pi = \pi^{\text{TSS}}$ has $\mathbf{1}$ as its global asymptotically stable equilibrium, and π^{TSS} is continuously differentiable so long as γ is. Moreover, ε_π can be chosen such that $\pi^{\text{TSS}}(F, \mu, o) \in \mathcal{P}^e(K)$ for all $(F, \mu, o) \in \Theta$. Propositions 5.1, 5.2 and 5.3 establish the existence of a global Lyapunov function, while Sections 5.3 and 5.4 verify Assumption (A4) and the drift conditions. Under the conditions of Theorem 5.1, the TSS iterates $\theta^t = (F^t, \mu^t, o^t)$ defined through (5.9), (5.10), (5.11) converge to $(F^*, \mu^*, \mathbf{1})$.

Because $\lim_{t \rightarrow \infty} o_k^t = 1$ with probability 1 for each $k \in [K]$, we have $\lim_{t \rightarrow \infty} \pi(\theta^t) = \lim_{t \rightarrow \infty} \gamma(\mu^t) = \gamma(\mu^*)$. The transition kernel P_{θ^*} is assumed to have $\pi_k \rho_k(x) dx C(dk)$ as its unique invariant density, and so $(X^t, \mathcal{K}^t) \Rightarrow \gamma_k \rho_k(x) dx C(dk)$.

Part II

Implementation

6 Windows, history forgetting, and multireplica

6.1 The windowed parametric invariant distribution

In this section we introduce the windowing system, which decomposes the global sampling of Λ into a collection of local sampling problems. This section parallels Section 2 of the paper, in that it introduces a family of extended probability densities depending on a pair of parameters π and F , and for each such pair defines ergodic sampling dynamics for which the corresponding density is invariant. These local dynamics entail local estimates of $\theta = (F, \mu, o)$, which will then have to be stitched together to provide global estimates across all of Λ ; in Section 6.2, we provide a procedure to determine global estimates based on the collection of window-local estimates.

We restate some of the definitions from the main paper, Section 3. The windowing system decomposes the parameter space Λ into a collection $\{W_j; 1 \leq j \leq J\}$ of connected overlapping windows $W_j \subset \Lambda$, and

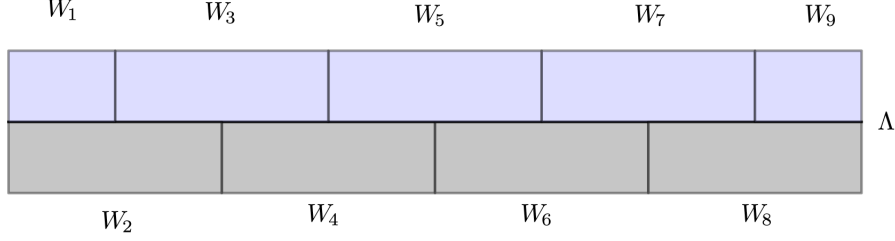


Figure 2: A configuration of nine windows which form a double cover of the space Λ ; two of the windows (W_1 and W_9) have half the window size to stagger the overlap between the windows.

separate TSS iterations are performed within each window. To link the windows together, it introduces window dynamics that allow the sampler to select a different window among those containing λ_k for the current rung k . Instead of sampling a pair $(X, \mathcal{K})$, we will thus now sample a triple $(X, \mathcal{K}, \mathcal{J})$, with the specification that the stationary density conditioned on $\mathcal{J} = j$ is equal to the original stationary density $p_{\pi, F}(x, k)$ defined in Section 2.1 of the main paper. We provide a more detailed description for the sake of concreteness, but note that other variants are possible. The important attributes that must be respected are that the windows cover the entire parameter space Λ and that the window dynamics are positively recurrent.

The windows are structured so that each $\lambda \in \Lambda$ belongs to exactly two windows, with the additional constraint that the symmetric matrix $\mathcal{W}$ with i, j entry equal to 1 if $W_i \cap W_j \neq \emptyset$ and otherwise equal to 0 be irreducible; see Figure 2 for an example.

If we define the set $\text{win}(k) = \{j \in [J] : \lambda_k \in W_j\}$, it follows that the cardinality of $\text{win}(k)$ is 2 for each rung $k \in [K]$. For each $j \in [J]$, quantities such as $F_{j,\cdot}$ and $\pi_{j,\cdot}$ (which are specified with a bullet in place of the second subscript) denote the sets of per-window quantities $\{Z_{j,k}; k \in [K] : \lambda_k \in W_j\}$ and $\{\pi_{j,k}; k \in [K]\}$, respectively. Quantities with no subscript denote the totality of components over all windows: $Z \equiv \{Z_{j,k}; j \in [J], k : \lambda_k \in W_j\}$, for example. Similar notation will be used for the per-window estimates. All per-window rung densities, such as $\pi_{j,\cdot}$, are defined so that $\pi_{j,k} \geq \varepsilon > 0$ for all $\lambda_k \in W_j$ and 0 for $\lambda_k \notin W_j$. The densities are also normalized, and thus $\sum_{k \in [K]} \pi_{j,k} = \sum_{k : \lambda_k \in W_j} \pi_{j,k} = 1$.

Next, we define the window process $\mathcal{J}^t$, which at any fixed time $t \geq 0$ is equal to one of the two possible indices in $\text{win}(\mathcal{K}^{t-1})$. The window $W_{\mathcal{J}^t}$ will be called the *active window*. The dynamics of $\mathcal{J}^t$ are determined by the transition density

$$\bar{P}_{\pi, F}^j(x', k', j' | x, k, j) = \delta(x - x') \delta(k - k') (1 - 1_{\{j\}}(j')) 1_{\text{win}(k)}(j'), \quad (6.1)$$

which, given k , selects the other window in $\text{win}(k)$ with probability 1. (If during the early stages of the simulation the estimates $\pi_{j,\cdot}, F_{j,\cdot}$ are not yet defined for some window W_j , neither the window nor rung is updated – that is, neither of the transition kernels (6.1) nor (6.2) are used – and instead (k', j') is set to (k, j) ; the rest of the algorithm continues normally, and $\pi_{j,\cdot}$ and $F_{j,\cdot}$ become defined in the following cycle.) This antithetic selection can of course be generalized to a random selection in which the other window is selected with probability $p \in (0, 1]$, but the need for efficient rung sampling (as discussed in Section 2 of the main paper) motivates us to take the most aggressive choice $p = 1$. The transition density for the triple $(X, \mathcal{K}, \mathcal{J})$ is defined by $\bar{P}_{\pi, F} = \bar{P}_{\pi, F}^k \circ \bar{P}_{\pi, F}^j \circ \bar{P}_{\pi, F}^x$, where

$$\bar{P}_{\pi, F}^k(x', k', j' | x, k, j) \propto \delta(j' - j) 1_{W_j}(\lambda_{k'}) P_{\pi_{j,\cdot}, F_{j,\cdot}}^k(x', k' | x, k) \quad (6.2)$$

has been restricted to only propose rungs within the active window, and

$$\bar{P}_{\pi,F}^x(x',k',j'|x,k,j) = \delta(j' - j) P_{\pi,F}^x(x',k'|x,k) \quad (6.3)$$

is unchanged. These dynamics and the irreducibility condition on the symmetric matrix $\mathcal{W}$ ensure each window will be visited asymptotically, and together with (6.1) determine the parametric invariant density to take the form

$$\bar{p}_{\pi,F}(x,k,j) = \bar{p}_{\pi,F}(j|x,k) \bar{p}_{\pi,F}(x,k) = \frac{1}{2} 1_{W_j}(\lambda_k) \bar{p}_{\pi,F}(x,k), \quad (6.4)$$

where $\bar{p}_{\pi,F}(j|x,k)$ denotes a conditional density and $\bar{p}_{\pi,F}(x,k)$ denotes a marginal density. Equation (6.4) states that at stationarity each rung k is assigned one of the two windows containing λ_k , with equal probability.

To complete the description of the parametric invariant density, we derive an explicit dependence of $\bar{p}_{\pi,F}(x,k)$ on π, F . To this end, set $\bar{p}_j \equiv \bar{p}_{\pi,F}(j)$ (the marginal density of $\mathcal{J}$), and let $\bar{\mathbb{P}}_{\pi,F}$ denote the distribution associated to the density $\bar{p}_{\pi,F}$ (with respect to $\mathrm{d}x \, C(\mathrm{d}j) C(\mathrm{d}k)$; see Section 1.1). The marginal density $\bar{p}_{\pi,F}(x,k)$ is relatable to the complementary marginal $\bar{p}_j$ by way of the conditional probability decomposition

$$\bar{p}_{\pi,F}(x,k) = \sum_{j \in [J]} \bar{p}_j \bar{p}_{\pi,F}(x,k|j) = \sum_{j=1}^J \bar{p}_j \frac{\pi_{j;k} e^{F_{j;k} - F_k^*}}{\sum_{\ell: \lambda_\ell \in W_j} \pi_{j;\ell} e^{F_{j;\ell} - F_\ell^*}} \rho_k(x). \quad (6.5)$$

Multiplying (6.5) by $1_{W_i}(\lambda_k)$, integrating over $x \in \mathcal{S}$, and summing over k implies, for $(\mathcal{X}, \mathcal{K}, \mathcal{J})$ distributed according to $\bar{\mathbb{P}}_{\pi,F}$ and each $i \in [J]$,

$$\bar{\mathbb{P}}_{\pi,F}(\lambda_{\mathcal{K}} \in W_i) = \sum_{j=1}^J \bar{p}_j \bar{\mathbb{P}}_{\pi,F}(\lambda_{\mathcal{K}} \in W_i | \mathcal{J} = j) = \sum_{j=1}^J \bar{p}_j \frac{\sum_{k: \lambda_k \in W_i \cap W_j} \pi_{j;k} e^{F_{j;k} - F_k^*}}{\sum_{k: \lambda_k \in W_j} \pi_{j;k} e^{F_{j;k} - F_k^*}}. \quad (6.6)$$

A different decomposition produces, for each $i \in [J]$,

$$\bar{p}_i = \bar{\mathbb{P}}_{\pi,F}(\mathcal{J} = i | \lambda_{\mathcal{K}} \in W_i) P(\lambda_{\mathcal{K}} \in W_i) + \bar{\mathbb{P}}_{\pi,F}(\mathcal{J} = i | \lambda_{\mathcal{K}} \notin W_i) \bar{\mathbb{P}}_{\pi,F}(\lambda_{\mathcal{K}} \notin W_i) \quad (6.7)$$

$$= \frac{1}{2} \bar{\mathbb{P}}_{\pi,F}(\lambda_{\mathcal{K}} \in W_i) + 0 \cdot \bar{\mathbb{P}}_{\pi,F}(\lambda_{\mathcal{K}} \notin W_i), \quad (6.8)$$

which together with (6.6) leads to the equation

$$\frac{1}{2} \sum_{j=1}^J \frac{\sum_{k: \lambda_k \in W_i \cap W_j} \pi_{j;k} e^{F_{j;k} - F_k^*}}{\sum_{k: \lambda_k \in W_j} \pi_{j;k} e^{F_{j;k} - F_k^*}} \bar{p}_j = \bar{p}_i, \quad \forall 1 \leq i \leq J. \quad (6.9)$$

Note that, for each $i \in [J]$, the family of disjoint subsets $\{W_i \cap W_j; 1 \leq j \leq J, j \neq i\}$ form a partition of W_i . It follows then that (6.9) asks for the invariant right-eigenvector of a left-stochastic matrix, which determines $\bar{p}_j$ (using $\sum_{j \in [J]} \bar{p}_j = 1$) and completes the description of the invariant distribution $\bar{p}_{\pi,F}(x,k,j) \mathrm{d}x \, C(\mathrm{d}j) C(\mathrm{d}k)$. This distribution is analogous to the invariant distribution $p_{\pi,F}(x,k) \mathrm{d}x C(\mathrm{d}k)$ defined in Section 2.1 of the paper, but now with windows.

6.2 Obtaining global estimates from local, per-window estimates

In this section we define the dependence of π and Z on the collection of per-window estimates $\theta \equiv \{\theta_{j,k}; j \in [J], k : \lambda_k \in W_j\}$, which requires defining the rung weights π_k , the partition functions Z_k , and the visit control partition functions Z_k° globally from the local estimates θ (defined for $j \in [J], k : \lambda_k \in W_j$). There are two results in this section: The first is the formulation in Section 6.2.1 of a convex optimization problem that allows us to produce, for a given θ , a set of global visit control free energies $\{F_k^\circ, k \in [K]\}$ that are used for sampling, analogously to those defined in the case of a single window in Section 2.2.1 of the main paper. These free energies are useful for driving the sampling, but are too noisy for the purpose of providing a final estimate. To address this issue, in Section 6.2.2 we also formulate a second, simpler optimization problem that provides a way to unite the free energy estimates across windows with a lower noise (albeit in a way only applicable near convergence) for the purpose of reporting a set of global free energies $\{F_k^{\text{TSS}}, k \in [K]\}$ to the user, which we call the *reported* free energies. The per-window recursions that define $\theta_{j,k}^{t+1}$ for $j \in [J]$ and $k : \lambda_k \in W_j$ based on $(\mathcal{X}^t, \mathcal{K}^t, \mathcal{J}^t)$ and θ^t will be given in Section 2.3.

We first describe an element common to both the visit control and reported free energies, which is the asymptotic rung density $\gamma \equiv \{\gamma_{j,k}; j \in [J], k : \lambda_k \in W_j\}$. Although the densities $\gamma_{j,\cdot}(\mu)$ could be defined to vary with j in an arbitrary fashion, in practice we take all these densities to be conditional densities of the common global density $\gamma(\mu)$ of components $\gamma_k(\mu)$ considered in Section 2.2.2 of the main paper. In what follows we only need the un-normalized functional dependence of $\gamma_k(\mu)$ on μ (e.g., of the kind expressed by the right-hand side of (17)), and its locality property that $\gamma_k(\mu)$ depends only on row k of the original $K \times M$ matrix μ . This locality implies that, for each $k \in [K]$, the function γ_k initially defined on matrices of size $K \times M$ can be naturally restricted to a function defined on matrices of size $1 \times M$ (i.e., rows), and hence we write $\gamma_k(\mu_{j,\cdot})$ to mean this restricted function evaluated on the k^{th} row of $\mu_{j,\cdot}$. This property allows for a simple definition of $\gamma_{j,k}$ as a function of the J matrices $\{\mu_{j,\cdot}, j \in [J]\}$ of sizes $K_j \times M$, where K_j is the number of rungs in W_j . We take

$$\gamma_{j,k}(\mu) = \frac{(1 - \varepsilon_\gamma)\gamma_k(\mu_{j,\cdot}) + \varepsilon_\gamma \mathcal{M}(\mu)}{\sum_{\ell: \lambda_\ell \in W_j} [(1 - \varepsilon_\gamma)\gamma_\ell(\mu_{j,\cdot}) + \varepsilon_\gamma \mathcal{M}(\mu)]} 1_{W_j}(\lambda_k), \quad \forall 1 \leq k \leq K, \quad (6.10)$$

where $\mathcal{M}(\mu) = \max_{k \in [K], j \in \text{win}(k)} \gamma_k(\mu_{j,\cdot})$ and $\varepsilon_\gamma \in (0, 1]$ is a positive regularization parameter that is small relative to 1. We note that the ε_γ regularization implies that $\gamma(\mu) \in \mathcal{P}^e(K)$ for all $\mu \in \mathbb{R}^{K \times M}$ and $\varepsilon_\gamma \geq K\varepsilon$. The parameter θ will thus be defined for each $j \in [J]$ and $k : \lambda_k \in W_j$ by a tuple of the form $\theta_{j,k} = (F_{j,k}, \mu_{j,k,1}, \dots, \mu_{j,k,M}, o_{j,k})$.

6.2.1 Global visit control free energies

To guide the reader in the definition of the global visit control free energies as a function of the per-window estimates $\theta = \{\theta_{j,k} : j \in [J], k : \lambda_k \in W_j\}$, we first sketch the general idea. In the single-window case we identified the visit control free energies explicitly in terms of the tilts, eliminating the need to refer to F° ; in the case of multiple windows, however, we will need F° as an intermediate in order to define visit control globally. The key step is to invoke the two-timescale separation described in Section 1.2 and impose the steady-state condition, thereby relating the tilts o and density γ to the density π and visit control F° , the latter two being used for sampling. This relationship allows us to obtain the marginal density for $\mathcal{J}$, and in turn the marginal density for $\mathcal{K}$. Finally, upon specifying the dependence of $\pi_{j,k}$ on θ through our particular choice of visit control (i.e., π_k^{TSS} as defined in Section 2.2.1 in the main paper), the steady state relation can be recast as a convex optimization problem, which leads to our definition of visit control free energies

$\{F_k^\circ, k \in [K]\}$. Notably, the complexity of the optimization problem scales with the number of windows, not the number of rungs (the latter typically being greater), which is important for efficiency.

To impose the steady-state condition, we begin by assuming that the tuple $(\mathcal{X}, \mathcal{K}, \mathcal{J})$ is drawn from the sampling density $\bar{p}_{\pi, F}(x, k, j)$ defined in (6.4) for fixed π, F , and that the tilts $o_{j\cdot}$ are at their steady state for each $j \in [J]$. We recall that this is exactly how visit control was defined in Section 2.2.1 of the paper. Analogously to the single-window case, the steady state condition applied to the marginal over all windows gives the relation

$$\sum_{j=1}^J p_j \frac{\gamma_{j,k}(\mu) o_{j,k}}{\sum_{\ell: \lambda_\ell \in W_j} \gamma_{j,\ell}(\mu) o_{j,\ell}} = \sum_{j=1}^J p_j \frac{\pi_{j,k} e^{F_{j,k} - F_k^\circ}}{\sum_{\ell: \lambda_\ell \in W_j} \pi_{j,\ell} e^{F_{j,\ell} - F_\ell^\circ}}, \quad \forall 1 \leq k \leq K. \quad (6.11)$$

with to-be-determined marginal rung probabilities $p_j, j \in [J]$ and global visit control free energies $F_k^\circ, k \in [K]$. As in the single-window case, when o is at its steady state for any fixed π, F , the relation (6.11) correctly implies $F_k^\circ = F_k^*$; otherwise, (6.11) defines the visit control free energy F_k° up to an additive constant. The equation itself restates the conditional probability decomposition $\bar{\mathbb{P}}_{\pi, F}(\mathcal{K}=k) = \sum_{j \in [J]} \bar{\mathbb{P}}_{\pi, F}(\mathcal{K}=k | \mathcal{J}=j) \bar{\mathbb{P}}_{\pi, F}(\mathcal{J}=j)$. Because the equality expressed by the equation holds at all times, we note (by multiplying by $1_{W_i}(\lambda_k)$ and summing over all $k \in [K]$, as in (6.5)-(6.9)) that the probabilities $\{p_j; 1 \leq j \leq J\}$ are the solution to the stochastic eigenvector problem

$$\frac{1}{2} \sum_{j \in [J]} \frac{\sum_{k: \lambda_k \in W_i \cap W_j} \gamma_{j,k}(\mu) o_{j,k}}{\sum_{k: \lambda_k \in W_j} \gamma_{j,k}(\mu) o_{j,k}} p_j = p_i, \quad \forall 1 \leq i \leq J, \quad (6.12)$$

which can be solved by, for example, the QR method (Davis 2011). When o is at its steady state, $F_k^\circ - F_k^*$ is constant over all $k \in [K]$, and in this case the $\bar{p}_j$ determined from (6.9) are equal to the p_j determined from (6.12).

We now use the specific form of $\pi_{j,k}^{\text{TSS}}$ as given by (13) in the main paper (applied separately to each individual window) to relate the left-hand side of (6.11) to its right-hand side, as follows

$$\pi_{j,k}^{\text{TSS}}(\theta) = \frac{\gamma_{j,k}(\mu) e^{\frac{\eta}{\eta+1}(F_k^\circ - F_{j,k})}}{\sum_{\ell: \lambda_\ell \in W_j} \gamma_{j,\ell}(\mu) e^{\frac{\eta}{\eta+1}(F_\ell^\circ - F_{j,\ell})}}, \quad \forall j \in [J], k \in \{\ell: \lambda_\ell \in W_j\}, \quad (6.13)$$

for some $\eta > 0$ and unknown visit control free energies $F_k^\circ, k \in [K]$. Combining everything, the relation (6.11) gives, for each $k \in [K]$ and unknowns $\{F_k^\circ, k \in [K]\}$

$$q_k \equiv \sum_{j \in \text{win}(k)} p_j \frac{\gamma_{j,k}(\mu) o_{j,k}}{\sum_{\ell: \lambda_\ell \in W_j} \gamma_{j,\ell}(\mu) o_{j,\ell}} = \sum_{j=1}^J p_j \frac{\gamma_{j,k}(\mu) e^{\frac{(F_{j,k} - F_k^\circ)}{(\eta+1)}}}{\sum_{\ell: \lambda_\ell \in W_j} \gamma_{j,\ell}(\mu) e^{\frac{(F_{j,\ell} - F_\ell^\circ)}{(\eta+1)}}}, \quad \forall 1 \leq k \leq K. \quad (6.14)$$

The first equality in (6.14) defines q_k as the left-hand side of (6.11), whereas the second equality defines a system of equations with unknowns F_k° .

As for the case of a single window, with $\pi_{j,k}^{\text{TSS}}(\theta)$ and $\gamma_{j,k}(\mu)$ on the rhs of the following equation given by (6.13) and (6.10), respectively, we re-define $\pi_{j,k}^{\text{TSS}}(\theta)$ through the ε_π -regularization

$$\pi_{j,k}^{\text{TSS}}(\theta) := (1 - \varepsilon_\pi) \pi_{j,k}^{\text{TSS}}(\theta) + \varepsilon_\pi \gamma_{j,k}(\mu), \quad \forall k: \lambda_k \in W_j, \quad (6.15)$$

where $0 < \varepsilon_\pi \leq 1$. We note that the ε_π -regularization leaves $\gamma_{j\cdot}(\mu)$ unchanged, if applied to it.

The solution of (6.14) would be readily provided by

$$F_k^\circ = (\eta + 1) \log \left(\frac{1}{q_k} \sum_{j \in \text{win}(k)} p_j \gamma_{j;k}(\mu) e^{\frac{(F_{j;k} - f_j)}{(\eta + 1)}} \right), \quad \forall 1 \leq k \leq K, \quad (6.16)$$

if only we knew the window free-energy offsets

$$f_j = (\eta + 1) \log \left(\sum_{k: \lambda_k \in W_j} \gamma_{j;k}(\mu) e^{\frac{(F_{j;k} - F_k^\circ)}{(\eta + 1)}} \right), \quad \forall 1 \leq j \leq J. \quad (6.17)$$

We note that any solution F° to (6.14) also minimizes the convex function

$$\sum_{j=1}^J p_j \log \left(\sum_{k: \lambda_k \in W_j} \gamma_{j;k}(\mu) e^{\frac{(F_{j;k} - F_k^\circ)}{(\eta + 1)}} \right) + \frac{1}{\eta + 1} \sum_{k=1}^K q_k F_k^\circ. \quad (6.18)$$

Except for an additive term equal to the Shannon entropy $-\sum_{k=1}^K q_k \log(q_k)$, the last expression is equal to

$$\frac{1}{\eta + 1} \sum_{j=1}^J p_j f_j + \sum_{k=1}^K q_k \log \left(\sum_{j \in \text{win}(k)} p_j \gamma_{j;k}(\mu) e^{\frac{(F_{j;k} - f_j)}{(\eta + 1)}} \right), \quad (6.19)$$

as follows by substituting (6.16) and (6.17) into (6.18). The minimization is carried out subject to the constraint $\sum_{j=1}^J p_j f_j = 0$, since (6.19) does not change if all offsets f_j are modified by the addition of one same number (details are provided in 7.2.2). Furthermore, only the components with $p_j \neq 0$ are uniquely determined. We set $f_j = 0$ for the others, since their precise values do not affect the algorithm. We note once again that the number of windows J is often much smaller than the number of rungs K , which helps mitigate the cost associated with this global step of the algorithm. Finally, we observe that the visit control free energies are heavily dependent on the tilts, which are estimated by counting events. Counting estimators are notoriously noisy, and thus we recommend that visit control free energies only be used for sampling, and that different estimates (to be defined in the following section) be reported to the user.

6.2.2 Global reported free energies

Windowed TSS evaluates a separate free energy $F_{j;k}^t$ for each window W_j containing λ_k , raising the question of how to combine the different estimates available for the same rung in a single global set of estimates. The visit control mechanism provides one possible answer in the form of the unique set of global free energy estimates $\{F_k^\circ; k \in [K]\}$, but these estimates are rather noisy, since they depend on the many tilts $\{o_{j;k}; j \in [J], k: \lambda_k \in W_j\}$, which are estimated by counting events. We can reduce the noise by assuming the algorithm has already converged, so the tilts have approached their asymptotic value of 1. The stationarity of the ODE for $o_{j;k}$ at the equilibrium value of 1 implies

$$o_{j;k} = \frac{e^{\frac{(F_{j;k} - F_k^\circ)}{(\eta + 1)}}}{\sum_{\ell: \lambda_\ell \in W_j} \gamma_{j;\ell}(\mu) e^{\frac{(F_{j;\ell} - F_\ell^\circ)}{(\eta + 1)}}} = 1, \quad \forall j \in [J], k: \lambda_k \in W_j. \quad (6.20)$$

This many equations cannot be accommodated just by selecting $\{F_k^\circ; k \in [K]\}$, which forces us to select $\eta = \infty$. With the tilts set to unity, (6.12) becomes

$$\sum_{j \in [J]} \left(\sum_{k: \lambda_k \in W_i \cap W_j} \gamma_{j;k}(\mu) \right) p_j = p_i, \quad \forall i \in [J], \quad (6.21)$$

which defines the stochastic eigenvector $(p_1, \dots, p_J)$. This eigenvector can be used to define a global estimate of the asymptotic rung density through the weighted average

$$\gamma_k^{\text{TSS}} \equiv \sum_{j \in \text{win}(k)} p_j \gamma_{j;k}(\mu), \quad \forall k \in [K]. \quad (6.22)$$

We refer to γ_k^{TSS} as the reported rung density. Similarly, by letting $\eta \rightarrow \infty$ in (6.16) with $q_k = \gamma_k^{\text{TSS}}$, we obtain the TSS reported free energies

$$F_k^{\text{TSS}} = \frac{1}{\gamma_k^{\text{TSS}}} \sum_{j \in \text{win}(k)} p_j \gamma_{j;k}(\mu) (F_{j;k} - f_j^{\text{TSS}}), \quad \forall 1 \leq k \leq K. \quad (6.23)$$

The window free energy offsets f_j^{TSS} are deduced by letting $\eta \rightarrow \infty$ in the system of equations obtained by setting the gradient of (6.19) to 0 and retaining the dominant terms. The offsets can be shown to solve the linear system

$$\sum_{j=1}^J (\delta_{ij} - t_{ij}) f_j^{\text{TSS}} = \sum_{k: \lambda_k \in W_i} \gamma_{i;k}(\mu) \left(F_{i;k} - \sum_{j \in \text{win}(k)} \frac{p_j \gamma_{j;k}(\mu)}{\gamma_k^{\text{TSS}}} F_{j;k} \right), \quad (6.24)$$

where δ_{ij} is the Kronecker symbol and t_{ij} are the entries of the right-stochastic matrix

$$t_{ij} = \sum_{k: \lambda_k \in W_i \cap W_j} \gamma_{i;k}(\mu) \frac{p_j \gamma_{j;k}(\mu)}{\gamma_k^{\text{TSS}}}. \quad (6.25)$$

As in the case of the reported free energies, once the offsets f_j^{TSS} are determined, the quantities F_k^{TSS} can be reported to the user by way of (6.23). We note that solution of (6.24) is not uniquely determined, and that one equation must be dropped out and replaced by $\sum_{j \in [J]} p_j f_j^{\text{TSS}} = 0$. Additional details relating to the implementation of (6.24) within the TSS code are available in the Section 7.2.3.

6.3 History forgetting and multireplica

In this section we describe two features of our implementation of TSS inspired by traditional statistical practices, namely burn-in and use of multiple parallel samplers, modified to be compatible/useful for on-the-fly estimation. We adapted burn-in as a *history forgetting mechanism*, which also enables the estimation of asymptotic error bars, by making use of the fact that TSS is formulated as an IIS estimator (as in (1.15)). The same formulation also enables the use of multiple conditionally independent samplers, which we call *multireplica* TSS.

We first describe the history-forgetting mechanism. At any time t , the history of the simulation is divided into two parts: the early history, corresponding to times s earlier than $\lfloor \alpha t \rfloor$ (the largest integer smaller than or equal to αt), and the recent history, corresponding to times s later than $\lfloor \alpha t \rfloor$. Here, $\alpha \in [0, 1)$ is a fixed fraction that determines the proportion of history that is “forgotten.” For the single-window IIS estimator (1.15), we can neglect the early history, and recast the estimator as follows

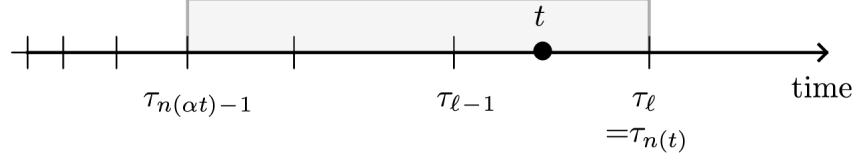


Figure 3: The epoch mechanism. The dot indicates the current time t , while the bars delineate the epochs. At time t , the current epoch is $n(t)$, and $t \in (\tau_{\ell-1}, \tau_\ell]$. The shaded region indicates the recent history of the simulation, and does not include time $\tau_{n(\alpha t)-1}$, but does include $\tau_{n(\alpha t)-1} + 1$.

$$Z_k^t \equiv \frac{1}{t - \lfloor \alpha t \rfloor} \sum_{s=\lfloor \alpha t \rfloor + 1}^t \frac{e^{-H_k(\mathcal{X}^s)}}{\sum_{\ell \in [K]} \gamma_\ell \frac{e^{-H_\ell(\mathcal{X}^s)}}{Z_\ell^{s-1}}}. \quad (6.26)$$

Note that, when $\alpha = 0$, (74) is exactly the IIS estimator (1.15). The influence of the early samples $\mathcal{X}^s$ on Z_k^t is now indirect, and can decay faster than the law t^{-1} otherwise allows. The problem with (74), though, is that it requires storing the entire state history in order to be able to evaluate the neglected part. We circumvent this shortcoming through the introduction of a system of *epochs*, which we define as follows:

Given a base multiplier $\phi > 1$, consider the exponentially increasing sequence of integers defined recursively by $\tau_0 = 0$, $\tau_1 = 1$, and $\tau_{l+1} = \lceil \phi \tau_l \rceil$ for $l \geq 2$. Here, $\lceil \phi \tau_l \rceil$ is the smallest integer greater than or equal to $\phi \tau_l$. For any integer $l \geq 1$, the semi-open interval $(\tau_{l-1}, \tau_l]$ is called the l -th epoch. For any real number $s \geq 0$, we let $n(s)$ be the smallest integer $l \geq 1$ such that $s \leq \tau_l$. The definition of $n(s)$ is such that s lies in the epoch $n(s)$. The current time t marks the beginning of a new epoch if and only if $t = \tau_{n(t)-1} + 1$, and the end of a new epoch if and only if $t = \tau_{n(t)}$. For any $\alpha \in [0, 1)$, the epoch index $n(\alpha t)$ increases by 1 at every cycle t for which $\lceil \alpha t \rceil$ crosses into a new epoch. Such crossing events happen if and only if simultaneously $\alpha(t-1) > 0$ and $n(\alpha t) > n(\alpha(t-1))$. We define the recent history of the simulation to be the sequence of state variables generated at the times $\tau_{n(\alpha t)-1} + 1, \tau_{n(\alpha t)-1} + 2, \dots, t$. See Figure 3 for details.

Having established this notation, we can now modify the stochastic approximation triple for use with the history forgetting mechanism, and simultaneously incorporate the use of multiple parallel samplers. In order to use the parallel samplers, we need to define θ^t using information across all $R \geq 1$ replicas, indexed by $r \in [R] = \{1, \dots, R\}$, which will allow us to define the rung density $\pi^t = \pi(\theta^t)$ across all windows. Each replica has its own state $(X, \mathcal{K}, \mathcal{J})$ and runs in parallel with all other replicas; the only data shared between the replicas are the sets $\pi^t = \{\pi_{j,k}^t, j \in [J]; k : \lambda_k \in W_j\}$ and $F^t = \{F_{j,k}^t, j \in [J]; k : \lambda_k \in W_j\}$. We let $(\mathcal{X}^t(r), \mathcal{K}^t(r), \mathcal{J}^t(r))$ denote the tuple for the r^{th} replica at time $t \geq 0$, which uses the transition kernel

$\bar{P}_{\pi^{t-1}, F^{t-1}}$ (defined above in (52)) conditionally independently from all other replicas given π^{t-1}, F^{t-1} for $t \geq 1$. A first step in defining the free energies with history forgetting is to compute and store the population counters

$$\mathcal{N}_j^{l,t} \equiv \sum_{s=\tau_{l-1}+1}^{\tau_l \wedge t} \sum_{r=1}^R 1_{\{j\}}(\mathcal{J}^s(r)), \quad (6.27)$$

representing the number of visits paid to window W_j during the l -th epoch by the R replicas. The number of visits paid to the window in the recent history of the simulation is

$$\mathcal{N}_j^t \equiv \sum_{l=n(\alpha)}^{n(t)} \mathcal{N}_j^{l,t}. \quad (6.28)$$

Only the epochs l with $n(\alpha) \leq l \leq n(t)$ are needed at the current time. The current epoch is typically incomplete, so the population counters with $l = n(t)$ change with time. During any time t that marks the beginning of a new epoch, new counters $\mathcal{N}_j^{n(t),t}$ are instantiated on the memory store. During any time t for which α crosses into a new epoch, the chronologically earliest counters $\mathcal{N}_j^{l,t}$, the ones with $l = n(\alpha) - 1$, are removed from the memory store. The number of counters committed to the store thus remains bounded during the simulation.

The next step for defining the free energies with history forgetting is to define estimates within each epoch, which we call the epoch free energies. For convenience, we define the importance sampling ratios:

$$\mathcal{R}_{j;k}^t(r) \equiv \frac{e^{-H_k(\mathcal{X}^t(r))}}{\sum_{\ell: \lambda_\ell \in W_j} \pi_{j;\ell}^{t-1} e^{F_{j;\ell}^{t-1} - H_\ell(\mathcal{X}^t(r))}} 1_{\{j\}}(\mathcal{J}^t(r)), \quad (6.29)$$

with the denominator set to 1 if $\mathcal{N}_j^t = 0$. For all $n(\alpha) \leq l \leq n(t)$, $k \in [K]$ and $j \in \text{win}(k)$, we then define the epoch free energies $F_{j;k}^{l,t}$ by

$$e^{-F_{j;k}^{l,t}} \equiv \frac{1}{\mathcal{N}_j^{l,t}} \sum_{s=\tau_{l-1}+1}^{\tau_l \wedge t} \sum_{r=1}^R \mathcal{R}_{j;k}^s(r), \quad (6.30)$$

where we use the convention that $F_{j;k}^{l,t} = 0$ if $\mathcal{N}_j^{l,t} = 0$. Despite the notation, at time t all epoch free energies remain unchanged, except for those with $l = n(t)$, which can be computed as follows: for all $j \in \cup_{r=1}^R \{\mathcal{J}^t(r)\}$ and k such that $\lambda_k \in W_j$

$$F_{j;k}^{l,t} = F_{j;k}^{l,t-1} - \ln\left(1 + \frac{1}{\mathcal{N}_j^{l,t}} [\mathcal{N}_j^{l,t-1} - \mathcal{N}_j^{l,t} + e^{F_{j;k}^{l,t-1}} \sum_{r=1}^R \mathcal{R}_{j;k}^t(r)]\right). \quad (6.31)$$

The overall free energy estimators with history forgetting $F_{j;k}^t$ are then obtained by straightforward averaging, as follows:

$$e^{-F_{j;k}^t} \equiv \frac{1}{\mathcal{N}_j^t} \sum_{l=n(\alpha)}^{n(t)} \mathcal{N}_j^{l,t} e^{-F_{j;k}^{l,t}}. \quad (6.32)$$

Recursions for $\mu_{j;ki}^{l,t} = \mathbb{E}_k[\psi_i(\mathcal{X})]$ and $o_{j;k}^{l,t}$ analogous to (6.31) are constructed along similar lines and combined over epochs as in (6.32). Together, these three recursions define the tuple $\theta_{j;k}^{l,t} = (F_{j;k}^{l,t}, \mu_{j;k}^{l,t}, o_{j;k}^{l,t})$ and the corresponding quantity averaged over epochs $\theta_{j;k}^t = (F_{j;k}^t, \mu_{j;k}^t, o_{j;k}^t)$. The computation of these recursions is laid out more explicitly in Section 7.

7 Programming the recursions

This section gives the mathematical details for the stochastic approximation procedure modified for use with the windowing system (Section 3.1 in the main text, Section 6.1) and history forgetting (Section 6.3). To

assist the reader interested in the implementation of the algorithm, we relate the equations to pieces of the code, the source code of which is available at <https://github.com/DEShawResearch/tss>.

Notation. The position $x \in \mathcal{S}$, rung $k \in [K]$ and window $j \in [J]$ of replica $r = 1, \dots, R$ at time t will be written as $\mathcal{X}^t(r)$, $\mathcal{K}^t(r)$ and $\mathcal{J}^t(r)$, respectively.

7.1 Per-window estimates

The recursions written here are for a single window W_j and multiple replicas $r = 1, \dots, R$. Define also the times τ_l introduced in 6.3. Given $\phi > 1$, we consider the exponentially increasing sequence of integers defined recursively by $\tau_0 = 0$, $\tau_1 = 1$ and

$$\tau_{l+1} = \lceil \phi \tau_l \rceil, l \geq 1. \quad (7.1)$$

Guidance on choosing $\phi > 1$ is provided in Section 9.

7.1.1 Epoch-based counting processes

Let $\alpha \in [0, 1)$ be the fraction of data that is dropped asymptotically, so that throughout the simulation, the fraction of stored samples is approximately $1 - \alpha$. The “recent history” is the data generated between time αt and t . At time t we only update epoch $n(t)$, and potentially drop or start a new epoch; all the intermediate (i.e., not the first nor the last) epochs are untouched.

The number of visits made to the window W_j by replica r in epoch l is

$$\mathcal{N}_j^{t,l}(r) = \sum_{s=\tau_{l-1}+1}^{\tau_l \wedge t} 1_{\{j\}}(\mathcal{J}^s(r)). \quad (7.2)$$

For each replica $r \in [R]$, the number of visits paid to the window W_j in the recent history of the simulation is

$$\mathcal{N}_j^t(r) = \sum_{l=n(\alpha t)}^{n(t)} \mathcal{N}_j^{t,l}(r) \quad (7.3)$$

which can be further combined over all replicas

$$\mathcal{N}_j^t = \sum_{r=1}^R \mathcal{N}_j^t(r) \quad (7.4)$$

Only the epochs l with $n(\alpha t) \leq l \leq n(t)$ are needed at the current time t . Notice that the current epoch is typically incomplete, so the population counters with $l = n(t)$ change with time. During any cycle t that marks the beginning of a new epoch, new counters $\mathcal{N}_j^{t,n(t)}$ are instantiated on the store and initialized to 0. During any cycle t for which αt crosses into a new epoch, the chronologically earliest counters $\mathcal{N}_j^{t,l}(r)$, the ones with $l = n(\alpha t) - 1$, are removed from the store. Within a cycle, both actions happen immediately after the execution of the Monte Carlo moves in the cycle. The number of counters committed to the store remains bounded during the simulation. We note that, by its very definition, $\mathcal{N}_j^{t,l}(r)$ needs to be incremented at the current time t only for $l = n(t)$ and only for those $j \in \cup_{r=1}^R \{\mathcal{J}^t(r)\}$.

7.1.2 Epoch-based free energy estimates

For each $j \in [J]$ and $k \in W_j$ and each replica $r = 1, \dots, R$, define the importance sampling ratio

$$\mathcal{R}_{j,k}^t(r) = 1_{\{j\}}(\mathcal{J}^t) \frac{e^{-H_k(\mathcal{X}^t(r))}}{d_j^t(r)}, \quad (7.5)$$

where

$$d_j^t(r) = \begin{cases} \sum_{k \in W_j} \pi_{j,k}^{t-1} e^{F_{j,k}^{t-1} - H_k(\mathcal{X}^t(r))}, & \text{if } \mathcal{N}_j^t \geq 1; \\ 1 & \text{else.} \end{cases} \quad (7.6)$$

The weights $\pi_{j,k}^{t-1}$ satisfy $\sum_{k \in W_j} \pi_{j,k}^{t-1} = 1$ and will be defined in the next section, whereas the overall estimates $F_{j,k}^{t-1}$ will be defined in Section 7.2.

For all $n(\alpha t) \leq l \leq n(t)$, $k \in [K]$ and $j \in \text{win}(k)$, we compute and store the epoch free energy estimates

$$e^{-\mathcal{F}_{j,k}^{t,l}} = \frac{1}{\mathcal{N}_j^{t,l}} \sum_{s=\tau_{l-1}+1}^{\tau_l \wedge t} \sum_{r=1}^R \mathcal{R}_{j,k}^s(r), \quad (7.7)$$

where we make the convenient but mathematically inconsequential convention that $\mathcal{F}_{j,k}^{t,l} = 0$ if $\mathcal{N}_j^{t,l} = 0$. At cycle t , all epoch free energies remain unchanged, except for those with $l = n(t)$. The **per-window, per-epoch** free energies for $l = n(t)$ are computed with the following recurrence: for $j = \text{win}(\mathcal{K}^t)$ and for all $k \in W_j$,

$$\mathcal{F}_{j,k}^{t,l} = \mathcal{F}_{j,k}^{t-1,l} - \log \left(1 + \frac{1}{\mathcal{N}_j^{t,l}} \left[\mathcal{N}_j^{t-1,l} - \mathcal{N}_m^{t,l} + e^{\mathcal{F}_{j,k}^{t-1,l}} \sum_{r=1}^R \mathcal{R}_{j,k}^t(r) \right] \right). \quad (7.8)$$

Equation (7.8) corresponds to `window_t::update_epoch_fe_estimates` in `window.cxx` in the source code in the GitHub repository.

We note that the convention $\mathcal{F}_{j,k}^{t,l} = 0$ if $\mathcal{N}_j^{t,l} = 0$ is indeed inconsequential, because the otherwise undefined $\mathcal{F}_{j,k}^{t-1,l}$ would have canceled out anyway on the right-hand side of equation (7.8). The **per-window, all-epoch** free energy estimates $F_{j,k}^t$ are defined by

$$e^{-F_{j,k}^t} = \sum_{l=n(\alpha t)}^{n(t)} \frac{\mathcal{N}_j^{t,l}}{\sum_{l'=n(\alpha t)}^{n(t)} \mathcal{N}_j^{t,l'}} e^{-\mathcal{F}_{j,k}^{t,l}} \quad (7.9)$$

for all $k \in [K]$ and $j \in \text{win}(k)$. Equation (7.9) corresponds to `window_t::estimate_all_epochs` in `window.cxx` in the source code.

7.1.3 Epoch-based rung distribution estimates

Much of the computation of the asymptotic rung distribution is implemented in `covariance_estimator.{cxx,hxx}`. The default TSS rung distribution is the coordinate-invariant one, which we recall is written in terms of the metric

$$g_{ij}(\lambda) = \mathbb{E} \left[\frac{\partial H_\lambda}{\partial \lambda_i}(X) \cdot \frac{\partial H_\lambda}{\partial \lambda_j}(X) \right] - \mathbb{E} \left[\frac{\partial H_\lambda}{\partial \lambda_i} \right] \mathbb{E} \left[\frac{\partial H_\lambda}{\partial \lambda_j} \right], \quad (7.10)$$

where the expectation is taken with respect to $p_\lambda(x)dx$. An unregularized form of the asymptotic TSS metric is given by

$$\frac{\sqrt{\det(g(\lambda_k))}\text{vol}(\lambda_k)}{\sum_{j \in [K]} \sqrt{\det(g(\lambda_j))}\text{vol}(\lambda_j)}, \quad (7.11)$$

where $\text{vol}(\lambda_k)$ is a volume element arising from the discretization of the space Λ into the set $\{\lambda_1, \dots, \lambda_K\}$. For instance, if $\Lambda = [0, 1]$, then we might choose the discretization $\{[k/K, (k+1)/K], k = 0, \dots, K-2\} \cup \{[1-1/K, 1]\}$, and then $\text{vol}(\lambda_k) = 1/K$ for all $k \in [K]$.

We will let G_λ denote the matrix whose i, j entry is $g_{ij}(\lambda)$, and G_k will denote G_{λ_k} . We can evaluate the partial derivatives using a second-order central difference,

$$\frac{\partial H_k}{\partial \lambda_i}(x) \approx \frac{H_{k+1}(x) - H_{k-1}(x)}{2}. \quad (7.12)$$

The central difference may be undefined for some of the rungs $k \in [K]$ such that λ_k lies on or near the boundary of Λ . In that case, we can use a first-order forward or backward difference, as needed. It is important that the approximation is tied to the original graph edges and is independent of the choice of windows. To avoid clutter, we set

$$\psi_k(x) = \nabla_\lambda H(x)|_{\lambda=\lambda_k}, \quad (7.13)$$

and note for the sake of generality that the estimation method described below is applicable to an arbitrary function $\psi(x)$.

As we have done for free energies, for all l such that $n(\text{out}) \leq l \leq n(t)$ and for all $k \in [K]$ and $j \in \text{win}(k)$, we introduce the epoch estimators Ψ by way of the formula

$$\Psi_{j;k}^{t,l} = \frac{1}{\mathcal{N}_j^{t,l}} \sum_{s=\tau_{l-1}+1}^{\tau_l \wedge t} \sum_{r=1}^R e^{F_{j;k}^{t,l}} \mathcal{R}_{j;k}^s(r) \psi_k(\mathcal{X}^s), \quad (7.14)$$

where the equality in (7.14) is meant componentwise whenever $\Psi_{j;k}^{t,l}$ is a vector. As before, we set $\Psi_{j;k}^{t,l} = 0$ if $\mathcal{N}_j^{t,l} = 0$. The definition of the epoch estimators is such that the epoch estimators $\Psi_{j;k}^{t,l}$ of a constant function $\psi_k(x)$ is the constant itself. The overall estimates are obtained by weighting the epoch estimates according to

$$\Psi_{j;k}^t = \sum_{l=n(\text{out})}^{n(t)} \frac{\mathcal{N}_j^{t,l} e^{-F_{j;k}^{t,l}}}{\sum_{l'=n(\text{out})}^{n(t)} \mathcal{N}_j^{t,l'} e^{-F_{j;k}^{t,l'}}} \Psi_{j;k}^{t,l}. \quad (7.15)$$

The estimates are left undefined if the denominator of (7.15) vanishes. As for the free energies, at the current time t the epoch estimates need to be updated only for the last epoch, the one with $l = n(t)$, as follows: for all $j \in \text{win}(\mathcal{K}^t)$ and $k \in W_m$,

$$\begin{aligned} \Psi_{j;k}^{t,l} &= e^{F_{j;k}^{t,l} - F_{j;k}^{t-1,l}} \\ &\times \left(\Psi_{j;k}^{t-1,l} + \frac{1}{\mathcal{N}_j^{t,l}} \left[(\mathcal{N}_j^{t,l} - \mathcal{N}_j^{t-1,l}) \Psi_{j;k}^{t-1,l} + e^{F_{j;k}^{t-1,l}} \sum_{r=1}^R \mathcal{R}_{j;k}^t(r) \psi_k(\mathcal{X}^t(r)) \right] \right). \end{aligned} \quad (7.16)$$

For each $j \in [J]$ and $k : \lambda_k \in W_j$, the estimates for the covariance matrices $G_{j;k}^t$ at step t are given by the formula

$$G_{j;k}^t = (\Psi_1)_{j;k}^t - (\Psi_2)_{j;k}^t ((\Psi_2)_{j;k}^t)^\top \quad (7.17)$$

where $(\psi_1)_{j;k}^t, (\psi_2)_{j;k}^t$ are the overall averages of $\psi_2(x) = \nabla_{\lambda} H_{\lambda}(x)|_{\lambda=k}$ and $\psi_1(x) = \psi_2(x)\psi_2(x)^{\top}$. Equation (7.17) is implemented in `covariance_estimator_t::update_epoch_estimates` in `covariance_estimator.cxx`.

The coordinate invariant distribution is regularized for any fixed $\varepsilon_{\gamma} \in (0, 1]$ as in (6.10), namely,

$$\gamma_{j;k}(\mu) = \frac{(1 - \varepsilon_{\gamma})\gamma_k(\mu_{j;\cdot}) + \varepsilon_{\gamma}\mathcal{M}(\mu)}{\sum_{\ell: \lambda_{\ell} \in W_j} [(1 - \varepsilon_{\gamma})\gamma_{\ell}(\mu_{j;\cdot}) + \varepsilon_{\gamma}\mathcal{M}(\mu)]} 1_{W_j}(\lambda_k), \quad \forall 1 \leq k \leq K, \quad (7.18)$$

where

$$\mathcal{M}(\mu) = \max_{k \in [K], j \in \text{win}(k)} \gamma_k(\mu_{j;\cdot}) \quad (7.19)$$

The regularized per-window rung distribution (7.18) is implemented in `window_t::normalized_weights` in `window.cxx`.

7.1.4 Epoch-based tilts

For all $n(\alpha) \leq l \leq n(t)$, $k \in [K]$ and $j \in [J]$, we compute and store

$$o_{j;k}^{t,l} = \frac{1}{\mathcal{N}_j^{t,l}} \sum_{s=\tau_{l-1}+1}^{\tau_l \wedge t} \sum_{r=1}^R 1_{\{j\}}(\mathcal{J}^t) \frac{1_{\{k\}}(\mathcal{K}^t)}{\gamma_{j;k}^{t-1}}. \quad (7.20)$$

We compute these estimates through the **per-window, per-epoch** recursion

$$o_{j;k}^{t,l} = o_{j;k}^{t-1,l} + \frac{1}{\mathcal{N}_j^{t,l}} \left[(\mathcal{N}_j^{t-1,l} - \mathcal{N}_j^{t,l}) o_{j;k}^{t-1,l} + \sum_{r=1}^R 1_{\{j\}}(\mathcal{J}^t) \frac{1_{\{k\}}(\mathcal{K}^t)}{\gamma_{j;k}^t} \right]. \quad (7.21)$$

Equation (7.21) corresponds to `window_t::update_tilts` in `window.cxx` in the code. We then provide combine the epoch estimates to obtain **per-window, all-epoch** tilt estimates $o_{j;k}^t$ through

$$o_{j;k}^t = \sum_{l=n(\alpha)}^{n(t)} \frac{\mathcal{N}_j^{t,l}}{\mathcal{N}_j^t} o_{j;k}^{t,l}. \quad (7.22)$$

Equation (7.22) corresponds to `window_t::overall_tilt` in `window.cxx` in the code.

7.2 Global estimates

7.2.1 Global rung weights

The C++ functions referred to in this section rely heavily on ideas and notation introduced in Section 3 of the main text and Section 6 of this document.

As a first step, we must estimate the window marginal probability $\bar{p}_j = \bar{P}_{\pi,F}(\mathcal{J} = j)$. Our estimate p_j of $\bar{p}_j$ is obtained by solving the eigenvector problem

$$Qp = p, \quad (7.23)$$

where $Q = (q_{ij})$, $1 \leq i, j \leq J$,

$$q_{ij} = \frac{1}{2} \frac{\sum_{k \in W_i \cap W_j} \gamma_{j;k} o_{j;k}}{\sum_{k: \lambda_k \in W_j} \gamma_{j;k} o_{j;k}}. \quad (7.24)$$

We note that early in the simulation, not all entries of the matrix are available, since only a limited number of windows W_j have been visited and thus have $\mathcal{N}_j^t \geq 1$. We restrict the eigenvector problem only to the visited windows, and set $p_j^t = 0$ (i.e., the estimate p_j at time t) if $\mathcal{N}_j^t = 0$. A column j of Q^t (the matrix Q at time t) such that W_j has been visited at least once may have non-zero components q_{ij}^t corresponding to non-visited windows W_i . Those components are summed up, with the sum added to the (j, j) diagonal entry. The non-visited components are then zeroed. We obtain a lower-dimensional left-stochastic matrix acting on the space of probability vectors defined over the space of visited states. We can use QR to find the invariant stochastic eigenvector of minimum L^2 norm, in the rare cases where the invariant stochastic vector isn't already unique. This function is implemented in `free_energy_estimator_t::window_weights_qr`.

Once the marginal window probabilities at time t , p_j^t , have been obtained through (7.23), we can compute the probability q_k^t of visiting rung $k \in [K]$ irrespective of the window as follows:

$$q_k^t = \sum_{j \in \text{win}(k)} p_j^t \frac{\gamma_{j;k}^t \sigma_{j;k}^t}{\sum_{\ell \in W_j} \gamma_{j;\ell}^t \sigma_{j;\ell}^t}. \quad (7.25)$$

Equation (7.25) corresponds to `free_energy_estimator_t::global_weights` in `free_energy_estimator.cxx` in the code.

7.2.2 Global visit control free energies

To determine the global visit control free energies, we must identify the offsets f_j^t introduced in 6.2.1. These offsets solve the following system of equations, for $j \in [J]$ and a visit control parameter $\eta > 0$

$$f_j^t = (\eta + 1) \log \left(\sum_{k: \lambda_k \in W_j} q_k^t \frac{\gamma_{j;k}^t e^{\frac{1}{\eta+1} F_{j;k}^t}}{\sum_{i \in \text{win}(k)} p_i^t \gamma_{i;k}^t e^{\frac{1}{\eta+1} (F_{i;k}^t - f_i^t)}} \right). \quad (7.26)$$

Equivalently, these unknown offsets f_j^t minimize the objective function

$$\frac{1}{\eta + 1} \sum_{j \in [J]} p_j^t f_j^t + \sum_{k \in [K]} q_k^t \log \left(\sum_{j \in \text{win}(k)} p_j^t \gamma_{j;k}^t e^{\frac{1}{\eta+1} (F_{j;k}^t - f_j^t)} \right), \quad (7.27)$$

which is a convex function in f_j^t .

There are multiple avenues for solving (7.26) or, equivalently, minimizing (7.27). We employ perhaps the simplest approach, which is to perform fixed point iteration, represented succinctly as

$$f = h(f), \quad (7.28)$$

where $f = (f_1, \dots, f_J)$ and $h(f) = (h_1(f), \dots, h_J(f))$ is defined componentwise by

$$h_j(f) = (\eta + 1) \log(g_j(f)), \quad j \in [J], \quad (7.29)$$

with

$$g_j(f) = \sum_{k \in W_j} q_k^t \frac{\gamma_{j;k}^t e^{\frac{1}{\eta+1} F_{j;k}^t}}{\sum_{i \in \text{win}(k)} p_i^t \gamma_{i;k}^t e^{\frac{1}{\eta+1} (F_{i;k}^t - f_i^t)}}. \quad (7.30)$$

This fixed point iteration is performed in `free_energy_estimator_t::improve_fes`.

7.2.3 Global reported free energies

The visit control free energies already define global estimators for the free energies, but these are η -dependent. Because they are functions of the empirical frequencies of visiting windows and of the tilts $o^t = \{o_{j;k}^t; j \in [J], k : \lambda_k \in W_j\}$, these estimators exhibit large error bars, which is typical of event counting. As described in Section 6.2.2 and mentioned in Section 3.1. of the main text, we can provide global estimators with lower error bars by letting $\eta \rightarrow \infty$; we call these the reported TSS free energies, and which we denote by F_k^{TSS} .

The limit $\eta \rightarrow \infty$ forces us to take $o_{j;k}^t = 1$ for all $j \in [J], k \in [K]$. Consequently, we can identify the global rung probabilities as in (7.25), this time with $o_{j;k}^t$ replaced by 1. We define the reported rung density to be

$$\gamma_k^{\text{TSS}} = \sum_{j \in [J]} p_j \gamma_{j;k}, \quad (7.31)$$

for each $k \in [K]$.

In the limit $\eta \rightarrow \infty$, we can compute the gradient of the objective (7.27) and set the limiting gradients to zero. The system of equations so-obtained is

$$\sum_{j \in [J]} (\delta_{ij} - t_{ij}) f_j^{\text{TSS}} = \sum_{k \in W_i} \gamma_{i;k}^t \left(F_{i;k}^t - \sum_{j \in \text{win}(k)} \frac{p_j^t \gamma_{j;k}^t}{\gamma_k^{\text{TSS}}} F_{j;k}^t \right), \quad (7.32)$$

where

$$t_{ij} = \sum_{k \in W_i \cap W_j} \gamma_{i;k}^t \frac{p_j^t \gamma_{j;k}^t}{\gamma_k^{\text{TSS}}}. \quad (7.33)$$

Once the offsets f_j^{TSS} are obtained, we can compute the reported TSS free energies for each $k \in [K]$ through

$$F_k^{\text{TSS}} = \frac{1}{\gamma_k^{\text{TSS}}} \sum_{j \in \text{win}(k)} p_j \gamma_{j;k} (F_{j;k} - f_j^{\text{TSS}}). \quad (7.34)$$

This procedure is carried out in `free_energy_estimator_t::solve_fes_infinite_eta`.

8 Jackknife error estimation

The task of computing error bars is made particularly easy by the fact that we compute and store epoch estimates. Thus, estimates for the errors are obtainable, for example, by way of the jackknife method. The m -th jackknife replicate $F_{j;k}^{t,(m)}$ is defined by

$$e^{-F_{j;k}^{t,(m)}} = \sum_{l=n(\alpha), l \neq m}^{n(t)} \frac{\mathcal{N}_{m;k}^{t,l}}{\sum_{l'=n(\alpha), l' \neq m}^{n(t)} \mathcal{N}_{j;k}^{t,l'}} e^{-F_{j;k}^{t,l}}, \quad (8.1)$$

which is equation (7.9) with the m -th epoch deleted. The use of replicates for the evaluation of error bars will be presented only for free energy differences, since it proceeds analogously for the other estimated quantities or functions of them. For instance, the m -th jackknife replicate $\langle \psi \rangle_{j;k}^{t,(m)}$ is obtained by deleting the m -th epoch in equation (7.15).

We assume that the TSS simulation is sufficiently advanced that all windows have been visited at least once, even when any one of the epochs is deleted. Otherwise, the analysis must be restricted to the set of all windows that have been so visited. Global reported free energies, as described in Section 6.2.2, can then be computed for each replicate m . The global m -th jackknife replicates are denoted by $F_k^{t,(m)}$.

For $l \in n(\alpha), \dots, n(t)$, the numbers

$$a_l = \frac{\tau_l \wedge t - \tau_{l-1}}{t - \tau_{n(\alpha)-1}} \quad (8.2)$$

are positive and sum to 1. Asymptotically, the error for $F_{e',k'}^t - F_{e,k}^t$ is normally distributed about a small bias, with the MSE estimated by

$$\widehat{\text{MSE}}(t; k, k') = \frac{1}{n(t) - n(\alpha)} \sum_{l=n(\alpha)}^{n(t)} \frac{(1 - a_l)^2}{a_l} \left[\left(F_{e',k'}^{t,(l)} - F_{e,k}^{t,(l)} \right) - \left(F_{e',k'}^t - F_{e,k}^t \right) \right]^2. \quad (8.3)$$

This MSE estimator is implemented in `free_energy_estimator_t::update_errors` in `free_energy_estimator.cxx`.

In our analysis, we do not use the debiased jackknife estimator. We assume the simulation is sufficiently well converged that the main source of error is the noise term, the variance of which is approximated by (8.3). The coefficients $(1 - a_l)^2/a_l$ have been chosen such that the estimator $\widehat{\text{MSE}}(t)$ has a chi-squared distribution in the limit $t \rightarrow \infty$ (assuming the central limit theorem holds).

Standard precautions apply for the estimation of the MSE to be accurate. The epochs must be long enough that the jackknife pseudo-values

$$a_l^{-1} \left[\left(F_{e',k'}^t - F_{e,k}^t \right) - (1 - a_l) \left(F_{e',k'}^{t,(l)} - F_{e,k}^{t,(l)} \right) \right] \quad (8.4)$$

can be assumed to be independent normal variables. At the same time, the number of epochs must be large enough that the MSE estimator defined by (8.3) has sufficiently small noise to approximate well the targeted MSE. For sufficiently large t , the instantaneous number of epochs $n(t) - n(\alpha) + 1$ fluctuates between two values n_{epochs} and $n_{\text{epochs}} + 1$, as new epochs are added and old epochs are eliminated. We recommend that the number of epochs be at least 32. The proper number of epochs can be dialed in by setting ϕ according to the formula

$$\phi = \alpha^{-1/n_{\text{epochs}}}. \quad (8.5)$$

As an example, for $\alpha = 0.19$ and $n_{\text{epochs}} = 32$, we compute $\phi = 0.19^{-1/32} \approx 1.0533$.

9 Putting it all together

In this section, we will use the diagram depicting the flow of the TSS algorithm from the main text (Figure 4) to discuss the order in which the elements of TSS we have introduced are employed and highlight some of their computational characteristics. We will then provide practical guidance regarding the selection of windows and other parameters.

The simulation begins with each replica having some state (x, k, j) . As shown in the upper-left-hand rectangular boxes in Figure 4, each replica first swaps to the other window containing its current rung k , after which the MD dynamics are performed by each replica separately, and the Hamiltonians are evaluated for each window occupied by a replica. Although the evaluation of the $H_k(x)$ can be computationally expensive (especially if $\Lambda \subset \mathbb{R}^d$ and $d > 1$), the most computationally expensive part of these boxes is generally the MD step.

Moving on to the stochastic approximation updates (upper-right-hand box), we note that the epoch-based, per-window estimator updates to $F_{j,:}^{t,t-1}$ (as defined in (6.31)), and the analogous updates to $\mu_{j,:}^{t,t-1}$, $o_{j,:}^{t,t-1}$, and $\mathcal{N}_j^{t,t-1}$ (as defined in Section 6.1), are computationally inexpensive operations relative to the cost of the MD step and the evaluation of the Hamiltonians for the active windows. The updates of f_j^{t-1} and $\pi_{j,:}^{t-1}$ are more computationally expensive than the epoch-based, per-window estimator updates, because they require the minimization of (6.19) to obtain the window free energy offsets $\{f_j^{t-1}; j \in [J]\}$. Notably, the convex optimization problem (6.19) is a global operation, and its cost increases with the number of windows.

As shown in Figure 4, each element of the state (x, k, j) is sampled sequentially, and neither the sampling of x nor j is dependent on the estimates (F^{t-1}, π^{t-1}) . One motivation for structuring the algorithm in this way is to take advantage of this independence, by computing the estimates (as outlined in Sections 2.1–2.3) on an external device simultaneously with the MD step and the evaluation of the energies $\{H_k(x), k \in W_{j'}\}$, and then receiving the estimates in order to perform the k dynamics, which is dependent on (F^{t-1}, π^{t-1}) . The energies $\{H_k(x), k \in W_{j'}\}$ are sent to the external device to update the estimates for the next cycle.

On certain computing architectures, such as the special-purpose molecular dynamics supercomputer Anton 3 (Shaw et al. 2021), the latency of communication with the external device (the time to send and receive a small packet of data) can be large relative to the MD step. To mitigate the cost of this communication, one can leverage the observation that self-adjustment reduces the variance of the free energy estimates, with the maximal improvement happening when the newly generated (x', k') state is stochastically independent of previous (x, k) states. We can approximate this stochastic independence by iterating a composition of the x - and k -transition kernels $v \geq 1$ times during a TSS cycle, without resampling the window index j' . The send and receive operations occur only once per cycle, and the v iterations use the same fixed estimate F^{t-1}, π^{t-1} received right before the first update to the state of k within a cycle. The strategy is effective because the variables $(X, \mathcal{K})$ are often strongly correlated, and would thus be poorly sampled if the x - and k -transition kernels in a cycle were not alternated before being composed.

Once the energy evaluations are available, the k dynamics (the center-left box following the send operation in Figure 4) can be performed for each replica, an inexpensive process compared to the computation of the energies. The updates to k do not require updates from the upper-right-hand box at every cycle, and thus do not incur any communication latency; they can be executed on the device that performs the sampling immediately after the energies are evaluated.

Finally, before proceeding to the next cycle $t + 1$, the epoch estimators must be updated. The epochs allow for the computation of error bars by means of a jackknife estimate. Although the computation of the error bars is computationally expensive (because for each epoch one needs to solve an optimization problem like the one for the reported free energies), this expense can be safely mitigated by computing the error bars infrequently, as they do not drive the simulation.

We now provide some guidance on choosing parameters that have been introduced in TSS: the visit control parameter $\eta \geq 0$, the history forgetting parameter $\alpha \in [0, 1)$ and base multiplier $\phi > 1$, the regularization parameters ε_γ and $\varepsilon_\pi \in (0, 1]$, as well as the selection of windows.

To begin, we recall that, at the extremes, the parameter $\eta \geq 0$ has the following binary behavior: When $\eta = 0$, visit control is “off,” and otherwise it is “on.” In practice, we find that $\eta = 2$ works well for a broad range of systems. A larger value of η may further accelerate convergence for some systems (though we recall that the choice does not affect the asymptotic variance of the estimates), but with increased risk of rung moves that lead to failure of the MD component, whereas a value of η smaller than 1 is never the optimal choice in our experience. Regardless of the particular value η is set to, the most important choice is to turn visit control on (i.e., to set $\eta > 0$). In Section 10, a range of values of η will be simulated, to illustrate the utility of $\eta > 0$ and the different rates at which the estimates reach their equilibrium value.

For the history forgetting parameter α , which determines the proportion of history that is “forgotten,” we recommend using the value 19%. We find this rate to be within a range of reasonable values for many systems, with the particular number chosen because the asymptotic statistical efficiency of using history forgetting relative to using the full trajectory, $\sqrt{1-\alpha} = 0.9$, is easy to remember. To determine ϕ , it is convenient to think in terms of number of epochs n_{epochs} and use the relation $\phi = \alpha^{-\frac{1}{n_{\text{epochs}}}}$; in general, we choose $n_{\text{epochs}} = 32$, which corresponds to $\phi \approx 1.0533$. Increasing the number of epochs will improve the precision of the error bars asymptotically, but will also require longer simulation times to ensure that the epochs are sufficiently decorrelated to use the jackknife error estimator.

The small regularization parameters $\epsilon_\gamma, \epsilon_\pi \in (0, 1]$ ensure that the visitation frequency is at least a fraction $\epsilon_\gamma \epsilon_\pi$ of the frequency of the most visited rung. If, for instance, there is a window $W_j \subset \Lambda$ for which $H_k(x)$ varies little with $\lambda_k \in W_j$, then the unregularized density π (which approaches γ asymptotically) may assign very little weight to that window (as it should). The downside is that the window may not be visited sufficiently often in the recent history of the simulation, which can increase the errors of the overall free energy estimates, despite the window contributing little to the overall estimate. To accommodate such degenerate cases, we recommend setting $\epsilon_\gamma = 0.01$ and $\epsilon_\pi = 0.001$, which we have found to be suitable for a wide range of simulations.

In most cases, TSS is robust to the choice of window size (see, for instance, Section 3), and the user should observe good convergence of the free energy estimates. In cases where convergence is poor, the behavior of the rung sampling process may help indicate whether a poor choice of window size is the source of the problem. If the process is becoming diffusive (i.e., the rung process $\mathcal{X}'$ is limited to moves between adjacent rungs, which can significantly increase the variance of the free energy estimates), or if the optimization problem becomes prohibitively expensive, the windows may be too small. One can reduce the computational expense resulting from a large number of windows, without increasing window size, by computing f_i less frequently than on each cycle. On the other hand, the windows may be too large if one observes poor pre-asymptotic behavior of the estimates: notably, the exponential slowdown discussed in Section 1.4. Overly large windows also allow the selection of unphysical rungs early in the simulation, when the estimates are poor, which can lead to failures in the MD step. Reducing the size of the windows will eliminate these problems, with the added benefit of reducing the computational expense of evaluating the Hamiltonians.

Part III

Numerics

10 Numerical example: calculation of free energy differences in aqueous solution

Here we illustrate the use of TSS for sampling and estimation on an example problem relevant to computational drug discovery, and in particular how certain TSS parameter choices that influence sampling in the Λ space affect free energy estimation in practice. Although the example we have chosen appears to be complex compared to typical models simulated with MCMC, it is a relatively straightforward instance of a problem that arises frequently in drug discovery (Rizzi et al. 2020): the calculation of free energy differences in aqueous solution.

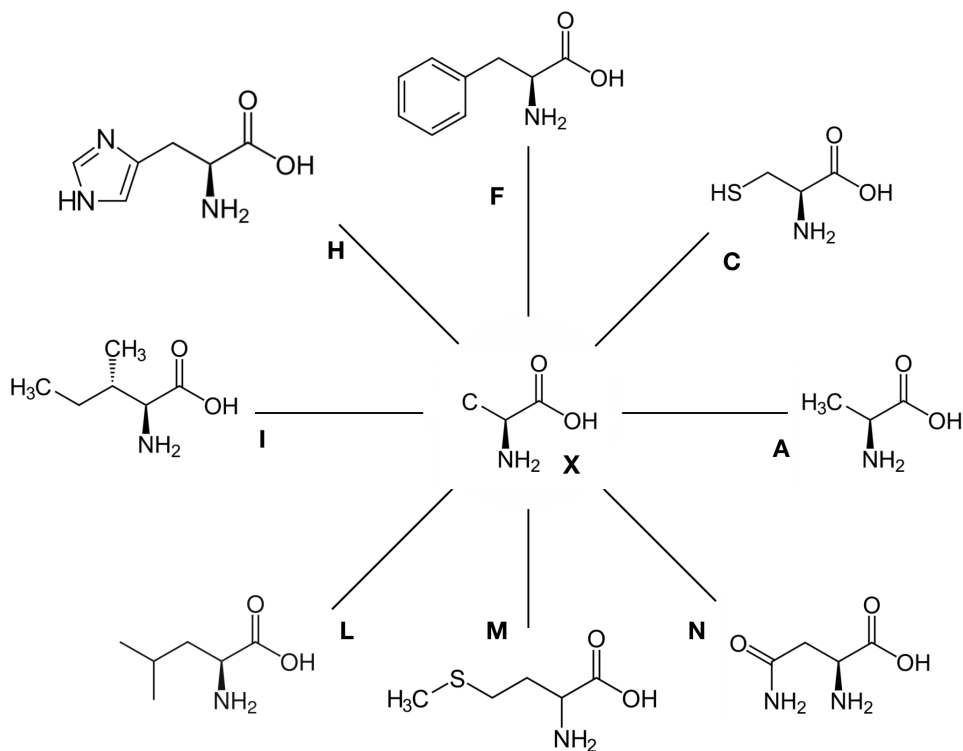


Figure 4: Collection of amino acids used in the simulations. The central substructure, X , is common to the eight amino acid structures.

10.1 Description of the chemical system

We simulated a collection of eight amino acids (A , C , F , H , I , L , M , and N) that have a common substructure (X), which is alanine (A) with three hydrogens removed (see Figure 4). To compute the free energy differences among the different structures using TSS, we pick a parameterized family of distributions $\rho_\lambda, \lambda \in \Lambda$ in which the distribution of each amino acid structure corresponds to a rung $\lambda \in \Lambda$, and for which ρ_λ varies smoothly with λ between the structures. There are standard approaches (broadly called “alchemical free energy perturbation”) in the MD literature for parameterizing such families of distributions, although we note that determining parameterizations that yield low-variance free energy estimates is itself an active area of research (Shobana et al. 2000; Jiang et al. 2019).

Broadly, the alchemical free energy perturbation parameterization approach we use is as follows: To compute the free energy difference between the X and A substructures, for instance, a one-dimensional parameter $\lambda \in [0, 1]$ is chosen such that the interaction energies of the three hydrogen atoms are “off” (equal to zero) when $\lambda = 0$ and “on” when $\lambda = 1$. As λ gradually moves from 0 to 1, terms in the Hamiltonian $H(x, \lambda)$ corresponding to bonded interactions (such as bond stretching or angle bending) and non-bonded interactions (such as the van der Waals and electrostatic interactions) are progressively scaled up and become equal to the strength they normally have in the A structure once $\lambda = 1$. We note that the usual van der Waals interaction potential has a singularity at the origin (i.e., when distance between two atoms is zero), and we

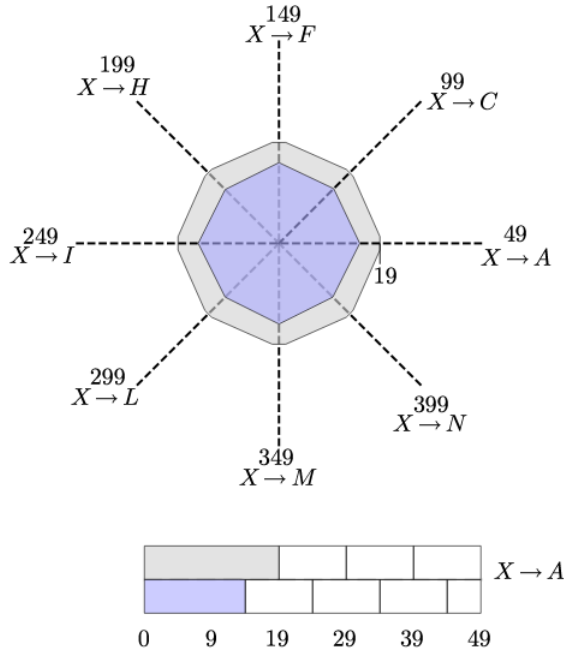


Figure 5: Default windowing system used in the amino acid simulations. (Top): The eight edges, illustrated with the vertex being the common substructure X . The two large central windows occupy the first 15 and 20 rungs of each edge, containing 120 and 160 rungs, respectively. Each edge contains 50 rungs. The smaller central window thus contains rungs with indices $50i + k, i = 0, \dots, 7, k = 0, \dots, 14$. Aside from the two central windows, the remainder of the window structure is identical for each edge. (Bottom): Window structure for one of the eight edges. The windows are offset to satisfy the irreducibility requirements of Section 2.1. All non-central windows contain either 5 or 10 rungs.

instead use a “softcore” van der Waals potential, which avoids this singularity (as described in Part 3, Section 7.2 of the SM). Additional information on the schedule and simulation details are provided in Part 3, Section 7.1 of the SM.

Each of the eight substructures is linked to the common X substructure through such a parameterization, and we take Λ to be a graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ consisting of nine vertices $\mathcal{V} = \{X, A, C, F, H, I, L, M, N\}$ and eight edges $\mathcal{E} = \{(X, A), \dots, (X, N)\}$. As shorthand, we refer to edge (X, C) simply as “edge C ”.

We now discretize the Λ space. We use a collection of 400 rungs, with 50 consecutive rungs for each edge $E_i \in \mathcal{E}, i = 0, \dots, 7$. We identify each E_i with the set $\{\lambda_{50i+k} \in \Lambda : k = 0, \dots, 49\}$, which corresponds to the parameter set that guides the morph according to the family of distributions $\rho_\lambda(x)dx, \lambda \in E_i$; the morph $X \rightarrow A$ for E_1 , for instance, is guided by the addition of the three missing hydrogens. The free energy difference of edge E_i is defined as $F_{50i+49} - F_{50i}$. An illustration of the discretization of Λ can be found in Figure 5.

10.2 Numerical simulations and results

In the next three sections, we study the effect on the free energy estimators of three features of TSS that affect the sampling in the Λ space: the windowing system, the visit-control mechanism, and the self-adjustment property (the former introduced in Section 2.1, and the latter two in Section 1.2). All the free energy estimates are reported in units of $k_B T$. We performed each simulation using 32 replicas (see Section 2.3 for discussion of replicas), with a per-sampler time of 250 ns (aggregate 8 μ s). The simulation length is much longer than needed to obtain useful estimates (as we will see below, for the best-performing setup the estimates stay within 0.5 $k_B T$ of their final values after just 1 ns of simulation). Long simulations do, however, enable a thorough examination of the effects of modifying the parameters, and also ensure that we obtain good estimates of the errors. The errors are estimated using the jackknife estimator (Section 2.3 and Part 2, Section 6 of the SM) with 32 epochs and a history forgetting parameter $\alpha = 0.19$. We used the regularization parameters $\epsilon_\gamma = 0.01$ and $\epsilon_\pi = 0.001$ and, except for the alternatives tested in Section 3.2.2, a default value of $\eta = 2$ for the visit control parameter. Similarly, we used a default windowing system illustrated in Figure 5, except for the simulations in which we tested alternative systems (Section 3.2.1). We used an MD time step of 2 fs, with each TSS update being performed at a default interval of 9.6 ps (unless otherwise specified). The transition kernel $\mathcal{T}_k(x'|x)$ was determined by the force field (Amber ff99SB*-ILDN (Lindorff-Larsen et al. 2010) with the TIP3P water model) and the ensemble (NPT; constant number of particles, pressure, and temperature) using the Desmond/GPU MD software (Bergdorf et al. 2021); further simulation details can be found in Part 3, Section 7.1 of the SM.

10.2.1 Windowing system

In this section we investigate the effect of different windowing systems on free energy estimation. As explained in Section 2.1, we expect that large windows will allow the process $\mathcal{K}^t$ to sample the rungs more effectively than will small windows, but that very large windows can lead to instabilities.

We simulated the system with three different window arrangements, labeled (A.1) through (A.3):

- (A.1): a single, very large window covering all 400 rungs
- (A.2): 392 windows of size 2, 8 windows of size 1 (each containing the last rung of an edge), and a single window of size 8 (containing the first rung of each of the eight edges)
- (A.3): our default setup: 58 windows, 48 of which have size 10, 8 of which have size 5, and the last 2 of which have sizes 160 ($= 8 \times 20$) and 120 ($= 8 \times 15$) that encompass the first 20 and 15 rungs, respectively, of every edge (see Figure 5)

First, simulation (A.1) failed after just a few moves of $\mathcal{K}^t$. The underlying cause was the instability of the free energy estimates, which led to $\mathcal{K}^t$ visiting a rung that is physically infeasible given $\mathcal{X}^{t-1}$, with atoms crashing into each other with unphysically large energies, ultimately leading to failure of the molecular dynamics integration scheme. We mention this result because it highlights the necessity of the windowing mechanism in practice.

At the other extreme, (A.2) used windows containing only two rungs. In this case there was little risk of visiting a physically infeasible rung, but the motion of $\mathcal{K}^t(r)$ for each sampler $r = 1, \dots, 32$ was severely restricted, as seen in Figure 6. Such behavior is unsurprising, because windows restrict the movement of the rung process $\mathcal{K}^t(r)$ to adjacent rungs, which promotes diffusive behavior of the rung process. Nevertheless, the simulation (A.2) successfully completed, and reported reasonable estimates of the free energy differences for each edge, with the caveat that the error bars for (A.2) tended to be larger for most edges than for (A.3),

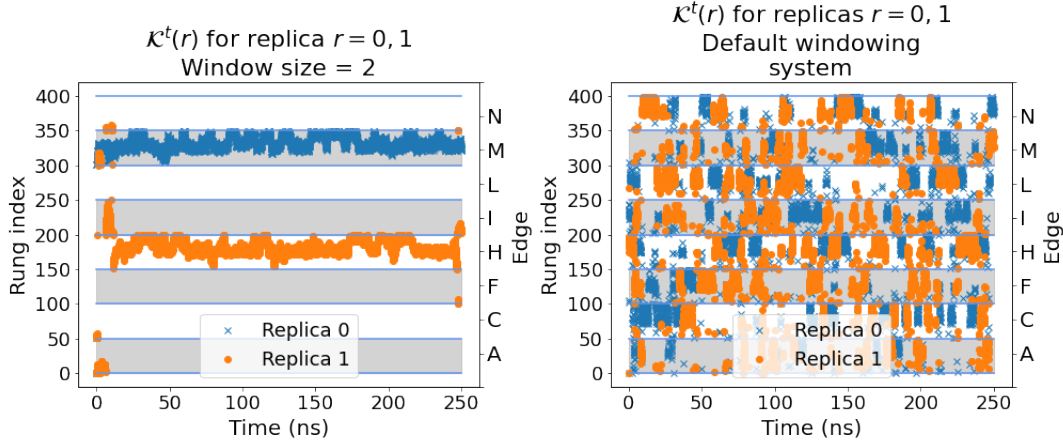


Figure 6: Small windows can cause diffusive rung dynamics. Plotted are simulations using windows of size 2 (A.2) and the default setup (A.3). As discussed in Section 2.4, the rung process can be monitored throughout the simulation, and the left-hand figure indicates that the window size of 2 is too small. (Left): The full rung trajectory for two of the replicas (with indices 0 and 1). The left y-axis labels indicate the rung index, and the right y-axis labels show which edge each range of rung indices corresponds to. The bands delineate the eight different edges. The rung trajectories show that the two samplers are confined to narrow regions of Λ , resulting in inefficient sampling of Λ ; the behavior of the other replicas (not shown) is similar. (Right): The full rung trajectory for two replicas with our default windowing system. Each of the two replicas visits each edge within 40 ns, which indicates much more effective sampling than with windows of size 2.

as shown in Figure 7. This comparison shows that the TSS free energy estimators are generally robust to the choice of windows, but that their selection affects the statistical efficiency of the simulation. As recommended in Section 2.4, the rung process can in practice be monitored, and if poor mixing properties are observed, the window size can be increased.

10.2.2 Visit control

In this section we explore the role of the visit control parameter $\eta \geq 0$ introduced in Section 1.2. Like the windowing system, visit control affects the dynamics of the rung process: Whenever $\eta > 0$, there is a bias toward under-visited rungs, which helps to prevent the rung process from getting trapped in a region of Λ . Unlike the windowing system, however, the choice of $\eta \geq 0$ does not affect the asymptotic variance of the estimates; it only modulates the speed at which the estimates approach a neighborhood of their equilibrium value. We present results from simulations with a range of η values:

- (B. η): a visit-control parameter of η ($\eta = 0, 1, 2, 4, 16, 64$) with our default windowing system.

(Note that (B.2), because it uses our default value $\eta = 2$, is equivalent to (A.3); when we refer to (B.2) in the discussion, it refers to the same simulation.) We first compare simulation (B.0) to (B.2). In 8, we see that the $\eta = 0$ simulation provides a significantly worse estimate of the free energies for each of the edges: The $\eta = 0$ trajectories all suffer from the exponential slowdown discussed in Section 1.4, and their error bars are

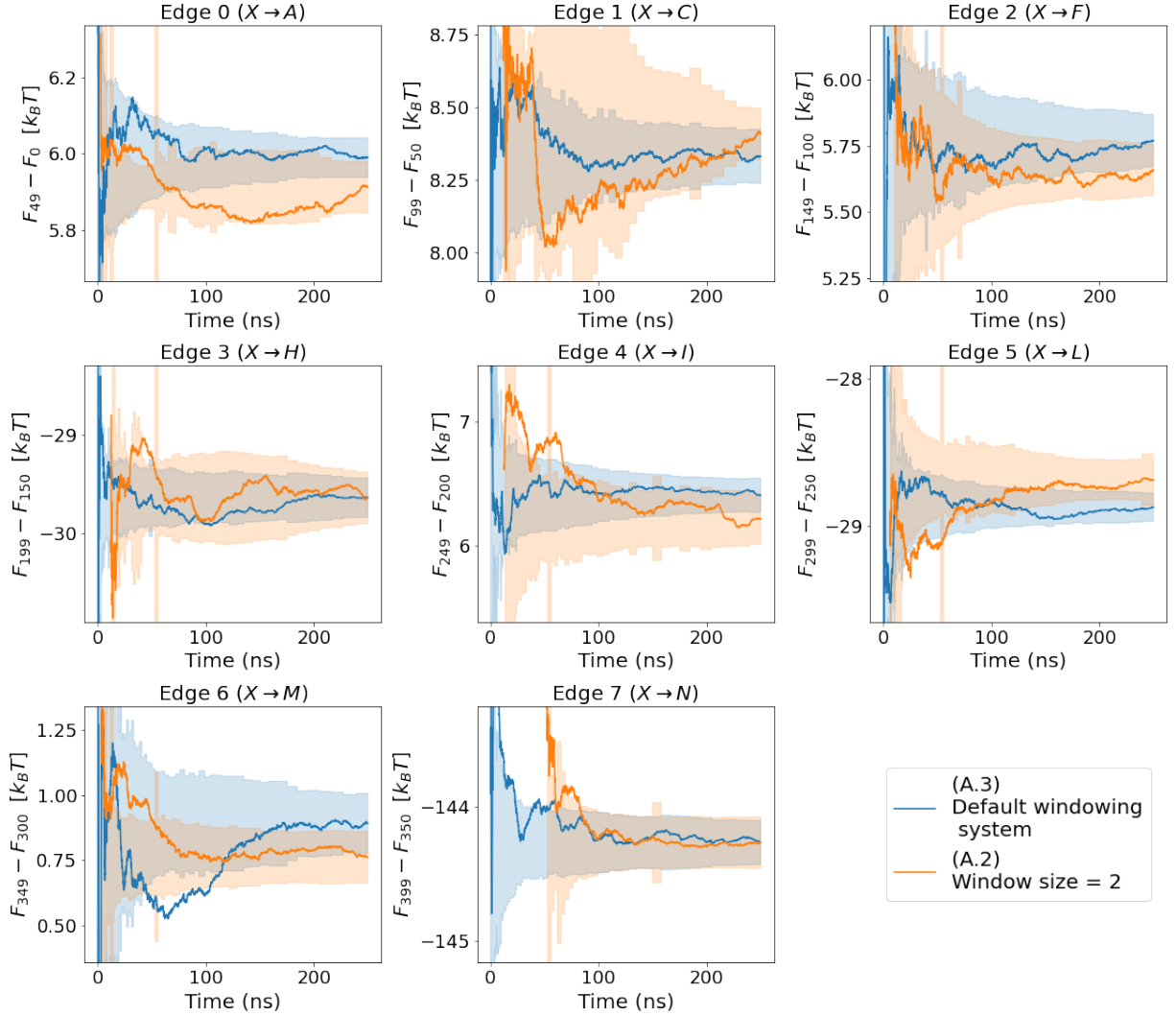


Figure 7: Window size influences the error bars of the free energy estimates. Plotted are simulations using windows of size 2 (A.2) and the default setup (A.3). The free energy difference estimates are shown for each of the eight edges along with their error bars. The error bars are larger and noisier for the simulation with windows of size 2 than when the default setup is used. It also takes longer to produce estimates for each edge with the size 2 windows than with our default windowing system; the delay is particularly notable in Edge 7, where it takes almost 50 ns before an estimate is available. This is due to the diffusive sampling of Λ , as explained in Figure 6; replicas do not sample the rungs associated to Edge 7 sufficiently to produce an estimate until after 50 ns.

noisier and an order of magnitude larger. The cause (and consequence) was the poor sampling of Λ : Without a mechanism to force them to explore the space Λ more broadly, each of the 32 TSS samplers was stuck in a narrow region of rungs for a significant period early in the simulation. A comparison of the rung processes $\mathcal{K}^t(r)$ is shown in Figure 9 for two of the 32 replicas. This example illustrates how the lack of visit control can make it infeasible to quickly approach convergence.

Next we compare (B.2) to (B. η) for $\eta \neq 0, 2$. As indicated in Section 1.2, if η is too large, the variance of $\frac{\rho_k}{\gamma_k}$ is large, and its fluctuations dominate the visit control rung distribution (33), creating an essentially random bias in the rung distribution. With $\eta = 64$, for instance, we see in Figure 13 that, early in the simulation, the estimates fluctuated strongly compared to $\eta = 2$. If η is too small, the bias towards under-visited rungs is not strong enough, and in Figure 13 we see that the exponential slowdown is present to some extent for $\eta = 1$. The same simulations are shown on longer timescales in Figure 14, where the estimates for $\eta = 1$ and $\eta = 64$ are visibly worse than those for $\eta = 2$ for approximately the first third of the simulation; neither $\eta = 1$ nor $\eta = 64$ was worse than $\eta = 0$ (8), however. It is harder to decide between $\eta = 2, 4$, and 16 based on visual inspection (see Figure 15 and 16), but based on experience of other test systems, we generally find that a broad range of η works well. The choice of η affects only how quickly we reach a neighborhood of the asymptotic value, but does not affect the asymptotic variance; see Table 1 for the error estimates for each edge and for all simulated values of η .

10.2.3 Self-adjustment

The variance of any free energy estimator decreases as one decreases the interval at which samples are provided to the estimator, with diminishing returns as the interval shrinks to 0, owing to correlation between the samples. Because TSS performs both the sampling and estimation, it is able to reduce the variance of the free energy estimates in ways that are not possible when the estimator is decoupled from the method by which the samples are generated. We showed in Section 1.4 that when independent samples are available from $\rho_\lambda(x)dx$ for each $\lambda \in \Lambda$, the sampling of the rung process $\mathcal{K}^t$ strongly influenced the variance of the free energy estimates, a property called self-adjustment. We take the same approach with the current example. Let $\bar{P}_{\pi,F}^{x,n}$ denote the transition kernel acting on $\mathcal{S} \times [K] \times [J]$ that applies n steps of the MD transition kernel $\mathcal{T}_k(x'|x)dx$, similarly to $P_{\gamma,Z}^x$ in equation (5), but augmented with the windowing variable j , as described below equation (48) in Section 2.1. Instead of using, at time $t \geq 0$,

$$\bar{P}_{\pi^t,Z^t} = \bar{P}_{\pi^t,Z^t}^j \circ \bar{P}_{\pi^t,Z^t}^k \circ \bar{P}_{\pi^t,Z^t}^{x,n},$$

which involves a single application of $\bar{P}_{\pi^t,Z^t}^k$ ($v = 1$) and n applications of $\mathcal{T}_k$ per estimator update (as was the case for all the (A) and (B) simulations), we use the “high-frequency” kernel $\bar{P}_{\pi^t,Z^t}^v$, defined for $v > 1$ and any $\pi \in \mathcal{P}^e(K)$ and $Z \in \mathbb{R}_{>0}^K$ by

$$\bar{P}_{\pi,F}^v = \bar{P}_{\pi,F}^j \circ \bar{P}_{\pi,F}^k \circ \bar{P}_{\pi,F}^{x,\frac{n}{v}} \circ \dots \circ \bar{P}_{\pi,F}^k \circ \bar{P}_{\pi,F}^{x,\frac{n}{v}} \quad (10.1)$$

$$= \bar{P}_{\pi,F}^j \circ (\bar{P}_{\pi,F}^k \circ \bar{P}_{\pi,F}^{x,\frac{n}{v}})^{(v)} \quad (10.2)$$

Here it is assumed that v divides n . In other words, rather than performing a single long MD trajectory followed by one sampling event of the rung k , we break the MD trajectory into v slices and follow each slice with a sampling event of the rung k . The total number of MD steps between updates of the estimates π^t, Z^t stays constant (v applications of $\frac{n}{v}$ steps), but there are v times as many intermediate updates to $\mathcal{K}^t$: $\mathcal{K}^{t,1}, \dots, \mathcal{K}^{t,v}$ with $\mathcal{K}^{t+1} = \mathcal{K}^{t,v}$. All the intermediate moves at time $t \geq 0$ are performed within the same window using the estimates π^t, Z^t , and are generally less computationally demanding than the MD steps.

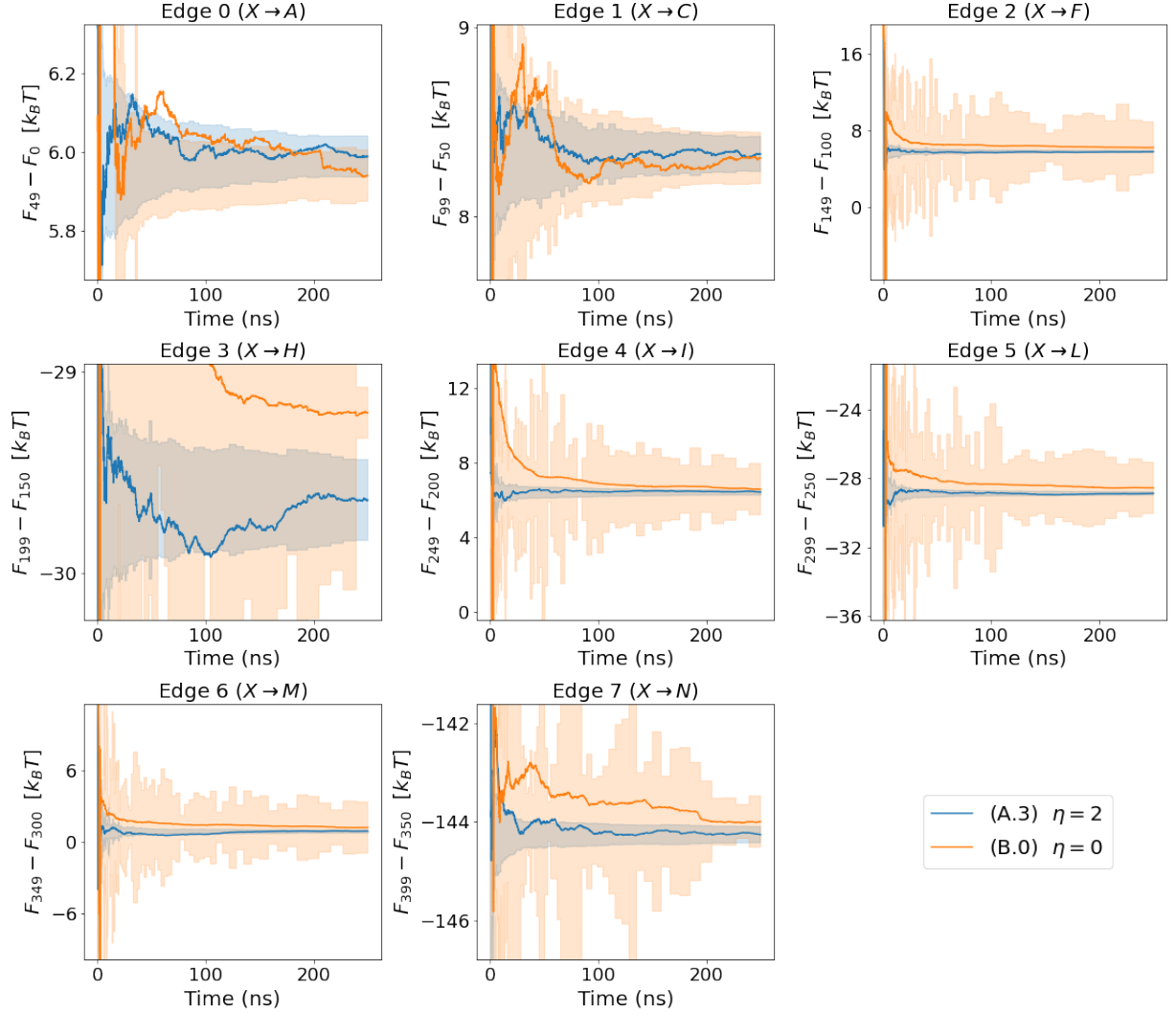


Figure 8: Visit control accelerates convergence of the estimates to a neighborhood of their equilibrium values. Simulations using $\eta = 0$ (B.0) and $\eta = 2$ (B.2) are shown. The free energy difference estimates are shown for each of the eight edges, along with their error bars. Use of visit control ($\eta = 2$) removes the exponential slowdown of the estimates apparent in its absence ($\eta = 0$), wherein the curves have a steady, small negative slope (see especially edges 2 through 7). Edges 0 and 1 rapidly converge to a neighborhood of the equilibrium even without visit control, because the substructure X is very similar to both A and L, making the estimation easier.

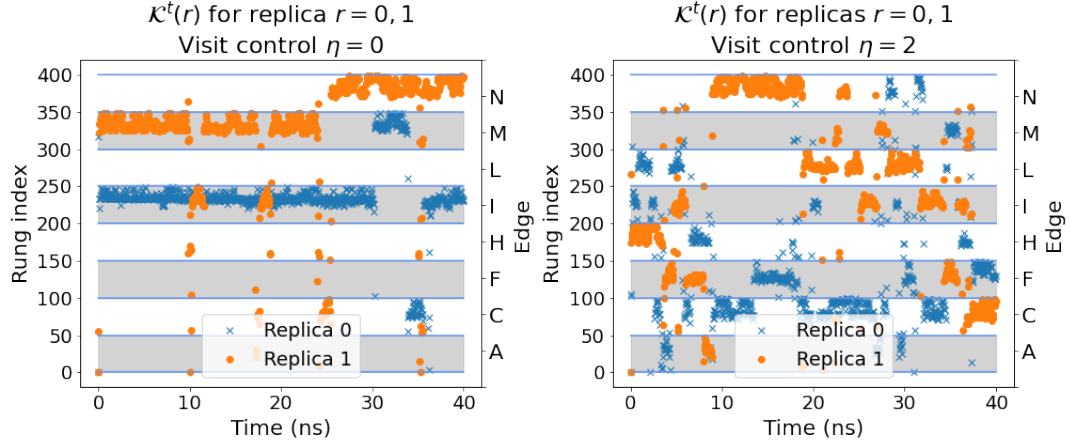


Figure 9: The visit control mechanism accelerates convergence in part by promoting efficient sampling of the Λ space. Plotted are simulations using $\eta = 0$ (B.0) and $\eta = 2$ (B.2). The rung trajectories for two of the replicas (with indices 0 and 1), which are representative of the behavior of the remaining 30 replicas, are shown for the first 40 ns of simulation. Early on, the rung process in the $\eta = 0$ simulation struggled to explore Λ . The $\eta = 2$ simulation, on the other hand, immediately started exploring across all eight edges, because the visit control mechanism pushes the replicas toward under-visited rungs. Past the 50 ns mark (not shown), the replicas for $\eta = 0$ began effectively exploring the rung space, and the visitation patterns became indistinguishable between $\eta = 0$ and $\eta = 2$ as the simulation progressed. Efficient exploration of the rung space is a necessary (but insufficient) condition for efficient estimation of the free energy differences (see Figure 8).

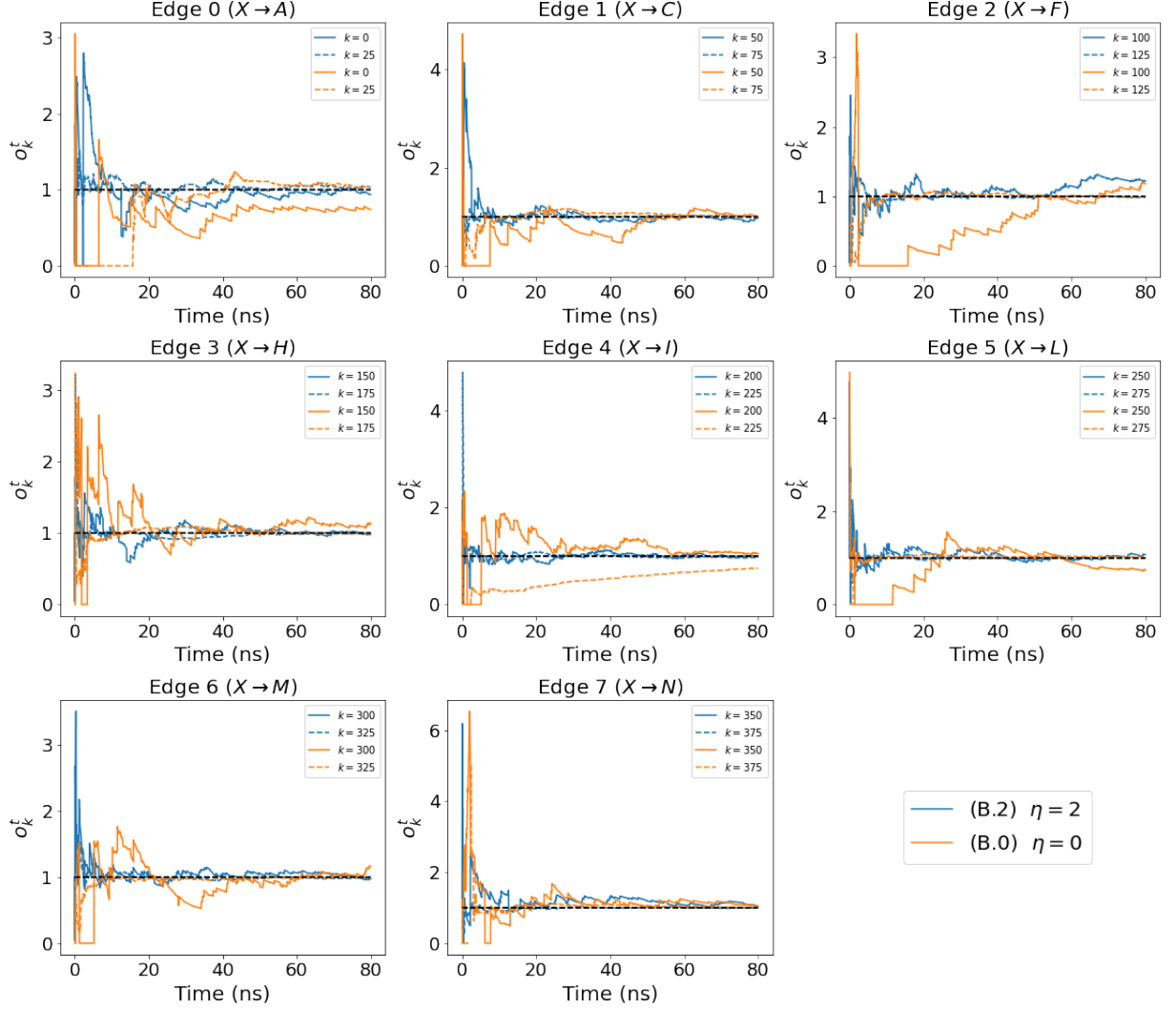


Figure 10: The visit control mechanism relies on estimates of the tilts o_k^t , which are plotted for two different runs per edge k for $\eta = 0$ (B.0) and $\eta = 2$ (B.2). Only the first 80 ns of the trajectory is shown. The tilts measure the proportion of empirical visits to their target visit density and converge to 1 asymptotically. When no visit control is used ($\eta = 0$), the proportion of visits can deviate greatly from 1.

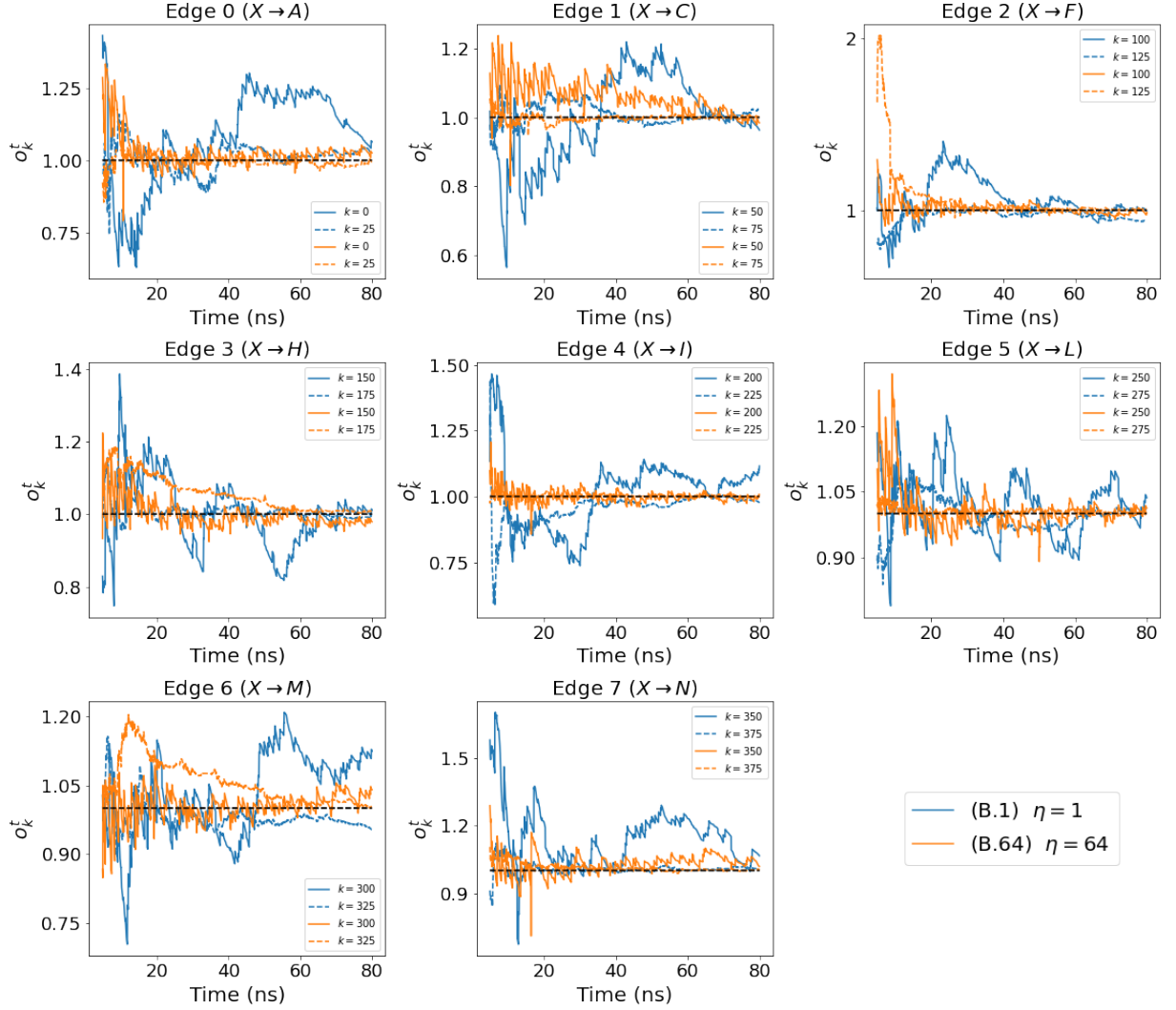


Figure 11: As in Figure 11, we plot the trajectories of the tilts o_k^l , this time for $\eta = 1$ (B.1) and $\eta = 64$ (B.64). The tilts are plotted on the time interval (5 ns, 80 ns), with the first 5 ns being dropped for visual clarity. When a strong visit control is used ($\eta = 64$), the tilts tend to stay closer to the target value of 1. However, given that the tilts provide a noisy estimate of the visit control free energies, this does not necessarily mean the free energy estimates are better (see Figure 14 for a comparison).

	A	C	F	H	I	L	M	N
(A.2)	5.911 (0.034)	8.409 (0.044)	5.658 (0.048)	-29.638 (0.130)	6.220 (0.104)	-28.686 (0.087)	0.761 (0.049)	-144.265 (0.096)
(A.3)	5.989 (0.026)	8.331 (0.046)	5.768 (0.049)	-29.636 (0.100)	6.407 (0.069)	-28.873 (0.048)	0.890 (0.058)	-143.66 (0.082)
(B.0)	5.940 (0.032)	8.310 (0.068)	6.194 (1.236)	-29.203 (0.064)	6.560 (0.686)	-28.552 (0.743)	1.188 (1.072)	-143.999 (0.259)
(B.1)	6.006 (0.029)	8.247 (0.051)	5.698 (0.061)	-29.663 (0.089)	6.305 (0.103)	-28.765 (0.057)	0.751 (0.063)	-144.291 (0.062)
(B.4)	6.016 (0.030)	8.380 (0.049)	5.702 (0.046)	-29.558 (0.077)	6.478 (0.087)	-28.756 (0.054)	0.943 (0.050)	-144.250 (0.077)
(B.16)	5.969 (0.030)	8.324 (0.061)	5.874 (0.053)	-29.513 (0.087)	6.370 (0.061)	-28.738 (0.058)	0.788 (0.058)	-144.115 (0.101)
(B.64)	6.036 (0.030)	8.391 (0.055)	5.730 (0.074)	-29.472 (0.121)	6.493 (0.071)	-28.755 (0.069)	0.827 (0.055)	-144.169 (0.103)
(C.1)	5.994 (0.018)	8.327 (0.033)	5.765 (0.036)	-29.636 (0.080)	6.453 (0.041)	-28.881 (0.047)	0.847 (0.032)	-144.235 (0.068)
(C.2)	6.010 (0.009)	8.334 (0.022)	5.728 (0.018)	-29.602 (0.047)	6.426 (0.022)	-28.773 (0.015)	0.843 (0.023)	-144.191 (0.051)

Table 1: Free energy difference estimate and root-mean-square error (in parentheses) at 250 ns for each edge, for each simulation. The expression used to compute the mean square error is provided in Section 8.

To study the effect of more rapid rung moves, as well as more rapid rung moves in combination with more rapid free energy updates, we performed simulations using the following setups:

- (C.1): we set $v = 25$ and the interval for free energy estimate updates to 9.6 ps; with our MD time step of 2 fs, this results in free energy updates every 4800 MD steps and rung moves every 192 steps.
- (C.2): we set $v = 1$ and the interval for free energy updates to 9.6 ps / 25 = 0.384 ps; this results in free energy updates and rung moves every 192 MD steps.

We compare (A.3), in which both free energy updates and rung moves occurred at intervals of 9.6 ps (4800 MD steps), to (C.1) and (C.2); these three simulations differ only in the intervals between free energy updates and rung moves. In Figure 12 we show the free energy estimates of simulations (A.3), (C.1), (C.2) along with their error bars. The simulations (C.1) and (C.2) have visibly lower variance than (A.3), with (C.2) having the lowest variance of all three (see Figure 12 and Table 1); indeed, (C.2) is statistically the best setup out of all our simulations. Notably, (C.1) has up to 1.8x lower estimated error than (A.3) depending on the edge, and (C.2) has up to 3.2x lower estimated error than (A.3) depending on the edge (see Table 1). These two comparisons suggest that a noteworthy portion of the error reduction from (A.3) to (C.2) is due to improved sampling of $\mathcal{K}^t$.

The significant variance reduction due to the improved rung sampling makes it relevant to consider the computational cost of an estimator update relative to the cost of a rung move. Estimator updates require information for all windows from all replicas, and thus incur a global communication cost that cannot be circumvented, on the top of the cost of solving the optimization problems described in 2.1 and 2.2. Rung moves, by contrast, require no communication between samplers, and can re-use estimates (all the moves

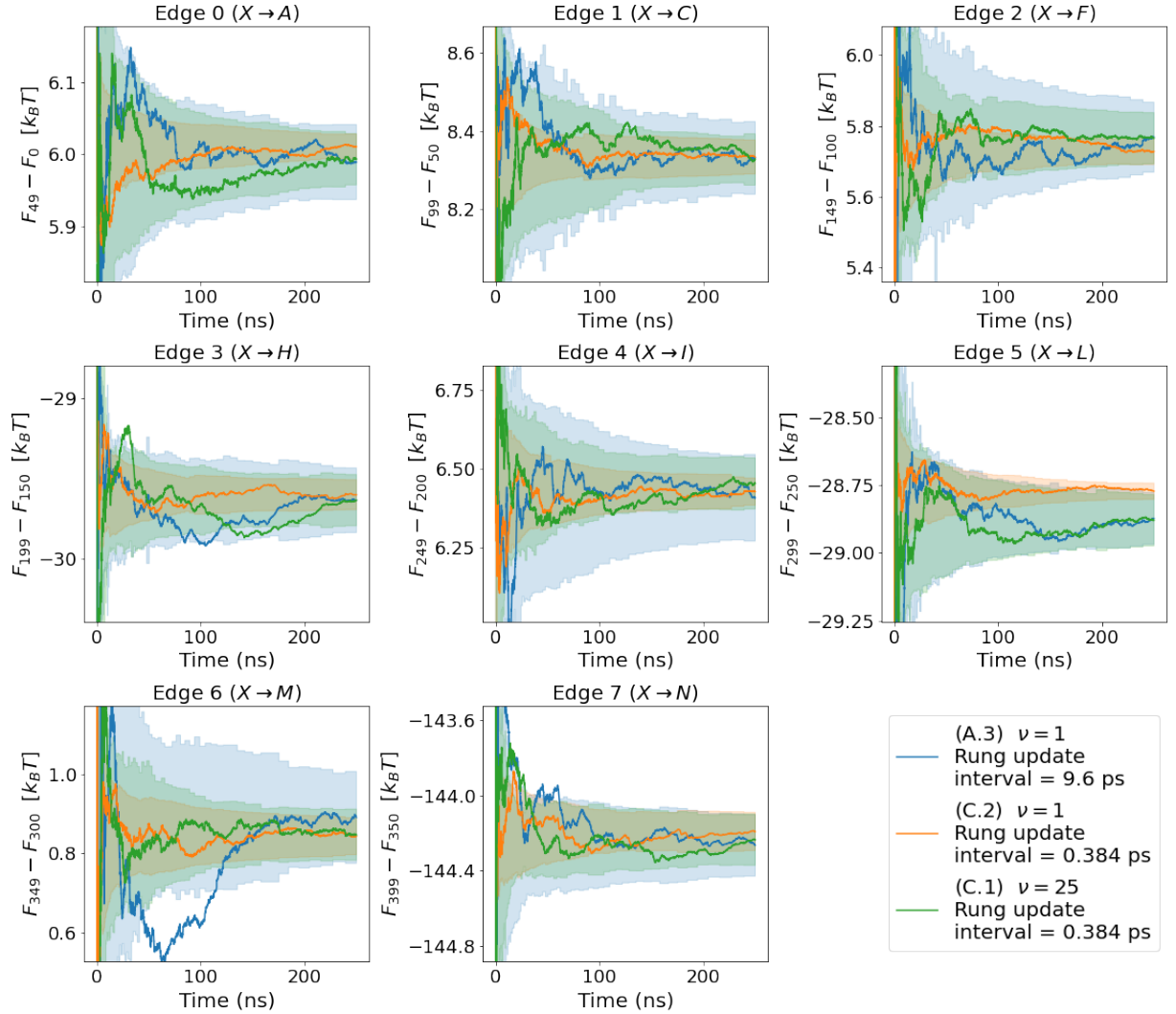


Figure 12: Increasing the frequency of rung updates can reduce the variance of the estimates, even if the number of estimator updates remains constant. The figure shows simulations that all use visit control with $\eta = 2$. Two simulations have $\nu = 1$ (i.e., one rung update per estimator update), one with the default interval between estimator updates of 9.6 ps (A.3) and one with a shorter interval of 0.384 ps (C.2); a third simulation (C.1) also has a short rung update interval of 0.384 ps but with $\nu = 25$, so estimator updates occur at the default interval. (C.2), which had high frequency rung and estimator updates performed the best out of all the simulations in terms of the estimated variance. Notably, (C.1) had smaller error bars than (A.3) for most of the edges, despite having the same estimator update frequency and only differing in rung update frequency. (Numerical values for the error bars are given in Table 1).

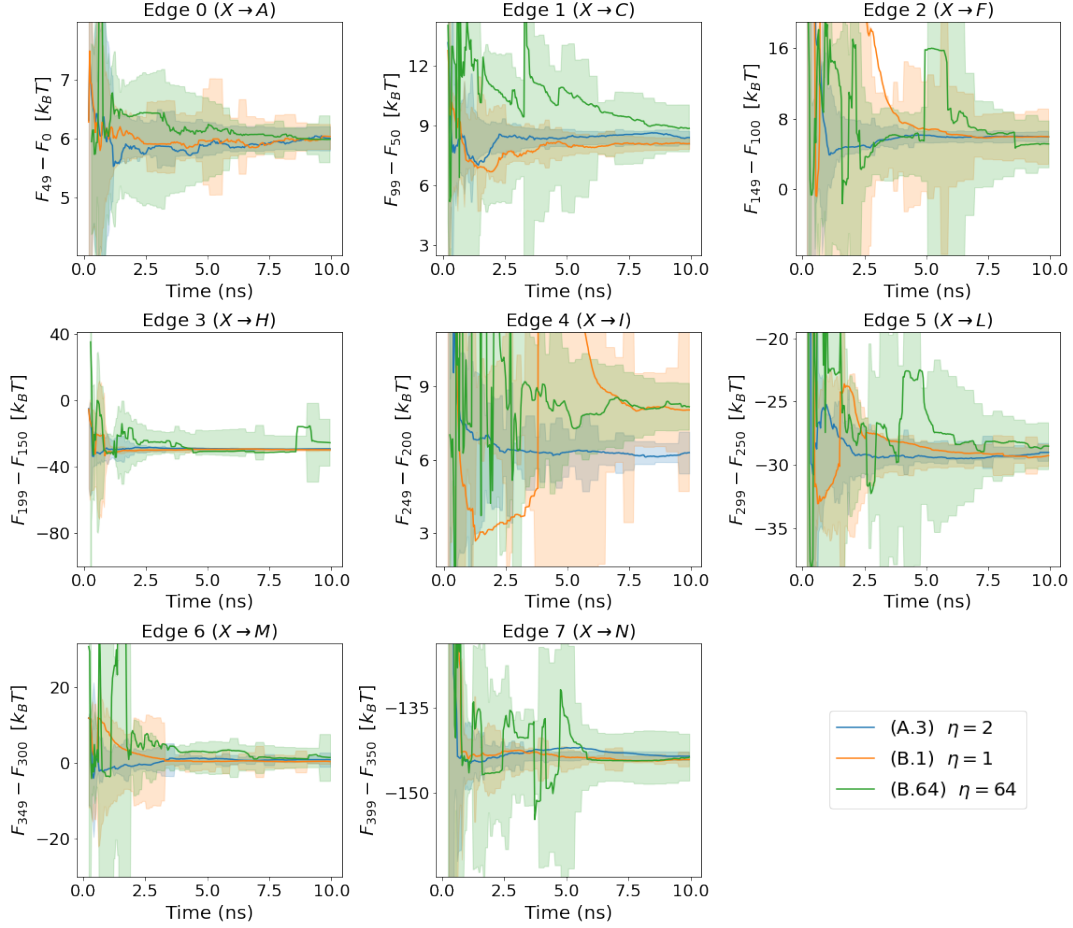


Figure 13: Visit control accelerates convergence to a neighborhood of the equilibrium values early in the simulation. Plotted are the first 10 ns of simulations using $\eta = 1, 2$, and 64 ((B.1), (B.2), (B.64), respectively, with (B.2) and (A.3) referring to the same simulation). The free energy difference estimates are shown for each of the eight edges, along with their estimated error bars. The $\eta = 1$ simulations suffer from the exponential slowdown described in Section 1.4, an effect which is particularly visible in Edges 2, 4 and 5. On the other hand, the $\eta = 64$ estimates tend to oscillate around the $\eta = 2$ estimates, which is particularly visible in Edges 2, 3, 5, and 7.

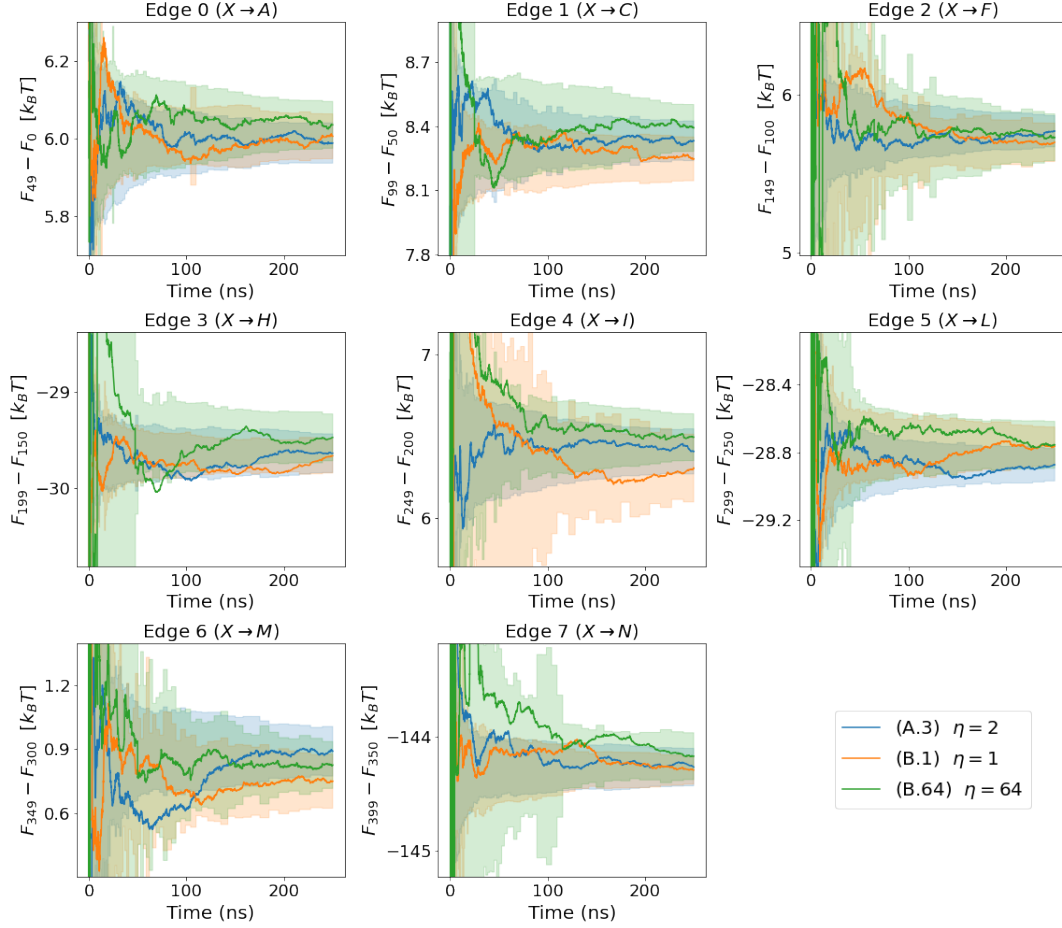


Figure 14: Visit control promotes efficient sampling and estimation for a broad range of $\eta > 0$, with decreasing efficiency at the extremes (when η is very small or very large). Plotted are simulations using $\eta = 1, 2$, and 64 ((B.1), (B.2), (B.64), respectively). The free energy difference estimates are shown for each of the eight edges, along with their estimated error bars. The error bars for the simulations with $\eta = 1$ and $\eta = 64$ are both large early in the simulation, and the estimates themselves sometimes appear oscillatory, especially within the first 50 ns. Afterwards, all three estimates and their error bars become comparable. It is worth noting that even these choices of η seem poor relative only to $\eta = 2$. The estimate for nearly every edge was within $1 k_B T$ by the 10 ns mark (the exceptions being Edge 3 for $\eta = 64$ and Edge 4 for $\eta = 1$); both $\eta = 1$ and $\eta = 64$ fare far better than $\eta = 0$, which corresponds to disabling visit control (see 8).

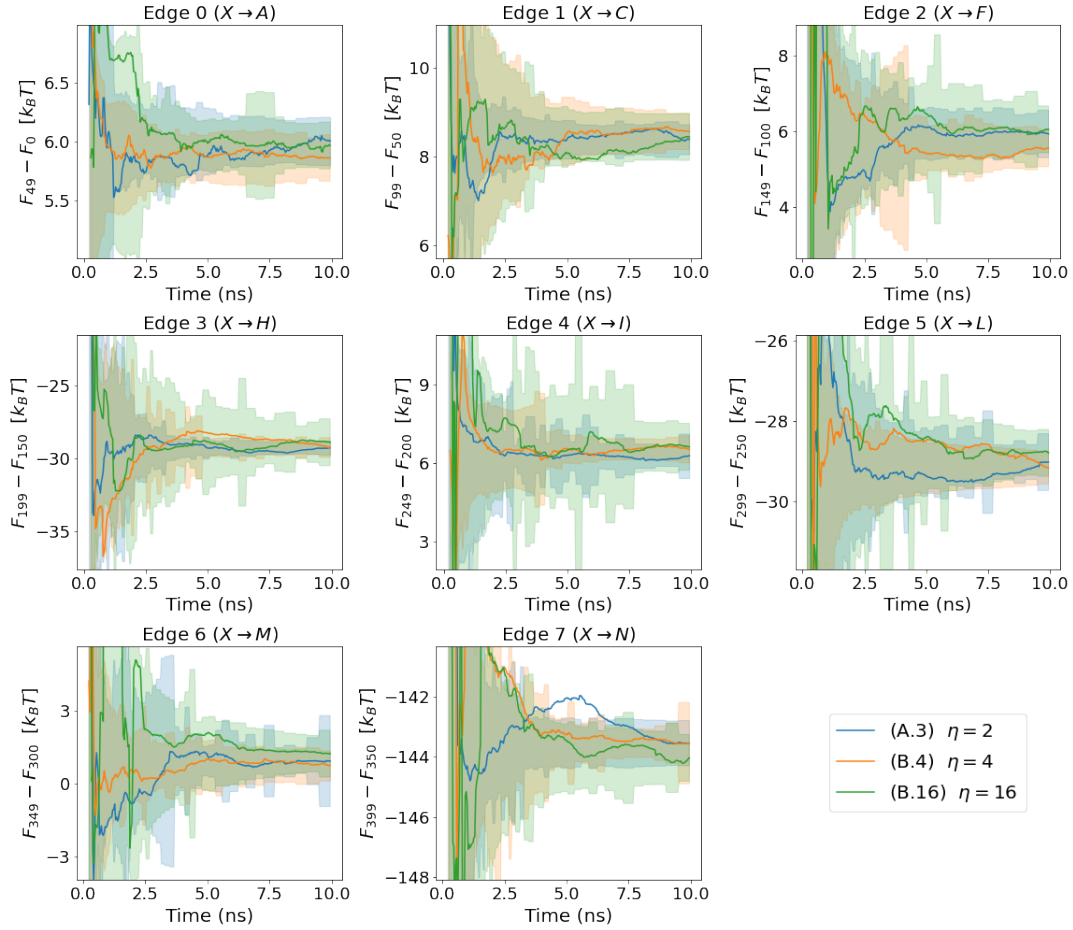


Figure 15: Visit control comparably accelerates convergence for a broad range of $\eta > 0$. Plotted are simulations using $\eta = 2, 4$, and 16 ((B.2), (B.4), (B.16), respectively). The free energy difference estimates are shown for each of the eight edges, along with their estimated error bars. It is difficult to distinguish between the three values of η after 5 ns, though we note that the most extreme value, $\eta = 16$, does appear to fluctuate more strongly prior to the 5 ns mark.

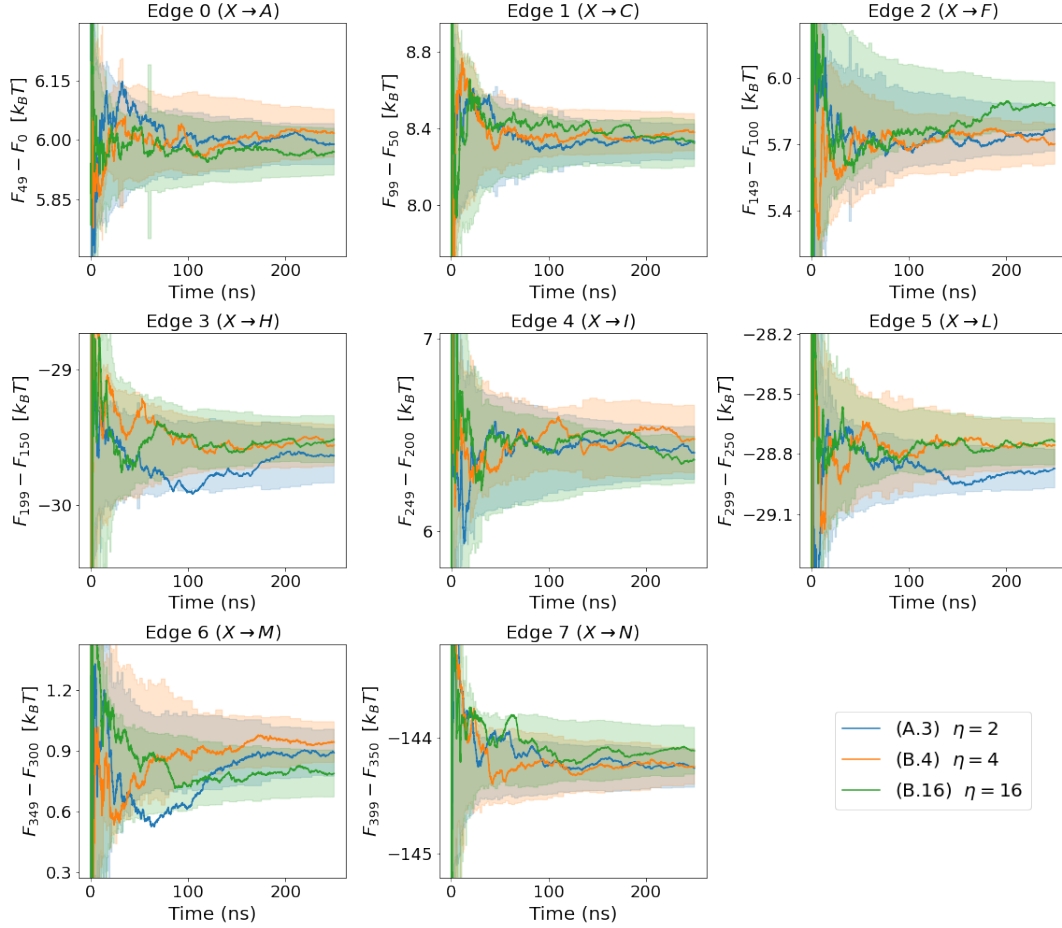


Figure 16: Visit control comparably accelerates convergence for a broad range of $\eta > 0$. Plotted are simulations using $\eta = 2, 4$, and 16 ((B.2), (B.4), (B.16), respectively). The free energy difference estimates are shown for each of the eight edges, along with their estimated error bars. Though the estimates for $\eta = 16$ tend to fluctuate more early in the simulation, on longer timescales the three choices of η are not distinguishable through their estimated error bars.

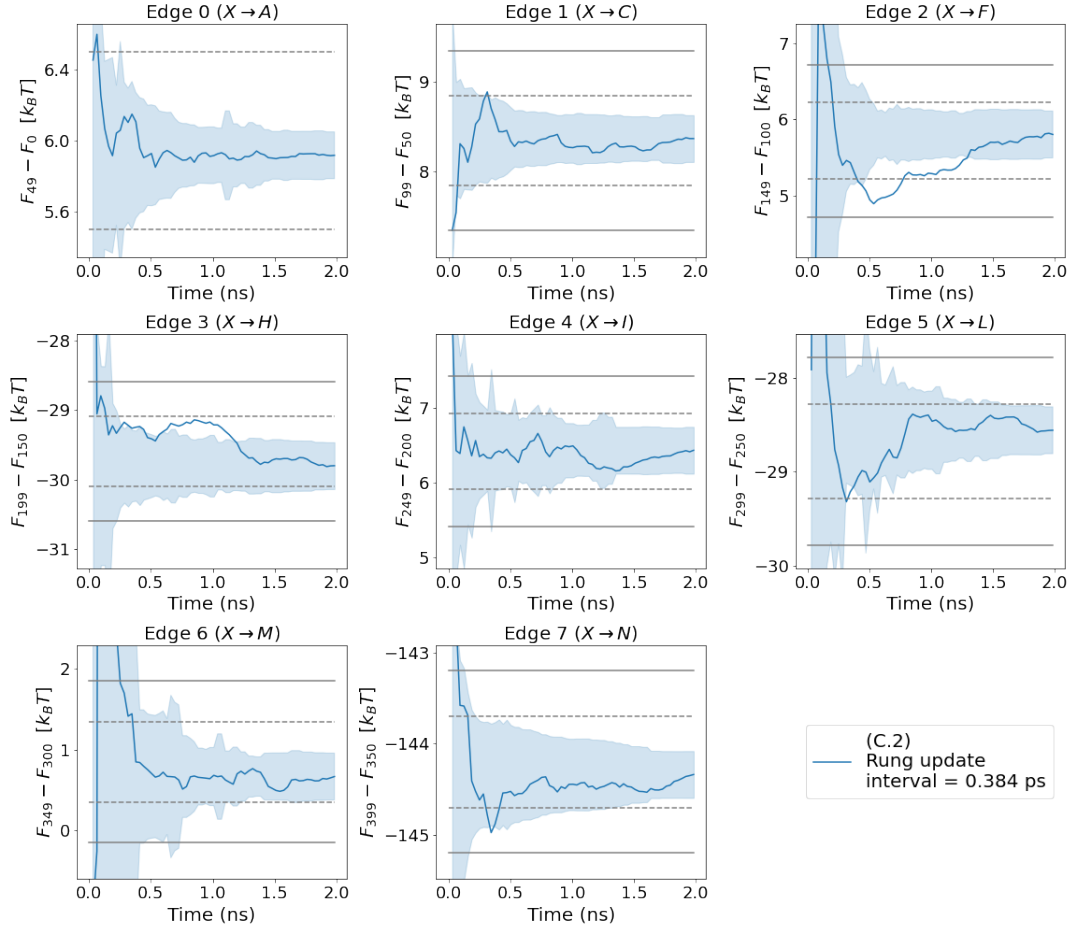


Figure 17: When applied in practice, TSS provides useful estimates quickly. We performed long timescale simulations for the solvation free energy calculations in order to study the asymptotic behavior of the algorithm, but in practice, one would typically run much shorter simulations. TSS provides estimates that reach a small neighborhood of the equilibrium values quickly. The figure shows the results of simulation (C.2), using $v = 1$ and a rung update interval of 0.384 ps, over the course of the first 2 ns of simulation, with two such neighborhoods delineated. The first neighborhood, delineated by dashed lines, indicates the range within $0.5 k_B T$ of the estimates at 250 ns; the second one, delineated by solid lines, indicates the range within $1 k_B T$ of the estimates at 250 ns. For this simulation, all the estimates stay within the first neighborhood for the remainder of the simulation after the 1 ns mark, and within the second neighborhood after just 0.3 ns.

in (6.14) use the same γ, Z), which is particularly important on architectures constrained by communication latencies. The effect of self-adjustment can be so dramatic that one can construct examples for which the asymptotic variance of the TSS free energy estimator is arbitrarily small relative to the asymptotic variance of a traditional estimator such as MBAR, for v large enough (see Section 4.4).

10.3 Simulation details

The molecular dynamics time step was 2 fs. The system was run in the NPT ensemble (constant number of particles, constant pressure, and constant temperature). The barostat and thermostat were applied using the multigrator [17]. The MTK barostat [19] was used, and the Langevin thermostat was used.

For the integration, a reversible reference system propagator algorithm (RESPA) scheme [30] was employed, with an inner time step of 2 fs and an outer time step (for the far electrostatics) of 6 fs.

The volume of the cell used for the simulation was approximately $51.5 \text{ \AA} \times 51.5 \text{ \AA} \times 51.5 \text{ \AA}$ throughout the simulation. The reference temperature was 298.15 K, and the reference pressure was 1.01325 bar.

We used a softcore potential, described in Section 10.3.1. For the electrostatic calculations, we used the u -series [23] with a cutoff radius of 12 $\text{\AA}$.

10.3.1 Softcore potential

The Lennard-Jones 6-12 potential is

$$V_{\text{LJ}}(r) = 4\epsilon_{\text{LJ}}\phi((r/\sigma_{\text{LJ}})^6), \quad (10.3)$$

where $\phi(z) = z^{-1}(z^{-1} - 1)$. The function ϕ has derivative

$$\phi'(z) = z^{-2}(-2z^{-1} + 1). \quad (10.4)$$

Let $r_s \in (0, \sigma_{\text{LJ}})$ be a softening ratio and $z_s = (r_s/\sigma_{\text{LJ}})^6$ the corresponding softening z -point location in $(0, 1)$. We replace $\phi(z)$ with the softened version

$$\tilde{\phi}(z) = \begin{cases} \phi(z_s) + \phi'(z_s)(z - z_s) & \text{if } z \leq z_s, \\ \phi(z) & \text{if } z > z_s. \end{cases} \quad (10.5)$$

The softened *and* tapered interparticle potential is

$$V_{ip}(r) = \begin{cases} 4\epsilon_{\text{LJ}}\tilde{\phi}((r/\sigma_{\text{LJ}})^6) - V_{\text{LJ}}(r_c) & \text{if } r < r_c, \\ 0 & \text{otherwise,} \end{cases} \quad (10.6)$$

where $r_c > r_s$ is the cutoff radius.

We can make this parametric potential into a softcore potential by multiplying it by $\lambda \in [0, 1]$ and setting z_s depending on λ . While the dependence of z_s on λ can be optimized, for the purposes of our simulation it is sufficient to use

$$z_s(\lambda) = \lambda b_{\min} + (1 - \lambda)b_{\max}, \quad (10.7)$$

with $0 \leq b_{\min} \leq b_{\max} \leq 1$. The value $b_{\min}$ is normally set to 0, but it can be set larger if only a portion of the range of the LJ potential is machine representable. The value $b_{\max}$ is determined experimentally, to minimize the error for free energy differences between two ends with the LJ interactions turned on and off, respectively. In our simulations, we used $b_{\max} = 1.0$.

10.4 Concluding remarks on the numerical simulations

Each of the three mechanisms investigated in this section—windowing, visit control, and self-adjustment—affects the sampling of the space Λ in a different way. The common theme is that, through their effect on $\mathcal{K}'$, they can improve (or worsen) the free energy estimates. Our best setup (C.2) performs both sampling and estimation steps frequently, and yields useful estimates after just 1 ns of sampling (see Figure 17). We expect that within the framework provided by TSS there are other, as yet unknown parameters that could be estimated on the fly and used to modulate the sampling of $\mathcal{K}'$ through the choice of rung distribution π . We also expect that one can find transition kernels with better mixing properties than the simulated tempering kernel $P_{\pi,F}^k$. Finally, we expect that efficient implementations of the rung move will lead to sampling schemes in which $v > 1$ would be preferred.

10.4.1 Acknowledgments

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References

- [1] C. Andrieu and J. Thoms. A tutorial on adaptive mcmc. *Stat. Comput.*, (18):343–373, 2008.
- [2] Christophe Andrieu, Eric Moulines, and Pierre Priouret. Stability of stochastic approximation under verifiable conditions. *SIAM Journal on Control and Optimization*, 44(1):283–312, 2005.
- [3] Y. Atchade, E. Moulines, G. Fort, and P. Priouret. *Adaptive Markov Chain Monte Carlo: Theory and Methods (in Bayesian Time Series Models)*. Cambridge University Press, 2011.
- [4] Y. Atchade and J. Rosenthal. On the ergodicity properties of some adaptive mcmc algorithms. *Annals of Applied Probability*, (16):1462–1505, 2006.
- [5] Charles Bennett. Efficient estimation of free energy differences from Monte Carlo data. *Journal of Computational Physics*, 22:245,268, 1976.
- [6] A. Benveniste, M. Metivier, and P. Prioruet. *Adaptive Algorithms and Stochastic Approximations*. Springer-Verlag, 1990.
- [7] Chuanhai Liu Faming Liang and Raymen Carroll. Stochastic approximation in Monte Carlo computation. *Journal of the American Statistical Association*, 102(477):305–320, 2007.
- [8] Charles Geyer. Estimating normalizing constants and reweighting mixtures in markov chain Monte Carlo. Technical Report No. 568, School of Statistics, University of Minnesota, 1994.
- [9] R. D. Gill, Y. Vardi, and J. A. Wellner. Large sample theory of empirical distributions in biased sampling models. *Annals of Statistics*, 16(3):1069–1112, 1988.
- [10] H. Haario, E. Saksman, and J. Tamminen. An adaptive metropolis algorithm. *Bernoulli*, 7(2):223–242, 2001.
- [11] Jack Hale. *Ordinary Differential Equations*. Krieger Publishing Company, Malabar, Florida, 1980.

- [12] Christopher Kellett. A compendium of comparison function results. *Math. Control Signals Syst.*, (26):339–374, 2014.
- [13] Christopher M. Kellett. Classical converse theorems in lyapunov’s second method. *Discrete and Continuous Dynamical Systems Series B*, 20(8), 2015.
- [14] A. Kong, P. McCullagh, X.-L. Meng, D. Nicolae, and Z. Tan. A theory of statistical models for Monte Carlo integration. *J. R. Statistic. Soc. B*, 65(3):585–618, 2003.
- [15] H. J. Kushner and G. G. Yin. *Stochastic Approximation Algorithms and Applications*. Springer New York, New York, 1997.
- [16] F. Liang. Trajectory averaging for stochastic approximation mcmc. *Annals of Statistics*, 38(5):2823–2856, 2010.
- [17] Ross A. Lippert, Cristian Predescu, Douglas J. Ierardi, Kenneth M. Mackenzie, Michael P. Eastwood, Ron O. Dror, and David E. Shaw. Accurate and efficient integration for molecular dynamics simulations at constant temperature and pressure. *J. Chem. Phys.*, 139(16), 2013.
- [18] Jun Liu, Augustine Kong, and Wing Hung Wong. Covariance structure of the gibbs sampler with applications to the comparisons of estimators and augmentation schemes. *Biometrika*, 81(1):27–40, 1994.
- [19] G. J. Martyna, D. J. Tobias, and M. L. Klein. Constant pressure molecular dynamics algorithms. *J. Chem. Phys.*, 101(5), 1995.
- [20] Jose L. Massera. Contributions to stability theory. *Annals of Mathematics, Second Series*, 64(1):182–206, 1956.
- [21] Sean Meyn and Richard Tweedie. *Markov Chains and Stochastic Stability*. Cambridge University Press, 2nd edition, 2009.
- [22] Giles Hooker Michael Shirts, Eric Bair and Vijay S. Pande. *Phys. Rev. Lett*, 91(14), 2003.
- [23] Cristian Predescu, Adam K. Lerer, Ross A. Lippert, Brian Towles, J.P. Grossman, Robert M. Dirks, and David E. Shaw. The u-series: A separable decomposition for electrostatics computation with improved accuracy. *J. Chem. Phys.*, 152(8), 2020.
- [24] Herbert Robbins and Sutton Monro. A stochastic approximation method. *Ann. Math. Statistics*, 22(3):400–407, 1951.
- [25] V. Roy, A. Tan, and J. Flegal. Estimating standard errors for importance sampling estimators with multiple Markov chains. *Statistica Sinica*, (28):1079–1101, 2018.
- [26] Michael Shirts and John Chodera. Statistically optimal analysis of samples from multiple equilibrium states. *The Journal of Chemical Physics*, 129, 2008.
- [27] Zhiqiang Tan. On a likelihood approach for Monte Carlo integration. *Journal of the American Statistical Association*, 99(468):1027–1036, 2004.
- [28] Zhiqiang Tan. Optimally adjusted mixture sampling and locally weighted histogram analysis. *Journal of Computational and Graphical Statistics*, 4(1):54–65, 2016.

- [29] Wolfgang Hahn (translator: Arne Baartz). *Stability of Motion*. Springer-Verlag New York, 1967.
- [30] M. Tuckerman, B. J. Berne, and G. J. Martyna. Reversible multiple time scale molecular dynamics. *J. Chem. Phys.*, (97), 1992.
- [31] Y. Vardi. Empirical distributions in selection bias models. *Annals of Statistics*, 13:178–203, 1985.
- [32] R.W. Zwanzig. High-temperature equation of state by a perturbation method. *J. Chem. Phys.*, (22):1420–1426, 1954.